Internal Medicine

General/ Geriatric

A 70-year-old man is brought to the clinic with an acute confusional state. On examination, he has postural hypotension and dry mucus membranes.
Test Result Normal Value
Calcium 3.41 2.15-2.62 mol/L

Which of the following is the most appropriate initial treatment?
A. I.V. 0.9% sodium chloride
B. I.V. sodium pamidronate
C. Oral prednisolone
D. I.V. furosemide

Answer: A

A 67-year-old man is brought to the Emergency Department by his family with confusion and a decreased level of consciousness. The family reported his being feverish and feeling unwell for 2 days that was also associated with diarrhea and poor oral intake. The patient has bronchial asthma, hypertension, and an old myocardial infarction. He has not been able to take his regular medications for 2 days due to the acute illness. On examination, the JVP is not visible, heart sounds are normal, no cardiac murmurs, and lungs are clear with no rhonchi or crepitation. The patient's extremities are cold and peripheral pulses feeble (see report).

Blood pressure 83/45 mmHg
Heart rate 120 /min (regular)
Respiratory rate 20 /min
Oxygen saturation 95% (room air)

ECG:
Sinus tachycardia and evidence of an old inferior myocardial infarction.

Which of the following is the most appropriate next step in management?
A. IV boluses of normal saline
B. Intravenous beta blockers
C. Nasogastric tube feeding
D. Dopamine infusion

Answer: A

A 68-year-old woman with history of heart failure, stroke and diabetes, admitted for hysterectomy. She was kept on normal saline due to poor oral intake. 2 days post-operation, the nurse noticed the patient pulse oxymetry was dropping with respiratory distress. On lung auscultation, there was bilateral basal crepitations Which of the following is the best action to prevent this complication?
A. Cardiology and pulmonary consultation
B. Oxygen by nasal cannula at 2 L/min
C. Furosemide 40 mg iv on post-op day 1
D. Daily reassessment of fluid status

Answer: D

A 78-year-old man presents with a 10-day history of intermittent colicky abdominal pain. The pain is low, central and seems to be worse after eating. He has no associated fever or vomiting but does feel nauseated when the pain is present. He says that prior to this episode he had hard stools once or twice a week that were difficult to pass. For the past several days, he has had only watery stools, several times a day. On examination, there is fullness in his left lower quadrant with nonspecific tenderness diffusely and no guarding or rebound. A urine dipstick is normal.
Which of the following is the most likely diagnosis?
A. Acute colitis
B. Constipation
C. Ureteric stone
D. Urinary tract infection

Answer: B
Which of the following is the best exercise to improve the function of a 68-year-old woman?
A. Walking
B. Swimming
C. Resistance training
D. Stationary bicycling

Answer: C
A 72-year-old man complains of difficulty in getting up from the chair for the last 3 months. History reveals he has controlled type 2 diabetes and long standing hypertension. Which of the following is the most leading type of injury in this patient?
A. Road traffic accidents
B. Fire-related injuries
C. Gunshot wounds
D. Falls

Answer: D

A 65-year-old man who is asymptomatic and has no known allergies is seen for a general examination. What is the best approach for pneumococcus immunization?
A. PPSV23
B. PCV13 followed by PPSV23
C. PPSV23 followed by PCV13
D. No evidence supports vaccinating this patient

Answer: B
A 64-year-old man presents to the clinic after having first episode of right knee pain for 2 weeks duration. He asks for a medicine to relieve his pain. He does not have any other symptoms apart from epigastric burning. He has no history of injury to the knee. Examination of knee is normal. Which is the most appropriate analgesic for this patient at this stage?
A. Aspirin
B. Codeine
C. Ibuprofen
D. Paracetamol

Answer: D
An 81-year-old woman reports low back pain that began after a forceful coughing episode. Lumbar radiographs reveal features of a lumbar compression fracture. Which of the following is the most appropriate initial step in her care?
A. Prescribe nasal calcitonin
B. Prescribe acetaminophen

C. Admit for vertebroplasty
D. Infuse zoledronate

Answer: B
A 77-year-old woman with stage 3 chronic kidney disease and pain from osteoarthritis of the first carpometacarpal joint. Which of the following is the best initial treatment for her pain?
A. Tendon-release surgery
B. Splinting and occupational therapy
C. Ibuprofen 800mg 3 times daily with food
D. Glucocorticoid injection into the first carpometacarpal joint

Answer: B
Which of the following physical maneuvers is most likely to assist the risk of future falls for a healthy 68-year-old man?
A. Test for nystagmus
B. Get-up-and-go test
C. Test for pulsus paradoxus
D. Lumbar spine flexibility test

Answer: B
A 75-year-old man with difficulty walking and left-sided leg weaknesses due to osteoarthritis of the hip. In which hand should a cane be used by this patient?
A. Left hand
B. Right hand
C. Either hand
D. Both hands simultaneously

Answer: B
A 71-year-old man is referred to the Outpatient Clinic for advice on his antihypertensive treatment. He has a history of stroke and longstanding uncontrolled hypertension. 6 blood pressure recordings at home and at his surgery within the past 2 weeks were 188-216/03-112 mmg. His therapy comprised ramipril, atenolol, and amlodipine (see lab results).
Test Result Normal Values
Sodium 138 134-146 mmol/L
Potassium 2.9 3.5-5.1 mmol/L
Urea 8 2.75-7.4 mmol/L
Creatinine 88 44-115 mol/L
Which of the following is most likely contributing to the treatment failure?
A. Antacids
B. NSAIDs
C. Table salt
D. Pseudoephedrine

Answer: B

Infectious disease

17. A 43-year-old patient man presents with headache and neck stiffness (see reports).

CSF Analysis:
Appearance cloudy
Glucose low
Protein high
White cells 100 /mm3 (70% polymorphs)

CT Brain:
Normal.

Which of the following is the most likely cause of the infection?
A. Cryptococcal
B. Tuberculous
C. Bacterial
D. Viral

Answer: C

18. A 22-year-old man presents with altered sensorium for the last 2 days and headaches for 5 days, with a history of fever for the preceding month. On examination, there is nuchal rigidity (see lab results).

Test Result Normal Values
Hb 100 130-170 g/L
WBC 11 4.5-10.5 x 109/L
Glucose, fasting 7.22 3.5-6.5 mmol/L

Cerebrospinal Fluid:
Test Result Normal Values
Appearance turbid clear
Cells 240 0-3 /uL
Total protein 3.6 0.22-0.33 g/L
Glucose 2.7 2.50-3.89 mmol/L
Neutrophils 23 0-1%
Lymphocytes 73 0-1%

Which of the following is the most likely diagnosis?
A. Septicemia
B. Tubercular meningitis
C. Phylogenetic meningitis
D. Viral meningoencephalitis

Answer: B

19. A 29-year-old woman comes to the Emergency Department with vomiting, profuse watery diarrhea and abdominal cramps. It started within 4 hours of eating a steak in a local restaurant.

Which of the following is the most likely cause?
A. Escherichia coli
B. Salmonella spp.
C. Campylobacter jejuni

D. staphylococcus aureus

Answer: D

1. A 72-year-old man presents to the Emergency Department with fever, night sweating, and cough for 2 weeks.
Which of the following is the most appropriate immediate action?
A. Bronchoscopy
B. Send sputum for Acid-fast bacilli
C. Isolate in a single, negative pressure room
D. Appointment in outpatient clinic after 2 weeks

Answer: C

A 35-year-old Indian man came to Saudi Arabia for Hajj. He presented with a 4-day history of intermittent fever, shortness of breath, drowsiness, convulsion and confusion. Which of the following is the most probable diagnosis?
A. Malaria
B. Influenza
C. Yellow fever
D. Tuberculosis

Answer: A

A 34-year-old male presents with chills, fever to 39 °C, retro-orbital pain, and severe muscle and joint pains. Which of the following is the most likely diagnosis?
A. Ebola
B. Chikungunya
C. Denque fever
D. Corona (MERS Cov)

Answer: C

Which of the following measures is recommended to avoid malaria infection in people traveling to an endemic country?
A. Vaccination
B. Immunoglobulin
C. Chemoprophylaxis
D. Avoiding highly endemic areas

Answer: C
Which of the following medications is the most likely that malaria parasites develop resistance to?
A. Malarone
B. Mefloquine
C. Chloroquine
D. Atorvaguone/proguanil

Answer: C
What is the most common diagnosis for travelers returning with fever from sub-Saharan Africa?
A. Malaria
B. Zika virus
C. Dengue fever
D. Travelers' diarrhea

Answer: A
What is the most common disease diagnosed among travelers returning with fever from Southeast Asia?
A. Malaria
B. Zika virus
C. Dengue fever
D. Travelers' diarrhea

Answer: C
which of the following is the peak biting period for Aedes aegypti mosquito (the primary vector of dengue fever)?
A. Early morning
B. Mid-day
C. Early night
D. Late night

Answer: A

A 20-year-old man is admitted as a suspected Ebola case. After few hours, the patient asked the treating physician to discharge him or he will leave the hospital. Which of the following is the most appropriate next action?
A. Call the hospitals security team
B. Consult the hospitals ethics committee
C. Refer the case to the infectious disease team
D. Ask the patient to sign the discharge against medical advice from

Answer: A

A 35-years-old man comes to the clinic complaining of fever. He just come back from a 2-weeks trip to a country in sub-Saharan Africa. Which of the following is the most likely cause of his fever?
A. Malaria
B. Zika virus
C. Dengue fever
D. Travelers' diarrhea

Answer: A
Which of the following preventive measures is recommended for travel diarrhea?
A. Avoid beverage with ice
B. Eat only in clean restaurants
C. Drink government-supplied water only
D. Avoid raw vegetables and eat fruits only

Answer: A
A 40-year-old man is planning to travel to Sudan for a short trip. Which of the following prophylactic medications is recommended for malaria prevention?
A. Chloroquine
B. Malaria vaccine
C. Atorvaguone/proguanil
D. No need for prophylaxis

Answer: C
Which of the following preventive measures is recommended for pilgrims from African meningitis belt countries in addition to quadrivalent (ACYW 135) vaccine against meningitis?

A. Strict surveillance for 6 days after arrival
B. Strict surveillance for 10 days after arrival
C. 1 tablet (500 mg) of ciprofloxacin chemoprophylaxis
D. Vaccination with a booster dose of the quadrivalent (ACYW 135) on arrival

Answer: C
According to the health regulations for travelers to Saudi Arabia for the pilgrimage (Haji), which of the following vaccines is required for pilgrims arriving from any country?
A. Meningitis
B. Yellow fever
C. Dengue fever
D. Poliomyelitis

Answer: A
A 40-year-old man comes to the clinic complaining of fever. He just arrived back from a short trip to a country in Southeast Asia. Which of the following is the most likely cause of his fever?
A. Malaria
B. Zika virus
C. Dengue fever
D. Travelers' diarrhea

Answer: C

A 71-year-old man who has been admitted due to clostriduim difficile infection last year. In order to search for possible risk factors that increase the risk of recurrent infection, his medications have been reviewed thoroughly. Which of the following medications is associated with increased risk of C. difficile infection?
A. Omeprazole
B. Ferrous sulfate
C. Calcium carbonate
D. Multivitamin tablet

Answer: A

A 20-year-old man presented with 3-months history of watery diarrhea, weight loss of 3 kg and abdominal cramps. He gave a history of travel to Mecca few months back. No other symptoms. Upper Gl endoscopy was normal. Smear from duodenal aspirate revealed many trophozoites (see lab results).
Test Result Normal Values Hb 109 130-170 g/L (Male) 120-160 g/L (Female)
Platelets count 459 150-400 x 109/L
Which of the following is the most likely diagnosis?
A. Salmonella
B. Giardia lambia
C. Clostridium difficile
D. Campylobacter jejuni

Answer: B
A 19-year-old man presented with 3-months history of watery diarrhea, weight loss of 3 kg and abdominal cramps. He gave a history of travel to India few months back. No other symptoms. A smear from duodenal aspirate revealed many trophozities (see lab results).
Test Result Normal Values
Hb 108 130-170 g/L (Male)
120-160 g/L (Female)

Platelets count 458 150-400 × 109/L
Which of the following is the most appropriate management?
A. Doxacyclin
B. Rifampicin
C. Metronidazole
D. Chloramphenicol

Answer: C
What is the best and simplest way to prevent giardiasis infection?
A. Hand washing
B. Avoid eating fruits
C. Avoid eating vegetables
D. Prophylactic antibiotics

Answer: A
A 67-year-old diabetic patient admitted to Intensive Care Unit with sepsis and urinary tract infection. Blood culture revealed methicillin-resistant staphylococcus aureus.
Which of the following is the most appropriate treatment option?
A. Cefuroxime
B. Vancomycin
C. Gentamycin
D. Flucioxacillin

Answer: B
What is the best and simplest precaution to prevent spread of MRSA infection?
A. Hand hygiene
B. Wearing gowns
C. Wearing face mask
D. Daily disinfection of the beds

Answer: A
A 25-year-old man developed fever, malaise and rash on his palm and soles. He gave a history of painless ulcer on his penis 6 weeks earlier. Examination was unremarkable a part from generalised lymphadenopathy. Which of the following is the most likely organism causing his presentation?
A. Coxiella burneti
B. Treponema pallidum
C. Staphylococcus aureus
D. Chylamdia trachomatis

Answer: B
A 33-year-old man presented with 2-weeks of lethargy, fever, generalised lymphadenopathy and cough. History revealed he is a business man travels frequently to Far East. On examination, he has oral candidiasis (see lab results).
Test Result Normal Values
RBC 4 4.7-6.1 x 1012/L (Male) 4.2-5.4 x 1012/L (Female)
Hb 13 130-170 g/L (Male)
120-160 q/L (Female)
Platelets count 145 150-400 x 109/L WBC 11 4.5-10.5 x 109/L Neutrophils 22 40-60% Lymphocytes 69 20-40 % Monocytes 9 2-8 %
Which of the following is the most likely diagnosis?
A. HIV
B. Syphilis
C. Brucellosis

D. Toxoplasmosis

Answer: A
A 56-year-old diabetic man presented with cough and fever. Chest-XR showed cavity at upper lobe of the left lung. What is the type of precaution required when dealing with this patient?
A. Droplet
B. Contact
C. Airborne
D. Standard

Answer: C

A 21-year-old man presents with paroxysms of coughing for the past week, following a mild flue illness. The paroxysms have been so severe that they have resulted vomiting and chest pain. He had been immunised for pertussis as a child. What is the best advice about the protection following immunisation?
A. Immunisation offers lifelong immunity against pertussis
B. Immunity following immunisation wanes within 3 years
C. Immunity following immunisation wanes within 10 years
D. Immunity following immunisation wanes after 25 years

Answer: B

A 22-year-old woman develops severe vomiting 4 hours after having lunch at a restaurant. What is the most likely causative organism?
A. Shigella
B. Campylobacter
C. Escherichia coli
D. Staphylococcus aureus

Answer: D

Which of the following stool electrolyte testing resuits would most represent a Vibrio cholerae?
A. Stool osmoles of 300 mOsm/kg
B. Stool osmoles of 410 mOsm/kg
C. Fecal osmotic gap of 30 mOsm/kg
D. Fecal osmotic gap of 110 mOsm/kg

Answer: C

A 35-year-old man presented with fever, rigors and weight loss for 10 days. He underwent Mitral valve replacement by prosthetic valve a month ago. Echo showed small vegetation. Blood pressure 90/70 mmHg, Heart rate 100 /min, Temperature 38.7 °C
Which of the following organisms is the most likely to be responsible for his presentation?
A. Coxiella burnetii
B. Staphylococcus aureus
C. Streptococcus viridans
D. Staphylococcus epidermidis

Answer: D

A 25-year-old man developed fever, malaise and rash on the palms of his hands as well as on the soles of both feet. 6 weeks earlier, he had a painless ulcer on his penis. He gave a history of recent travel and extramarital unprotected sex. On examination, he has generalized lymphadenopathy. Blood pressure 110/70 mmHg, Heart rate 87 /min, Temperature 38.6 °C
Which of the following is the most likely diagnosis?
A. HIV
B. Syphilis
C. Herpes zoster
D. Chlamydia trachomatis

Answer: B

A 43-year-old veterinarian man presents with 2-months history of progressive fatigue, back pain and fever. He also noticed a recent history of headache, mood /behaviour changes and paraesthesia. He is does not smoke with no history of alcohol or other significant medical history. On examination, he was uncooperative and agitated. Abdomen examination showed mild hepatosplenomegaly and tender right sacroiliac joint (see lab results). Blood pressure 110/70 mmHg, Heart rate 88 /min, Temperature 38.6 °C Test Result Normal Values
RBC 4 4.7-6.1 x 1012/L (Male) 4.2-5.4 x 1012/L (Female)
Hb 93 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 120 150-400 x 109/L WBC 9 4.5-10.5 x 109/L
Which of the following is the most likely diagnosis?
A. Syphilis
B. Brucellosis
C. Tuberculosis
D. Toxoplasmosis

Answer: B
A 41-year-old veterinarian man presents with 3-months history of progressive fatigue. back pain and fever. He also noticed a recent history of headache, mood /behaviour changes and paraesthesia. He does not smoke and with no history of alcohol or other significant medical history. On examination, he was uncooperative and agitated. Abdomen examination showed mild hepatosplenomegaly and tender right sacroiliac joint (see lab results).
Blood pressure 110/70 mmg
Heart rate 86 /min
Temperature 38.6 'C
Test Result Normal Values
Hb 98 130-170 g/L. (Male)
120-160 g/L (Female)
Platelets count 122 150-400 x 109/L
WBC 8 4.5-10.5 x 109/L.
Which of the following is the most appropriate next step in management?
A. Liver function
B. X-ray sacroiliac joint
C. Tuberculin skin test
D. Tube agglutination test

Answer: D
A 43-year-old man was recently diagnosed with neurobrucellosis. He is started on rifampicin (600 mg/day) and doxycycline (100 mg twice daily orally). which of the following is the recommended treatment duration?
A. 6 weeks

B. 8 weeks
C. 4 months
D. 6 months

Answer: D
A 22-year-old man known case with AIDS on active antiretroviral therapy (HART) presented with history of headache and weakness at left side of the body. On examination, he is found to have left hemiparesis, hemianopia, aphasia, ataxia, and 6/7 cranial nerve palsies. Investigation showed a rising titers of anti- toxoplasma immunoglobulin G (IgG) antibodies and positive immunoglobulin M (IgM) antibody (see report).
Brain MRI scans: Demonstrated multiple hypodense lesions in white matter and basal ganglia with mass effects.
Which of the following is the best treatment option?
A. Rifampicin and cefuroxime
B. Doxacyclin and clindamycin
C. Cholramphenicol and leucovorin
D. Sulfadiazine and pyrimethamine

Answer: D

A 25-year-old woman presented with 4-days history of confusion, lethargy and fever. On examination, she was drowsy with mild neck stiffness (see report).
CT scan brain: Showed area of low attenuation at right tempo-parietal region.
Which of the following is the most likely diagnosis?
A. Brain abscess
B. Premococcal meningitis
C. Herpes simplex encephalitis
D. Listeria meningoencephalitis

Answer: C

A 22-year-old woman previously healthy was admitted with 3-days history of fever, myalgia and pharyngitis. She gave history of 1-day macular rash over the trunk after she received amoxicillin from private clinic. On examination, she has generalised lymphadenopathy. Which of the following is the most likely diagnosis?
A. Diphteria
B. Hodgkin's disease
C. Infectious mononucleosis
D. Cytomegalovirus infection

Answer: C

An 82-year-old woman admitted to the hospital for meningitis. Her CSF was obtained and Gram stain showed Gram negative cocco-bacilli. Which chemo-prophylaxis should be offered to her contacts?
A. Rifampicin
B. Ceftriaxone
C. Ciprofioxacin
D. Azithromycin

Answer: A

A 35-year-old man presented to Emergency Room with recurrent attacks of high-grade fever with chills for 1 week. He was on a work trip to Sudan 3 weeks ago. Each attack begins with chills for few hours followed by fever up to 40.5 degree, which resolves with profuse sweating. The attacks comes every other day. On examination, mild splenomegaly was detected. Blood smear at Emergency Room was negative for parasites. Blood pressure 110/70 mmH, Heart rate 120 /min, Temperature 39 °C
Which of the following is the most appropriate next step in diagnosis?
A. Repeat thin blood smear
B. Repeat thick blood smear once more
C. Repeat blood smear during the attack
D. Repeat blood smear every 8 hours for 2 days

Answer: D
A 51-year-old woman presented to the hospital with 2-days history of malaise and headache. The headache became more intense and was associated with pain in her neck. Her husband reported that she had also fever. She was previously well with no significant medical history. On examination, she looked unwell confused, febrile and with positive Kernig's sign. CT scan brain was normal. Lumbar puncture done (see lab results).
Test Result Normal Values
Colour Turbid colourless
Cells 200 0-3 /pL
Total protein (Women) 0.9 0.15-0.45 g/L
Glucose 1.6 2.50-3.89 mmol/L
Which of the following is the most likely causative organism?
A. Escherichia coll
B. Listeria monocytogenes
C. Streptococcus pneumonia
D. Staphylococcus pyogenes

Answer: C

A 42-year-old man with no medical problems presents with a 2-day history of profuse bloody diarrhea, severe abdominal cramping and fever. He has recently returned from a week-long trip to Indonesia. His stool culture is positive for Campylobacter. Which of the following is the most appropriate treatment?
A. Metronidazole
B. Ciprofloxacin
C. Azithromycin
D. Amoxicillin

Answer: C
A 31-year-old man complains of persistent watery diarrhea and has noticed some weight loss. He returned 5 weeks ago from Egypt. He submits a stool sample that came back negative. Which of the following is the most likely diagnosis?
A. Giardiasis
B. Shigellosis
C. Amoebiasis
D. Ulcerative colitis

Answer: A
A 38-year-old man presented to the Emergency Department with 5-day history of fever and a productive cough (see lab results and reports)
Blood pressure 100/65 mHg
Temperature 39 °C

Test Result Normal Values
WBC 12 4.5-10.5 x 109/L
ESR 26 2-10 mm/h (Male)
3-15 mm/h (Female)
Chest X-ray: Right middle lobe infiltrate.
Sputum culture: Pending.
Which of the following is the most appropriate immediate therapy?
A. Moxifloxacin 400mg IV once daily
B. Ceftazidime 1 gm IV every 8 hours
C. Meropenem 1 gm IV every 8 hours
D. Piperacillin /Tazobactam 4.5 gm every 6 hours

Answer: A
A 27-year-old man presented to the Emergency Department with 2-days history of confusion and 5-hours history of seizures. Positive history of multiple unprotected sexual contact for the last 6 years. On examination, he was disoriented to time; place and person (see lab results and report.
Temperature 39 °C
Test Result Normal Values
WBC 1.3 4.5-10.5 x 109/L
Serum Cryptococcal Antigen: Positive.
Which of the following is the most appropriate diagnostic test?
A. HIV serology
B. CSF toxoplasma PCR
C. CSF culture for syphilis
D. Herpes Simplex Polymerase chain reaction (PCR)

Answer: A
A 22-year-old man recently diagnosed with HIV presented to the clinic for counselling. Whom should he disclose his HIV status to?
A. Wife
B. Parents
C. Brother
D. Employer

Answer: A
A 53-year-old man recently diagnosed with HIV came to the clinic for counselling. Who should be offered HIV testing?
A. Parents
B. Ex-wife
C. 28-year-old son
D. Office colleagues

Answer: B
A 69-year-old man known to have diabetes mellitus, presented to Emergency department with 4-day history of productive cough and tachypnea. On examination, oriented to time place and person. Chest exam revealed right lower lobe crepitation (see lab results and report).
Respiratory rate 23 /min
Temperature 38.5 °C
Test Result Normal Values
WBC 12 4.5-10.5 x 109/L
Urea 5 2.75-7.4 mmol/L
Chest X-Ray: Right lower lobe infiltrate.
Which of the following is the most appropriate management approach?

A. Admit to general ward and start IV antibiotics
B. Admit to Intensive care unit and start IV antibiotics
C. Start oral antibiotics and observe in Emergency Room for 24 hours
D. Start oral antibiotics and see in outpatient clinic in 7-days

Answer: A
A 70-year-old woman known to have hypertension on medication. Presented to the Emergency Department with 4-days history of productive cough, shortness of breath and tachypnea. Chest exam, right lobe zone crepitation. Temperature 38.5 °C
Which of the following is the most appropriate next diagnostic test to confirm the diagnosis?
A. Chest X-ray
B. Elevated WBC
C. Sputum culture
D. Echocardiogram

Answer: A
A 54-year-old man known to have diabetes and hypertension presented to Emergency Department with 9-days history of cough, headache, shortness of breath and fever. Cough was productive of white sputum, which has become yellowish lately. On chest examination, wheezing, and right side crepitation (see lab results and reports).
Temperature 38 °C
Test Result Normal Values
Reticulocyte 8 0.2-1.2 %
WBC 12 4.5-10.5 x 109/L
Cold agglutinin titre: 1:256
Chest X-Ray: Bilateral shadowing both lungs with more density on the right middle lobe. Which of the following is the most likely diagnosis?
A. Bronchial asthma exacerbation
B. Streptococcal pneumonia
C. Atypical pneumonia
D. Exacerbation of heart failure

Answer: C
A 28-year-old woman referred to the clinic with 3-weeks history of generalized fatigability, malaise, arthralgia, back pain and fever. Positive history of raw milk ingestion. Abdominal examination, mild hepato-splenomegaly (see lab results and report).
Temperature 39 °C
Test Result Normal Values
Hb 10.5 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 120 150-400 x 109/L
Blood culture: Gram negative cocco-bacilli.
Which of the following is the first line in treatment option?
A. Doxycycline and clindamycin
B. Doxycycline and streptomycin
C. Ciprofloxacin and trimethoprim sulfamethoxazole
D. Rifampin and trimethoprim sulfamethoxazole

Answer: B
A 28-year-old woman referred to the clinic with 3-week history of generalized fatigability. malaise, arthralgia, back pain and fever. Positive history of raw milk ingestion. Abdominal examination, mild hepato-splenomegaly (see lab results and reports).
Temperature 38.5 °C

Test Result Normal Values
Hb 10.5 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 120 150-400 x 109/L
Blood culture: Gram negative cocco-bacilli
MRI of the back: Sacroiliac joint spondylitis.
Which of the following is the optimal duration of therapy?
A. 3 weeks
B. 6 weeks
C. 12 weeks
D. 24 weeks

Answer: C
A 45-year-old housemaid from Southeast Asia, recently came to Saudi Arabia, presented to the Emergency Department with 2-weeks history of productive cough and shortness of breath. Positive history of hemoptysis. Chest exam showed right upper zone crepitations (see lab results and report.
Temperature 38.5 °C
Test Result Normal Values
WBC 14 4.5-10.5 x 109/L
ESR 73 2-10 mm/h (Male)
3-15 mm/h (Female)
Chest X-Ray: Right upper lobe infiltrate and cavitation.
Which of the following is the most appropriate next step?
A. IV ceftriaxone
B. Bronchoscopy
C. Sputum for acid fast bacilli
D. Isoniazid, rifampin, pyrazinamide and ethambutol

Answer: C
A 45-year-old man admitted to the hospital with history of productive cough and shortness of breath. He was found to have cavitatory pulmonary tuberculosis (TB. He was started on 4 anti- TB drugs. Few days later, he noticed that the urine color is red. Detailed history revealed family history of urethral stones (see lab resuits).
Test Result Normal Values
Hb 11 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 110 150-400 x 109/L
Which of the following is the most likely cause of his new symptom?
A. Urethral stones
B. Renal tuberculosis
C. Medication side effect
D. Hematological abnormalities related to TB

Answer: C
A 45-year-old man admitted to the hospital with history of productive cough and shortness of breath. He was found to have positive acid fast bacilli (AFB) in sputum. He was started on isoniazid, rifampin, pyrazinamide and ethambutol. He came back to the clinic for follow up after 3- weeks. Cough was better and no other symptoms (see lab results)
Admission lab results: Test Result Normal Values
Total bilirubin 5 3.5-16.5 mol/L
Aspartate aminotransferase 28 12-40 IU/L Alanine aminotransferase 32 5-40 IU/L
ESR 75 2-10 mm/h (Male)
3-15 mm/h (Female)

Follow-up lab results: Test Result Normal Values
Total bilirubin 0.7 3.5-16.5 mol/L
Aspartate aminotransferase 75 12-40 IU/L
Alanine aminotransferase 80 5-40 IU/L
ESR 35 2-10 mm/h (Male)
3-15 mm/h (Female)
Sputum AB Negative
Which of the following is the best management approach?
A. Continue anti-TB and repeat LFT in 1 week
B. Stop all anti-TB and repeat LFT in 1 weeks
C. Stop pyrazinamide and repeat LFT in 2 weeks
D. Stop isoniazid and rifampin and repeat LFT in 2 weeks

Answer: A; LFTs increased by less than 3 folds.
A 60-year-old man known to have hypertension and post aortic valve replacement on lisinopril 20 mg. amlodipine 5 mg and warfarin 3 mg once daily. He was admitted to the hospital with history of productive cough and shortness of breath, and was found to have positive acid fast bacilli (AFB) in sputum. First line four anti-TB drugs were started.
Which of the following regarding medications should be done?
A. Lisinopril dose needs to be decreased
B. Amlodipine needs to be stopped
C. Warfarin needs to be increased
D. Rifampin should be stopped

Answer: C
A 60-year-old man from India presented to the Emergency Department with headache and low-grade fever for 3-weeks. The headache became worse over the last 2-days and was associated with vomiting. In the emergency room, he developed generalized seizures (see lab results and reports),
Test Result Normal Values
WBC 19 4.5-10.5 x 109/L
Pressure 280 200-250 mmH20 (Male)
300-400 mmH20 (Female)
Cells 380 0-3 /ML
Total protein (Men) 3 0.22-0.33 g/L
Glucose 1.2 2.50-3.89 mmol/L
CSF differential WBC count: 73% lymphocytes and 13 % neutrophils.
CSF culture: Pending.
Which of the following is the most likely diagnosis?
A. Neurocysticercosis
B. Tuberculous meningitis
C. Bacterial meningitis
D. Viral meningitis

Answer:B
A 53-year-old woman known diabetic, hypertensive and cardiac arrhythmia on amiodarone, presented to Emergency Department with 3-days history of progressive redness and pain on her right leg. Examination showed a sharply demarcated red lesion, which was tender with no ulcer, Temperature 38.8 °C. Which of the following is the most likely diagnosis?
A. Erysipelas
B. Erythema nodosum
C. Side effect of Amiodarone
D. Necrobiosis lipodica diabeticorum

Answer: A
A 53-year-old woman known diabetic presented to the Emergency Department with 5-days history of progressive redness and pain on her right thigh. On examination, right thigh showed 15 cm red lesion, which was tender with no ulcer (see report).
Temperature 38.3 °C
Ultrasound: Showed inflammatory changes and 10 x 8 cm deep abscess collection. Which of the following is the best management approach?
A. Admit to ward and start IV antibiotics
B. Admit to ICU and start IV antibiotics
C. Discharge home on oral antibiotics with 1-week follow up
D. Admit to ward, start IV antibiotics, and surgical consultation

Answer: D
A 53-year-old woman recently returned from Egypt, known diabetic, presented to the Emergency Department with 3-days history of progressive redness and pain on her right shin. Positive history of allergy to penicillin. On examination, right leg has red lesion, which was tender with no ulcer. She was started on oral antibiotics. Next day she complained of abdominal pain, watery diarrhoea and fever. Temperature 37.8 °C
Which of the following is the most likely diagnosis?
A. Worsening of diabetic gastroparesis
B. Allergy to antibiotic
C. Clostridium difficile infection
D. Salmonellosis

Answer: C
A 63-year-old woman known diabetic for the last 20-years on insulin, presented to the Emergency Department with 3-days history of right knee pain and swelling. Right knee examination revealed, red, swollen and tender with severely limited range of movement. Other joints were normal (see lab result and report). Temperature 39 °C
Test Result Normal Values
WBC 17 4.5-10.5 × 109/L
Synovial fluid: WBC: 55.000
Crystals: Pending
Culture: Negative
Which of the following is the most likely diagnosis?
A. Charcot joint
B. Septic arthritis
C. Gout arthritis
D. Pseudo-gout arthritis

Answer: B
A 32-year-old man presents to the Emergency Department with 3-days history of generalized body aches, sore throat, runny nose, productive cough and shortness of breath. Chest exam shows right lung crepitations. The patient was admitted and according to the results of throat swab was started on oseltamivir (see report). Temperature 39 °C.
Chest X-Ray: Right middle lobe infiltrate.
Which of the following isolation precautions should be considered?
A. Contact
B. Standard
C. Airborne
D. Droplet

Answer: D

A 33-year-old woman presents with headache and myalgia for 5 days. 2 days ago, she started to have fever and noticed a skin rash on her face. She returned from Jeddah 10-days ago (see lab resuits).
Test Result Normal Values
Hb 8 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 80 150-400 x 109/L
Which of the following is the most appropriate next step in management?
A. Steroid
B. IV antiviral
C. IV antibiotics
D. Supportive management

Answer:D
A 27-year-old soldier going to a military mission at the Southern border of Saudi Arabia. His medical history is insignificant apart from a positive history of depression, which is controlled by medication. He came for counselling regarding malaria prevention. Which of the following is the best malaria chemoprophylaxis regimen?
A. Atovaquone-proguanil
B. Doxycycline
C. Mefloquine
D. Chloroquine

Answer: A
A 30-year-old man presents to the Emergency Department with history of fever and headache. He recently came from a trip to Sudan, where he stayed there for 10-days. Abdominal examination, palpable spleen 3 cm below costal margin (see lab results).
Temperature 40 °C
Test Result Normal Values
Hb 9 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 90 150-400 x 109/L
Aspartate aminotransferase 88 12-40 IU/L
Alanine aminotransferase 94 5-40 IU/L
Which of the following is the first line treatment option?
A. Quinine
B. Mefloquine
C. Chloroquine
D. Artemisinin based combination therapy

Answer: D

A 45-year-old man, heavy smoker, presents to the Emergency Department with 5-days history of fever and skin rash and 3-days of dyspnea and dry cough. Skin examination shows vesicular lesions over the truck and extremities, some of the lesions were crusted. Chest examination shows bilateral crepitation's (see report).
Respiratory rate 28/min
Temperature 39 °C
Oxygen saturation 82 %
Chest X-Ray: Diffuse bilateral infiltrate.
Which of the following is the most appropriate intravenous treatment option?
A. Ceftriaxone and azithromycin
B. Acyclovir
C. Steroids

D. Furosemide

Answer: B
A 22-year-old man recently returned from Umrah, presents to Emergency Department with 1- day history of fever and headache. Upon arrival to the Emergency Department, he started to have seizures on the right extremities. On examination, positive Kering sign (see lab results and report).
Temperature 39 °C
Test Result Normal Values
Cells 380 0-3 /pL
Total protein (Men) 1.2 0.22-0.33 g/L
CSF analysis: WBC differential count: 87% neutrophils and 5% lymphocytes.
Which of the following is the most appropriate intravenous treatment option?
A. Ceftriaxone, vancomycin and steroid
B. Ceftriaxone and vancomycin
C. Ceftriaxone and IV acyclovir
D. Acyclovir and steroid

Answer: A
A 22-year-old man presents to Emergency Department with 4-days history of fever, headache and confusion. On examination, Kemig's sign is present (see lab results and reports).
Temperature 39 °C
Test Result Normal Values
Cells 380 0-3 /pL
Total protein (Men) 0.8 0.22-0.33 g/L
Glucose 3.7 2.50-3.89 mmol/L
CSF analysis: WBC differential count: 5% neutrophil and 90% lymphocytes. CSF Culture: Pending.
Which of the following is the most appropriate intravenous treatment option?
A. Ceftriaxone, vancomycin and steroid
B. Ceftriaxone and vancomycin
C. Ceftriaxone
D. Acyclovir

Answer: D
A 22-year-old man recently returned from Hajj, presents to Emergency Department with 1-day history of fever and headache. On examination, Kerig's sign is present. Skin examination showed petechial rash on extremities and trunk (see lab results and report).
Temperature 39 °C
Test Result Normal Values
Cells 380 0-3 /pL
Total protein (Men) 0.9 0.22-0.33 g/L
Glucose 1.2 2.50-3.89 mmol/L
CSF analysis:
WBC differential: 87% neutrophils and 5% lymphocytes.
Which of the following isolation precautions should be considered?
A. Contact
B. Droplet
C. Standard
D. Airborne

Answer: B
A 27-year-old man recently returned from Umrah. Presents to Emergency Department with 1- day history of fever and headache. On examination, positive Kernig's sign and petechial rash on extremities and trunk. He was isolated and started on antibiotics (see lab results and report).

Temperature 39 °C
Test Result Normal Values
Cells 380 0-3 /pL
Total protein (Men) 0.9 0.22-0.33 g/L
Glucose 1.2 2.50-3.89 mmol/L
CSF analysis: WBC differential 87 % neutrophil and 5 % lymphocytes.
How many hours after starting antibiotics should isolation be discontinued?
A. 12
B. 24
C. 48
D. 72

Answer: B
A 64-year-old diabetic woman was recently started on hemo-dialysis through a femoral central line. During dialysis, the nurse notices that she is shivering. Detailed history reveals a 3-day history of fever. On examination, central line site was red with pus discharge. Which of the following is the most appropriate management approach?
A. Withdraw blood culture and start IV antibiotics
B. Start IV antibiotics and hold dialysis for 3 days
C. Withdraw blood culture, change the line and resume dialysis
D. Withdraw blood culture, start antibiotics and remove the line

Answer: D
A 64-year-old diabetic woman is receiving hemo-dialysis through a femoral central line. She develops line sepsis. Blood culture grows MRSA (methicillin resistant staphylococcus aureus). She was started on IV vancomycin 1000 mg every 72 hours (to be infused over 20 min). Immediately after starting vancomycin, she started to have flushing, redness and itching over the face, neck and trunk.
Which of the following is the most appropriate management approach?
A. Stop vancomycin and label her allergic to vancomycin
B. Continue vancomycin but a lower dose
C. Continue vancomycin but slower infusion rate
D. Change vancomycin to cefazolin 1000 mg every 24 hours

Answer: C
A 38-year-old woman came for routine follow up 2-months after delivery. She has no symptoms and her baby is well and lactating normally. Upon reviewing her labs, a positive urine culture: >100,000 fu/ml E. coli was noticed which was sensitive to trimethoprim/sulfamethoxazole, ciprofioxacin, and nitrofurantoin. Which of the following is the best treatment option?
A. Oral ciprofioxacin
B. Oral nitrofurantoin
C. Oral trimethoprim/sulfamethoxazole
D. No treatment needed

Answer: D
A 38-year-old woman, 28-weeks pregnant came for routine follow up. She has no symptoms and her pregnancy is all well. Upon reviewing her labs, a positive urine culture: > 100,000 fu/mi E.coli was found which was sensitive to ciprofioxacin, nitrofurantoin, trimethoprim/sulfamethoxazole and ceftriaxone Which of the following is the best treatment option?
A. Oral ciprofloxacin
B. Oral nitrofurantoin
C. Oral trimethoprim/sulfamethoxazole
D. No treatment needed

Answer: B
A 45-year-old diabetic woman presented to Emergency Department with fever, flank pain and dysuria. She is allergic to penicillin. Examination shows flank tenderness (see lab result and report).
Temperature 39 °C
Test Result Normal Values
WBC 17 4.5-10.5 x 109/L
Urine culture: > 100.000 cfu/ml E.coli.
Which of the following is absolutely contraindicated?
A. Piperacillin tazobactam
B. Ciprofloxacin
C. Meropenem
D. Ceftriaxone

Answer: A
A 45-year-old woman presented to Emergency Department with fever, flank pain and dysuria (see lab result and report).
Test Result Normal Values
WBC 14 4.5-10.5 x 109/L
Urine culture: > 100,000 cfu/mi E.coll
Which of the following is the best empirical treatment option?
A. Piperacillin tazobactam
B. Nitrofurantoin
C. Meropenem
D. Ceftriaxone

Answer: D; Pyelonephritis
A 48-year-old diabetic woman presented to Emergency Department with dysuria (see lab result and report).
Test Result Normal Values
WBC 14 4.5-10.5 x 109/L
Creatinine 230 44-115 mol/L
Urine culture: > 100,000 cfu/ml E.coli
Which of the following medications is contraindicated?
A. Nitrofurantoin
B. Ciproflaxacin
C. Piperacillin tazobactam
D. Trimethoprim / sulfamethoxazole

Answer: A; Because of renal impairment
A 62-year-old man presented with 2-weeks history of tiredness, dyspnea and 5-days history of fever. No previous surgical history. Cardiac exam shows pan-systolic murmur at the apex. Hands exam shows splinter hemorrhage on nails.
Temperature 38.8 °C
Which of the following is the most likely causative organism?
A. Staphylococcus epidermidis
B. Enterococcus species
C. Klebsiella pneumoniae
D. Streptococcus species

Answer: D
A 58-year-old man presented with 2-weeks history of tiredness, dyspnea and 5-days history of fever. No previous surgical history. Cardiac exam shows pan systolic murmur at the apex. Hands exam

shows splinter hemorrhage on nails Temperature 38.5 °C
Which of the following is the most appropriate empirical therapy?
A. Ceftriaxone
B. Gentamicin
C. Piperacillin tazobactam
D. Ceftriaxone and vancomycin

Answer: D
A 24-year-old woman is 10-weeks pregnant. She has dysuria and is diagnosed with a urinary tract infection. Which of the following antibiotics is contraindicated?
A. Nitrofurantoin
B. Ciprofioxacin
C. Amoxacillin
D. Ceftriaxone

Answer: B

Pulmonology

10. A young man comes into the clinic with intermittent hemoptysis for 2 months, associated with some blood in his urine. He is a non-smoker and was previously in excellent health. He also describes a weight gain of 20 kg over the past 3 weeks, with "puffy eyes" in the morning. He has an elevated serum creatinine concentration (see report).

Lung Biopsy: Serum immunoglobulin on basement membranes.

Which of the following is the most likely diagnosis?
A. Acute myocardial infarction
B. Goodpasture syndrome
C. DiGeorge syndrome
D. Graves' disease

Answer: B

11. Which of the following sites is the best to aspirate a sample of right pleural effusion with a syringe and needle?
A. Midclavicular line 6th intercostal space
B. Midaxillary line 6th intercostal space
C. Midaxillary line 9th intercostal space
D. Parasternal 6th intercostal space

Answer: C

12. A 45-year-old man presents with difficulty in breathing and a nocturnal cough for the last 3 weeks. Spirometry results show a forced vital capacity and the FEV/FVC ratio lower than normal.
Which of the following is consistent with these findings?
A. Reduced strength of inspiratory muscles
B. Decreased lung compliance
C. Increased lung compliance
D. Increased airway resistance

Answer: D

13. What is the optimal treatment for community-acquired pneumonia in an otherwise healthy patient?
A. Fluoroquinolone
B. Azithromycin
C. Vancomycin
D. Dicloxacillin

Answer: B

Explanation: Fluoroquinolone for comorbid pt.

14. What is the optimal duration of anticoagulation following an initial episode of pulmonary embolism?
A. 6 weeks
B. 3 months
C. 6 months
D. 12 months

Answer: C

15. In which of the following conditions is interstitial pneumonitis a characteristic histopathologic finding?
A. Viral pneumonia
B. Lobar pneumonia
C. Bronchopneumonia
D. Secondary tuberculosis

Answer: A

16. Which of the following pneumoconiosis will commonly predispose to the development of pulmonary tuberculosis?
A. Silicosis
B. Asbestosis
C. Anthracosis
D. Farmer's lung

Answer: A

17. A 38-year-old woman with pleurisy requires a thoracentesis. The needle should be placed on the midaxillary line of the affected side.
Which rib level should the needle penetrate through?
A. At the 4th
B. At the 5th
C. Between the 6th and 7th
D. Between the 8th and 10th

Answer: D

18. A 72-year-old man presents to the Emergency Department with fever, night sweats, and cough for 2 weeks (see report).

Chest X-ray:
Right upper lobe consolidation.

Which of the following is the best next step in management?
A. Perform bronchoscopy
B. Admit the patient to the Intensive Unit
C. 2 weeks follow up in outpatient
D. Isolate the patient in a single, negative pressure room

Answer: D

19. Which of the following can improve survival in patients with severe chronic obstructive pulmonary disease?
A. Inhaled corticosteroids
B. Supplemental oxygen
C. Smoking cessation
D. Beta agonist

Answer: B

20. A 25-year-old patient with moderate persistent asthma is currently on a low-dose, inhaled corticosteroid but needs to use a beta-2 agonist rescue inhaler at least once a day.
Which of the following should be added to the current treatment?
A. Theophylline to steroid regimen
B. Leukotriene inhibitor to the steroid regimen
C. Long-acting beta-2 agonist to steroid regimen
D. Ipratropium bromide to corticosteroid regimen

Answer: C

21. A 55-year-old patient with chronic obstructive pulmonary disease presents with shortness of breath. The patient is in severe respiratory distress and hypoxemic with decreased breath sounds and a prolonged expiration phase. There is no improvement after bronchodilators. The patient is conscious with a pH of 7.3.

Which of the following is the best next step in management?

A. Aminophylline IV
B. Methylprednisolone IV
C. Repeat bronchodilators
D. Non-invasive positive pressure ventilation

Answer: D

22. Which of the following is the most effective treatment for adults with obstructive sleep apnea?

A. Modafinil
B. Maintain a BMI less than 30 kg/m2
C. Mandible-positioning oral appliances
D. Continuous positive airway pressure

Answer: D

23. A 28-year-old patient with mild intermittent asthma has daytime symptoms about once to twice a week and none at night.

Which of the following initial treatment should be given when necessary?

A. Inhaled short-acting beta-2 agonist
B. Inhaled high-dose corticosteroids
C. Inhaled leukotriene modifiers
D. Oral aminophylline

Answer: A

24. A 28-year-old woman has had mild intermittent asthma for years. She uses a short-acting bronchodilator, as needed. Over the past 2 months, she has noticed that she is using her inhaler at least 3 times a week, and on occasions has had to wake up in the middle of the night to use her inhaler.

Which of the following treatments should be added?

A. Leukotriene receptor antagonist
B. Long-acting beta-agonist
C. Inhaled corticosteroid
D. Cromolyn

Answer: C

25. A 55-year-old man presents with shortness of breath and a productive cough. The patient has a history of smoking. On auscultation, there is wheezing throughout the chest (see reports).

Chest X-ray: Normal.

Arterial Blood Gases:

Normal.

Which of the following is the most likely diagnosis?

A. Pulmonary fibrosis
B. Occupational lung disease

C. Idiopathic pulmonary hemosiderosis
D. Chronic obstructive pulmonary disease

Answer: D

26. A 25-year-old patient suddenly develops chest pain and dyspnea. Vocal and tactile fremitus are reduced, and breath sounds are diminished (see report).
Chest X-ray:
It shows absent lung markings on the left side.

Which of the following is the most likely diagnosis?
A. Spontaneous pneumothorax
B. Pulmonary tuberculosis
C. Pulmonary embolism
D. Emphysema

Answer: A

27. Which of the following is the most likely cause of lobar pneumonia in adults?
A. Klebsiella pneumonia
B. Haemophilus influenzae
C. Staphylococcus pyogenes
D. Streptococcus pneumonia

Answer: D

28. A 45-year-old man is admitted with general tiredness and fever. There is a past history of a chest infection of 4-week duration, for which he was given oral antibiotics. He felt better initially, but never fully recovered and is now becoming worse (see report).

Chest X-ray:
It shows a right-sided pleural effusion.

Which of the following is the most likely diagnosis?
A. Lung cancer
B. Tuberculosis
C. Pulmonary embolism
D. Para-pneumonic effusion

Answer: D

29. A 45-year-old nurse working in the pulmonary clinic complains of fever and a productive cough for the last 3 months. She has received 2 courses of oral antibiotics but showed no improvement (see image).

Which of the following is the most likely diagnosis?
A. Pulmonary tuberculosis
B. H. Influenza pneumonia
C. Infectious mononucleosis
D. Streptococcal pneumonia

Answer: A

30. A 55-year-old patient presents to the clinic with shortness of breath and a cough productive of yellow sputum for the past 2 days. There has also been blood-stained sputum for the last 3 hours. The patient has smoked 2 packs of cigarettes a day for the past 35 years. On examination, there are rhonchi, wheezing, and crepitatons heard over the right hemithorax (see report).

Blood pressure 110/70 mmHg,
Heart rate 84 /min,
Respiratory rate 22 /min,
Temperature 38.8° C
Oxygen saturation 95% on room air

Chest X-ray:
It shows a dense infiltrate in the right lower lobe.

Which of the following is the most likely diagnosis?
A. Community-acquired pneumonia
B. Acute bronchitis
C. Tuberculosis
D. COPD

Answer: A

31. A 27-year-old woman with asthma has not had a severe attack for the last 3 months. However, she has a history of cough and wheezing 2-3 times each week.
Which of the following is the most appropriate management?
A. Ipratropium bromide
B. Inhaled corticosteroid
C. Salbutamol
D. Salmeterol

Answer: B

1. A 21-year-old woman comes to the clinic with episodes of shortness of breath on exertion for one year. She has to stop after 15-20 minutes of exercise, because of dyspnea and coughing, which get better after about one hour of rest. There are no symptoms at rest. Physical examination and spirometry are normal.

Blood pressure 125/75 mmH
Heart rate 80 /min
Respiratory rate 12 /min
Temperature 37.0° C

Which of the following is the most appropriate inhaled treatment before exercise?
A. Salmeterol
B. Salbutamol
C. Budesonide
D. Ipratropium

Answer: B

2. A 42-year-old man with asthma presents with daily symptoms despite regularly using a long-acting beta-agonist inhaler and an inhaled high-dose corticosteroid. He also uses a short-acting beta-agonist inhaler as "rescue" medication. These episodes also wake him from sleep 2-3 times a week. Physical examination shows bilateral (see table).
Blood pressure 140/85 mmH
Heart rate 90 /min
Respiratory rate 18 /min
Temperature 36.7° C
Forced Expiratory Volume in 1 Second Normal value
Before bronchodilator 65 % 25-75 %
After bronchodilators 91 %

Which of the following is the most appropriate next step in management?
A. Add theophylline
B. Add a leukotriene inhibitor
C. Observe the patient using his inhaler
D. Treat for gastroesophageal reflux disease

Answer: C

3. A 67-year-old patient presents to the Emergency Department with chest pain, shortness of breath, and hemoptysis following a car trip for more than 5 hours (see report).
ECG:
Right axis deviation, an S1-Q3-T3 pattern, and a right bundle branch block.

Which of the following is the most likely diagnosis?
A. Acute myocardial infarction
B. Bronchogenic carcinoma
C. Pulmonary embolism
D. Pericarditis

Answer: C

4. A 55-year-old smoker presents to a clinic with progressive shortness of breath (see report).

Chest X-ray:
It reveals a massive right pleural effusion.

Which of the following is the best next step in management?
A. Thoracentesis
B. High-resolution CT scan
C. Intercostal tube under water seal
D. Repeated bronchodilator inhalations

Answer: A

5. A 28-year-old patient develops the sudden onset of dyspnea and tachycardia 5 days post a knee surgery.

Which of the following is the most likely diagnosis?
A. Pulmonary embolism
B. Myocardial infarction
C. Ventricular tachycardia
D. Pulmonary emphysema

Answer: A

6. A patient with stage IIIB small cell lung cancer complains of new onset of back pain for the past 12 hours. Initial neurological examination is normal.

Which of the following is the most appropriate immediate action to take?
A. MRI
B. Radiation therapy
C. Take no immediate action
D. High dose IV corticosteroids and MRI

Answer: D

7. On a follow-up consultation of a patient with a squamous cell carcinoma, some abnormalities are noted in blood analysis (see lab results).

Test Result Normal Values
Parathyroid hormone 1.0 1.1-5.3 pmol/L
Calcium 3.30 2.15-2.62 mmol/L
Calcium ionized 1.6 1.1-1.3 mmol/L
Sodium 140 134-146 mmol/L
Potassium 4.3 3.5-5.1 mmol/L

Which of the following is the most likely cause of the metabolic abnormality?

A. Renal failure
B. Primary hyperparathyroidism
C. Secondary hyperparathyroidism
D. Secretion of PTH-related peptide by the tumor

Answer: D

8. A 78-year-old man and current smoker, with a 50 pack-year smoking history, complains of shortness of breath, dry cough and fatigue progressing over the past 8 weeks. The patient also reports a significant weight loss, corresponding to approximately 15% of body weight over the last 12 weeks. Physical examination is normal.

Which of the following is the most likely diagnosis?
A. COPD
B. Pneumonia
C. Lung cancer
D. Interstitial lung disease

Answer: C

A 40-year-old man presents to the clinic complaining of numbness of the feet for 1 month. He was diagnosed with pulmonary tuberculosis 4 months ago. He has been taking isoniazid and rifampin and has 2 months of treatment remaining. He has well-controlled diabetes mellitus for the last 3 years. The physical examination is normal.

Blood pressure 110/70 mmHg
Heart rate 80 /min
Respiratory rate 18 /min
Temperature 37.7° C

Which of the following drugs should be given?
A. Thiamine
B. Pyridoxine
C. Vitamin B12
D. Amitriptyline

Answer: B

Which of the following is the main method for prevention of tuberculosis?
A. BCG vaccine
B. Chemoprophylaxis
C. Health education
D. Treatment of the patients with tuberculosis

Answer: D

A 55-year-old man who is a heavy smoker is being evaluated for increasing bilateral lower limb swellings. Chest examination confirms resonant lungs with reduced breath sounds and occasional rhonchi. The apex beat could not be easily located. No cardiac murmur is heard, but P2 is markedly loud. The JVP is elevated, and bilateral pitting edema was confirmed.

Which of the following is the most likely cause of the lower limb swelling?
A. Liver cirrhosis
B. Bilateral DVT
C. Cor pulmonale
D. Nephrotic syndrome

Answer: C

A 40-year-old man is brought to the Emergency Department with progressive difficulty in breathing. History reveals, he is being treated for bronchogenic carcinoma. On examination, the JVP is elevated; lungs clear and heart sounds are very quiet.
Blood pressure 88/50 mmHg
Heart rate 125 /min (regular)
Respiratory rate 24 /min
Temperature 36.8 C°

Which of the following is the most appropriate test to confirm the diagnosis?
A. Chest X-ray
B. Echocardiogram
C. Electrocardiogram
D. Arterial blood gases

Answer: B

A patient with active rheumatoid arthritis presents with pleural effusion. Which of the following is typical finding in the pleural fluid analysis?
A. Hemorrhagic
B. Transudative
C. High pH, > 7.64
D. Glucose content of 1 mmol/L

Answer: D

A 40-year-old woman presents with recurrent attacks of dyspnea and cough when exposed to dust and perfumes. Physical examination is normal except for mild expiratory wheeze on lung auscultation. Which of the following tests has the highest diagnostic value?
A. Spirometry
B. Chest X-ray
C. Arterial blood gases
D. Methacholine challenge test

Answer: A

A 44-year-old obese woman presents with sudden onset dyspnea. No cough, chest pain orthopnea or PND. History is significant for hemorrhoidectomy 2 weeks ago (see image, lab results and report).
Blood pressure 110/70 mmHg
Heart rate 105 /min
Test Result Normal Values
ABG HCO_3^- 20 22-28 mmol/L

ABG PCO2 4.5 4.7-6.0 kPa
pH 7.5 7.36-7.45
ABG PO2 10 10.6-14.2 ka
Chest x-ray: Normal except for the obliterated left costophrenic angle. Which of the following has the highest diagnostic value?
A. D-dimer
B. Spirometry
C. Spiral CT-scan
D. Echocardiogram

Answer:C
A 30-year-old woman with rheumatoid arthritis presents with progressive dyspnea and dry cough for 3 months. No fever or weight loss. Physical examination revealed cyanosis, digital clubbing and bibasilar end-inspiratory crackles (velcro type) on lung auscultation.
Which of the following is the most likely diagnosis?
A. Lung abscess
B. Bronchiectasis
C. Bronchogenic carcinoma
D. Interstitial pulmonary fibrosis

Answer: D
A 44-year-old man with poorly controlled epilepsy presented with fever, cough and left side chest pain for 6 weeks. The cough is productive of copious foul-smelling sputum. Physical examination revealed digital clubbing and course crepitations in the right infra-clavicular area. Heart examination is normal (see image and report).
Blood pressure 110/70 mmHg Heart rate 100 /min Respiratory rate 25 /min Temperature 39 °C
Sputum culture: Mixture of aerobic and anaerobic organisms. What is the most likely mechanism for the underlying condition?
A. Aspiration
B. Vegetation embolism
C. Bronchogenic carcinoma
D. Direct extension from infected chest wall

Answer: A
A 30-year-old woman with bronchial asthma developed worsening dyspnea and cough following paint inhalation from neighborhood apartment. On examination, she is in respiratory distress, cannot complete a sentence and cyanosed. Chest auscultation revealed reduced breath sounds in both sides with short inspiratory and exploratory rhonchi (see lab results).
Blood pressure 110/70 mmHg Heart rate 130 /min Respiratory rate 20 /min Oxygen saturation 85 %
Test Result Normal Values
ABG HCO3- 24 22-28 mmol/L
ABG PCO2 7.8 4.7-6.0 kPa
pH 7.3 7.36-7.45
ABG P02 7.5 10.6-14.2 kPa
Which of the following is the best immediate action in management?
A. Intravenous magnesium infusion
B. Intravenous aminophylline infusion
C. Intubation and mechanical ventilation
D. Nasal continuous positive airway pressure ventilation

Answer:C
A patient with poorly controlled bronchial asthma developed hypertension, which of the following antihypertensive medications is contraindicated?

A. Beta blocker
B. Calcium channel blocker
C. Angiotensin II receptor blocker
D. Angiotensin converting enzyme inhibitor

Answer:A
A 60-year-old man with advanced squamous cell bronchogenic carcinoma presents with massive malignant right pleural effusion. He has been admitted several times with similar findings for which he undergoes several therapeutic pleural taps. Which of the following is the next best option in management?
A. Diuretics
B. Ultrafiltration
C. Chemical pleurodesis
D. Low salt human albumin

Answer: C
A 40-year-old woman presents with progressive dyspnea, dry cough and reddish nodular skin lesion over both shins. On examination, she has palpable cervical lymph nodes and bilateral diffuse fine crepitations in both lungs (see reports).
Chest radiograph: Diffuse bilateral interstitial infiltrates with bilateral hilar lymph nodes enlargement.
Purified protein derivative (PPD): Negative.
Cervical lymph node biopsy: Non-caseating granuloma.
Which of the following is the next best option in management?
A. Montelukast
B. Glucocorticoids
C. Anti-tuberculosis
D. Salmeterol inhaler

Answer: B
A 60-year-old man with post pulmonary tuberculosis bronchiectasis presents with massive hemoptysis. Physical examination shows course crepitations in the right scapular area (see image). Which of the following is the most appropriate initial management?
A. Sedation
B. Pneumonectomy
C. Chest physiotherapy
D. Positioning on right lateral decubitus

Answer: D
A 78-year-old woman is admitted with fever and confusion due to pneumonia. The doctors decided to start her IV antibiotics. However, her son, who is the next of kin, disagreed and argued that antibiotics will cause more harm. What is the best way to deal with this situation?
A. Ignore the opinion of the son and give the antibiotics to save the patient
B. Counsel the son in the presence of the consultants
C. Respect the opinion of the next of kin
D. Take the opinion of the patient

Answer:B
A young woman with long-standing rheumatoid arthritis is diagnosed to have non-specific interstitial lung disease. The physician decided to start her on oral glucocorticoid. However, the patient is concerned about the side effects related to this treatment. What is the best approach to deal with this situation?
A. Stress the importance of the treatment and give the patient time to decide
B. Inform the patient that refusing this treatment might cause death

C. Inform the patient that there is no alternative treatment
D. Tell the patient that most of the side effects are not serious

Answer: A
An elderly patient is diagnosed with pulmonary tuberculosis and started on anti-TB treatment. He developed jaundice, which progressed to liver failure for which he was admitted to the Critical Care Unit. His son was very angry and decided to complain against the treating team. What is the appropriate response to the son?
A. Tell him that this incident can happen to anyone
B. Inform him that the doctors offered the right management
C. Advise him against complaining as there is no medical error
D. Calm him down and inform him that this incident will be investigated

Answer:D
An elderly patient who has poorly controlled bronchial asthma with repeated admission to the hospital is found to be non-compliant with his medications. Which of the following is the most appropriate action?
A. Tell the patient that it is not acceptable to disrespect the advice of doctors
B. As the patient is noncompliant, just keep him in the hospital
C. Involve the next of kin to help to give his medications as prescribed
D. Tell the patient that he might die if he does not comply with the medications

Answer: C

A 69-year-old man with chronic obstructive pulmonary disease on oxygen therapy for the last 5 years. Which of the following markers is most useful for monitoring the progression of this patient?
A. Oxygen saturations
B. Number of exacerbations per year
C. FEV1/FVC ratio
D. FEV1

Answer: D

A 31-year-old man with asthma complaining of increasing wheeze in the past few hours. He rarely attends asthma clinic and his previous best peak flow readings has been 400 L/min. Which of the following would suggest acute severe asthma?
A. 02 saturation <95%
B. Respiratory rate >25
C. Carotid pulse > 100 bpm
D. Peak flow rate <300 L/min

Answer: B
A 38-year-old woman presents for annual asthma review. Her symptoms are very well controlled and she has not had any attacks since her last review a year ago and has not used her salbutamol inhaler once in that year. She is prescribed an inhaled beomethasone steroid twice daily in addition to her salbutamol inhaler. What is the most appropriate next step in modifying her medication?
A. Add a leukotriene receptor antagonist
B. Double the dose of the inhaled corticosteroid
C. Stop both the salbutamol and becomethasone inhalers
D. Half the beclomethasone dose and review in 6 months

Answer: D

A 38-year-old man is admitted with a sudden onset dyspnoea and pleuritic chest pain. He is a smoker but has no history of respiratory disease. He considers himself healthy and regularly plays football. On admission, he has a chest x-ray that shows a pneumothorax with a 3cm rim of air. Aspiration is successful and he is discharged. A follow-up chest x-ray 2 weeks later shows a complete resolution. What is the single most important piece of advice to reduce his risk of further pneumothoraces?
A. Stop smoking
B. Avoid flying for 12 months
C. Avoid sports for 12 months
D. Arrange a course of respiratory physiotherapy pnemothorax

Answer: A
A 28-year-old wood worker presents to the clinic complaining of episodic shortness of breath. He describes feeling wheezy and having a tendency to cough. A diagnosis of occupation asthma is suspected. Which of the following is the most appropriate diagnostic investigation?
A. Serial peak flow measurements at work and at home
B. Specific IgE measurements
C. Skin prick test
D. Patch testing

Answer: A
A 66-year-old man presents with shortness of breath. He smokes 20 cigarettes/day but is otherwise well. Spirometry shows a restrictive picture. Which of the following is the most likely diagnosis?
A. Pulmonary fibrosis
B. Bronchiectasis
C. Heart failure
D. Asthma

Answer: A
A 31-year-old man came for smoking cessation support as his father has recently been diagnosed with lung cancer. He has a past medical history of epilepsy, which is well-controlled on carbamazepine. Which smoking cessation medication would be contraindicated for this patient?
A. Nicotine replacement therapy
B. E-Cigarettes
C. Varenicline
D. Bupropion

Answer: D

A 51-year-old woman with asthma complaining of increasing shortness of breath and wheeze in the past few hours. She has a long-standing diagnosis of asthma but her medication history suggests a lack of compliance. Her best peak flow measurements in previous consultations have been 300 L/min.
Which of the following features suggests acute severe asthma in this patient?
A. Heart rate 100/min
B. Respiratory rate >20/min
C. Peak flow rate 200 L/min
D. Inability to complete sentences in one breath

Answer: D
A 31-year-old man with asthma complaining of increasing wheeze in the past few hours. His medication history suggests a lack of compliance. His best peak flow measurements in previous

consultations have been 600 L/min. Which of the following features suggests acute severe asthma in this patient?
A. A peak flow rate of 250 L/min
B. Respiratory rate >20/min
C. Heart rate 100/min
D. SpO2 95%

Answer: A
A 22-year-old man, whose asthma is uncontrolled on beta agonist therapy, is commenced on a steroid inhaler. Which of the following is the most likely adverse effect?
A. Dysphonia
B. Dental abscess
C. Peptic ulceration
D. Thrombocytopenia

Answer: A
A 58-year-old man presents with a 6-month history of breathlessness. He has been smoking all his adult life, he smokes 10-15 cigarettes per day. On examination, he has lost the nail fold angle of his fingers. Which of the following is the most likely diagnosis?
A. Chronic obstructive pulmonary disease
B. Bronchial carcinoma
C. Aspergillosis
D. Asthma

Answer: B
A 36-year-old man with multiple health problems presents to the Emergency Department with an episode of bronchospasm. He has a history of asthma and hypertension. He is undergoing a medication review. Which of the following medications is most likely to be responsible?
A. Salmeterol
B. Atenolol
C. Ramipril
D. Aspirin

Answer: B

A 58-year-old man with chronic obstructive pulmonary disease (COPD) is ready to be discharged home after acute exacerbation of the disease. Which of the following parameters indicate offering long-term oxygen therapy?
A. 1 arterial blood gas measurement with p02 < 7,7 kPa
B. 2 arterial blood gases measurements with p02 < 7.3 kPa
C. 1 arterial blood gas measurement with p02 < 8.0 kPa
D. 2 arterial blood gases measurements with p02 < 6.3 kPa

Answer: B

A 22-year-old woman with cystic fibrosis presents with a 4-month history of weight loss, diarrhoea and abdominal bloating. She has pale stools. There are no specific abnormalities on examination of the abdomen other than she is clubbed with scattered chest crackles.
Which of the following is the most likely diagnosis?
A. Acute pancreatitis
B. Pancreatic carcinom

C. Pancreatic pseudocys
D. Chronic pancreatitis

Answer: D

A 59-year-old man comes to the clinic, with a 45 cigarette pack-year history, complaining of breathlessness on exertion and a productive cough of white sputum over the last 6 months. Cardiological investigations are normal (see report).
Spirometry: Shows an FEV1/FVC of 60% with minimal reversibility after a 2-week trial of oral steroids
Which of the following is the most likely diagnosis?
A. Asthma
B. Lung fibrosis
C. Left ventricular failure
D. Chronic obstructive pulmonary disease

Answer: D

A 62-year-old woman is seen in the clinic with a history of weight loss, loss of appetite and shortness of breath for the last 6 weeks. On examination, trachea is shifted to left side; there is reduced air entry and duliness to percussion on the lower to midzones of the right lung.
Blood pressure 110/70 mmH
Heart rate 102 /min
Respiratory rate 22 /min
Temperature 37 °C
Oxygen saturation 91 %
Which of the following is the most likely diagnosis?
A. Right-sided pleural effusion
B. Right middle lobe pneumonia
C. Pulmonary embolism
D. Right pneumothorax

Answer: A

A 19-year-old man known to have asthma. He has never been admitted to hospital with an asthma exacerbation and experiences symptoms once or twice a week but he is not on any medication. Today he come to discuss the treatment options and he prefer to start medications only if needed during the attacks. His peak expiratory flow reading is currently 85% of the normal predicted value expected for his age and height.
Which of the following is the most appropriate first step in treatment?
A. Short-acting beta-2 agonist Inhaler
B. Long-acting beta-2 agonist inhaler
C. Leukotriene receptor antagonists
D. Low-dose steroid inhaler

Answer: A

A 40-year-old man diagnosed with pulmonary TB who started treatment 3 months ago, presents to the clinic with a week's history of pins and needles in his hands and feet with associated numbness. He tells that his symptoms started since he stopped taking the vitamins given to him
at the start of his TB treatment. Which of the following drugs is responsible for the symptoms described by the patient?
A. Pyrazinamide

B. Ethambutol
C. Rifampicin
D. Isoniazid

Answer: D

A 44-year-old woman with heart failure diagnosed 9 months ago is evaluated for 3-month history of non- productive cough. No history of chest pain, dyspnoea or orthopnoea. She has never smoked. Her current medications are lisinopril, carvedialol and fruosemide. Unremarkable cardiac and chest examination (see reports). Blood pressure 100/70 mmHg
Heart rate 76 /min
Respiratory rate 18/min
Oxygen saturation 95 %
Chest X-ray: Normal
ECHO: Showed left ventricular ejection fracture 40%.
Which of the following is the most appropriate management?
A. Discontinue lisinopril
B. Order pulmonary function test
C. Increase the dose of fruosemide
D. Start inhaled B2- adrenergic agonist

Answer: A; In other recalls A is “stop Furosemide”so we went with B.

A 45-year-old man has been diagnosed as a case of pulmonary tuberculosis. which of the following investigations is essential prior initiation of the therapy?
A. Vitamin B6
B. PPD skin test
C. Plasma glucose
D. Liver function test

Answer: D
A 39-year-old man came to the clinic for follow-up. He is recently diagnosed as pulmonary tuberculosis and was started on rifampicin, Isoniazid, pyrazinamide and ethambutol for 4 weeks. Basic lab workup were normal. LFT test was repeated (see lab results).
Test Result Normal Values
Direct bilirubin 56 1.5-6.5 mol/L
Total bilirubin 76 3.5-16.5 mol/L
Aspartate aminotransferase 465 12-40 IU/L
Alanine aminotransferase 620 5-40 IU/L
Alkaline phosphatase 720 39-117 IU/L
Gamma glutamyltransferase 387 6 to 37 1U/L
Which of the following is the most appropriate next step?
A. Stop Isoniazid
B. Stop rifampicin
C. Stop pyrazinamide
D. Stop all medications

Answer: D; Increased LFTs by five folds

A 54-year-old man presents with shortness of breath and fever for 1 week. On physical examination, the patient is febrile and dyspneic. His oxygen's saturation is 89% on room air. Chest examination shows absent breath sounds and dullness to percussion over right lung field. Chest radiography

confirms a large right-sided pleural effusion. Which of the following is the most useful to establish a diagnosis of exudative pleural effusion?
A. Pleural fluid protein/ serum protein < 0.5
B. Pleural fluid pLDH/ serum LDH > 0.6
C. Pleural fluid LDH >1/3 normal upper limit for serum
D. Pleural fluid white cell count > 1000

Answer: B
A 54-year-old man presents with shortness of breath and fever for 1 week. On physical examination, the patient is febrile and dyspneic. His oxygen's saturation is 89% on room air Chest examination shows absent breath sounds and dullness to percussion over right lung field. Chest radiography confirms a large right-sided pleural effusion. Which of the following factors indicating the need for therapeutic thoracentesis?
A. Loculated pleural fluid
B. Pleural fluid pH more than 7.30
C. Pleural fluid glucose more than 80 mg/di
D. Negative gram stain of the pleural fluid

Answer:A

A 38-year-old man who is smoker for several years presents to the clinic with progressive shortness of breath. Pulmonary function test (PFT) done which results in the following parameters: forced expiratoy volume (FEV1/forced vital capacity (FVC) ratio < 0.7 with minimal improvement after bronchodilators low diffusing capacity for carbon monoxide (DLCO).
Which of the following is the most likely diagnosis?
A. Emphysema
B. Bronchial asthma
C. Chronic bronchitis
D. Interstitial lung disease

Answer: A

A 50-year-old man presents to the clinic with chronic history of productive cough and shortness of breath. A diagnosis of chronic obstructive pulmonary disease (COPD) is suspected. Pulmonary function test is ordered. Which of the following pulmonary function tests is most likely to represent COPD?
A. Increased total lung capacity (TLC), decreased vital capacity (VC), decreased FEV1/FVC ratio
B. Increased total lung capacity (TLC), decreased vital capacity (VC), increased FEV1/FVC ratio
C. Decreased total lung capacity (TLC), increased vital capacity (VC), decreased FEV1/FVC ratio
D. Increased total lung capacity (TLC), normal vital capacity (VC), decreased FEV1/FVC ratio.

Answer: A
A 35-year-old-man presented to the Emergency Department with 5-day history of fever, shortness of breath and productive cough. Lung examination reveals duliness to percussion to approximately half of the lung field on the right and his breath sounds are markedly diminished. Chest radiography, shows right lower and middle lobes consolidation with moderate pleural effusion.
Which of the following would indicate the need for tube thoracostomy?
A. Pleural fluid glucose below 60 mg/di
B. Pleural fluid PH more than 7.20
C. Pleural fluid with more than 90% polymorphonucleocytes (PMNs)
D. Pleural fluid/ serum protein less than 0.4

Answer: A

A 36-year-old man presents to Emergency Department with 5-day history of fever, productive cough and shortness of breath. On examination, he is febrile with temperature 38.5 C and his lung examination reveals dullness to percussion with markedly diminished breath sound over half of the lung field on the right side. Chest radiography shows right lower and middle lobes consolidation and moderate pleural effusion. Which of the following is the most appropriate management?
A. Start azithromycin and ceftriaxone
B. Start azithromycin, ceftriaxone and oseltamivir
C. Start azithromycin, ceftriaxone and do thoracocentesis
D. Start azithromycin, ceftriaxone and do CT scan of the chest

Answer: C
A 36-year-old man presents to Emergency Department with 4-week history of low-grade fever, non-productive cough and shortness of breath. On examination, he is febrile with temperature 37.9 °C and his lung examination reveals duliness to percussion with markedly diminished breath sound over half of the lung field on the right side. A diagnosis of tuberculous pleural effusion is suspected. Chest radiography shows moderate pleural effusion (see lab results).
Test Result Normal Values
Leukocytes 4500 <1000/pL
Total Protein 6 10-20 g/L
Glucose 1.6 >2.2 mmol/L
LDH 650 <200/UL
Which of the following tests is the most sensitive to establish the diagnosis?
A. Positive pleural fluid AB culture
B. Exudative pleural fluid with lymphocytosis
C. Pleural fluid with very low glucose concentration
D. Pleural fluid adenosine deaminase more than 45 IU/L

Answer: D
A patient with a diagnosis of advanced chronic obstructive pulmonary disease (COPD) who is discussing with his doctor the therapeutic benefits of different therapies on his survival rate Which of the following therapies is most likely to improve his survival?
A. Theophylline
B. Corticosteroids
C. Mucolytics
D. Oxygen

Answer: D
A 25-year-old man presents to the Emergency Department with progressive shortness of breath, non- productive cough and wheezes for 1 day. On examinations, he has obvious difficulty in breathing and his lungs show with diffuse wheezes but no crackles. Blood pressure 120/70 mmHg, Heart rate 100 /min, Respiratory rate 28 /min, Temperature 36.6 °C, Oxygen saturation 93 %
Which of the following is the most effective initial treatment at this time?
A. Epinephrine
B. Inhaled steroids
C. Inhaled B2 agonists
D. Intravenous magnesium

Answer: C
A 25-year-old woman non-smoker presents to Emergency Department with sudden onset of severe chest pain and shortness of breath. Her past medical history is significant for caesarean section 4 days ago. Her examinations show a distressed woman with oxygen saturation 92% but otherwise unremarkable. Chest radiography is unremarkable.
Which of the following is the most appropriate management?

A. Start warfarin
B. Start thrombolytic drugs
C. Start low molecular heparin
D. Insert inferior vena caval filter

Answer: C

A 76-year-old man with chronic obstructive pulmonary disease (COPD) comes to the clinic for a follow-up. He is currently taking salmeterol, using albuterol rescue inhaler 2 to 3 times per day for the past 3 months. He has been hospitalized for COPD exacerbations twice in the past 6 months. Which of the following medications should be added to the maintenance regimen?
A. Montelukast
B. Prednisone
C. Tiotropium
D. Formoterol

Answer: C

A 30-year-old woman presents with recurrent attacks of dyspnea and cough when exposed to dust and perfumes. Physical examination is normal except for mild expiratory wheeze on lung auscultation (see report).
Spirometry: Reversible bronchospasm.
Which of the following is most appropriate to initiate at this stage?
A. Inhaled tiotropium
B. Inhaled salbutamol
C. Inhaled budesonide
D. Montelukast

Answer: B
A 60-year-old man with a history of 20-pack-year smoking presents with exertional dyspnea and productive cough for 3 months. He had similar attack last year. On examination, he has central cyanosis and expiratory wheeze on lung auscultation (see lab resuits)
Test Result Normal Values
ABG HCO3- 36 22-28 mmol/L
ABG PCO2 7.7 4.7-6.0 kPa
pH 7.35 7.36-7.45
ABG P02 8.4 10.6-14.2 kPa
Which of the following is the most likely diagnosis?
A. Bronchial asthma
B. Chronic bronchitis
C. Pulmonary embolism
D. Interstitial pulmonary fibrosis

Answer: B
A 70-year-old man heavy smoker, presents with left sided chest pain and hemoptysis for 2 days. Recently his son has noticed change in the quality of his father's voice. On examination, he has early digital clubbing and both of his wrists are painful and tender.
Blood pressure 110/70 mmHg
Heart rate 85 /min
Respiratory rate 22 /min
Temperature 36.6 °C

Oxygen saturation 92 %
Which of the following is the most likely diagnosis?
A. Tuberculosis
B. Bronchiectasis
C. Pulmonary embolism
D. Bronchogenic carcinoma

Answer: D
In a patient with acute asthmatic attack, which of the following is the most recommended tool to assess the response to therapy?
A.Chest radiography
B. Arterial blood gas
C. Pulsus paradoxus
D. Peak flow meter

Answer: D
A 65-year-old smoker is being evaluated for pleural effusion. He has cough, dyspnea and hemoptysis. On examination, signs of massive pleural effusion are present on right side but the trachea and apex beat are not shifted.
Which of the following explains these clinical findings?
A. Heart failure
B. Tubercular pleural effusion
C. Effusion due to esophageal rupture
D. Carcinoma, obstructing ipsilateral main bronchus

Answer: D
A 45-year-old man is diagnosed to have pulmonary tuberculosis for which he was started on 4 anti-tuberculous drugs 2 months ago. During routine follow up, his blood work shows hyperuricemia (see lab result).
Test Result Normal Values
Uric acid 600 200-410 mol/L (Male)
140-360 mol/L (Female)
What anti-TB drug is responsible for this finding?
A. Isoniazid
B. Rifampicin
C. Ethambutol
D. Pyrazinamide

Answer: D
A 32-year-old man is seen in the clinic for evaluation of primary infertility. Past medical history is significant for recurrent sinusitis. Sperm analysis shows normal number of sperms, but they are immotile (see report).
Chest radiography: shows situs inversus.
Which of the following is the most likely diagnosis?
A. Cystic fibrosis
B. Kartagener's syndrome
C. Wegner's granulomatosis
D. Good pasture's syndrome

Answer: B
A 40-year-old man presents to the Emergency Department complaining of progressive dyspnea over 5 days. Past medical history is significant for road traffic accident 2 week ago. Lung auscultation

revealed absent breath sounds in the right side (see image and lab results).
Blood pressure 100/70 mmH
Heart rate 90 /min
Respiratory rate 24 /min
Temperature 36.6 °C
Oxygen saturation 90 %
Test Result Normal Values
Hb 90 130-170 g/L (Male)
120-160 g/L (Female)
WBC 6 4.5-10.5 x 109/L
Which of the following is the most appropriate management?
A. Antibiotics
B. Anticoagulation
C. Chest tube insertion
D. Salbutamol nebulizer

Answer: C
A male farmer with chronic liver disease secondary to repeated infections of schistosoma mansoni complains of progressive dyspnea. Examination revealed elevated JVP, clear lung fields and left parasternal heave.
What is the likely complication that has occurred?
A. Bronchiectasis
B. Pulmonary embolism
C. Constrictive pericarditis
D. Pulmonary hypertension

Answer: D
A 45-year-old man presents with 5-year history of cough, productive of large volume of sputum associated with dyspnea and hemoptysis. On examination, he has clubbing and course crackles in the right base. Which of the following tests has the highest diagnostic value?
A. X-ray chest
B. Bronchoscopy
C. High resolution CT chest
D. Pulmonary function tests

Answer: C
A 50-year-old woman with pulmonary hypertension is being evaluated for a possibility of chronic thromboembolic disease. Which of the following investigations has the highest diagnostic value?
A. Echocardiogram
B. Electrocardiography
C. Spiral CT chest with contrast
D. Ventilation-perfusion lung scan

Answer: D
Which of the following drugs is a leukotriene receptor antagonist?
A. Infliximab
B. Tenofovir
C. Montelukast
D. Voriconazole

Answer: C
According to the World Health Organization classification of pulmonary hypertension, which of the following conditions falls under group 3?

A. Portal hypertension
B. Left ventricular failure
C. Chronic thromboembolic disease
D. Lung fibrosis with chronic hypoxemia

Answer: D
A 55-year-old man presents with progressive dyspnea and cough for 6 weeks. He is a heavy smoker. 2 weeks ago, he started having headaches and noticed swelling of his face. Physical examination reveals prominent non-pulsating neck veins, puffiness of the face and edema of the upper limbs (see report).
Chest X-ray: Shows a prominence of the right hilum.
Which of the following types of lung cancer is the most commonly associated with this clinical presentation?
A. Adenocarcinoma of the lung
B. Carcinoid tumor of the lung
C. Squamous cell lung cancer
D. Small cell lung cancer

Answer: D
An asymptomatic 60-year-old man, non-smoker presents for a routine annual physical check- up. A chest X-ray showed an opacity, which is less than 3 cm in the left upper lobe.
which of the following is the next best step in the management?
A. Surgical resection
B. Review old chest X-ray
C. CT chest with IV contrast
D. CT-guided transthoracic needle biopsy

Answer: B
A 55-year old man with stable COPD, presents with sudden onset left-sided pleuritic chest pain and exertional dyspnea. On examination, he is not in distress. the trachea is central and breath sounds are equal in both sides with occasional mild expiratory wheezes (see report).
Blood pressure 130/70 mmHg
Heart rate 80 /min
Respiratory rate 22 /min
Temperature 36.6 °C
Oxygen saturation 93 %
Chest X-ray: Shows a left sided pneumothorax measuring 2 cm between the lung and chest wall.
Which of the following is the most appropriate next step in management?
A. Needle aspiration
B. Chest tube insertion
C. Perform pleurodesis
D. Oxygenation and observation

Answer: D; Because the patient is vitally stable, and Apex-Cupola distance is less than 3 cm.
A 20-year-old woman presents to the Out-Patient Department with mild dyspnea, malaise, low-grade fever and arthralgia. Physical examination showed tender erythematous nodules measuring 2-3 cm in both shins. Chest auscultation revealed fine basal crackles. Which of the following is the next best action in management?
A. Blood culture
B. Oral steroids
C. Chest X-ray
D. Skin biopsy

Answer: C; Beware that most diagnostic is Biopsy from the most accessible lesion (eg. skin)
Which of the following is the best oxygen delivery system in a patient with Chronic Obstructive Pulmonary Disease?
A. Simple face mask
B. Rebreathing bag
C. Nasal cannula
D. Venturi mask

Answer: D
A 65-year-old man known to have COPD presented to the Emergency Department with increasing dyspnea, cough, expectoration and fever. Initial assessment revealed fully conscious patient with mild distress, central cyanosis and bilateral wheezing on chest auscultation. The patient was started on high flow oxygen and salbutamol nebulizer but 1 hour later, his level of consciousness is reduced (see lab result)
Test Result Normal Values
ABG HCO3. 36 22-28 mmol/L
ABG PCO2 9 4.7-6.0 kPa
pH 7.18 7.36-7.45
ABG PO2 15 10.6-14.2 kPa
Which of the following is the best initial step in management?
A. Start mechanical ventilation
B. Reduce oxygen flow
C. Order brain CT scan
D. Give naloxone

Answer:B
A 30-year-old postpartum woman is admitted to the General Ward with diagnosis of right leg DVT. She was started on enoxaparin 80 mg bid, but in the same day of admission, she developed sudden onset dyspnea and right pleuritic chest pain. On examination, she was dyspneic and apprehensive. Heart sounds showed loud P2 and lungs were clear to auscultation (see lab results and report).
Blood pressure 125/70 mmHg Heart rate 100 /min Respiratory rate 22 /min Temperature 36.6 °C Oxygen saturation 95 %
Test Result Normal Values
ABG HCO3- 22 22-28 mmol/L
ABG PCO2 4.2 4.7-6.0 kPa
pH 7.48 7.36-7.45
ABG PO2 12 10.6-14.2 kPa
Spiral CT chest: Thrombus in the right lower pulmonary artery. Which of the following is the most appropriate step in management?
A. Change enoxaparin to sodium heparin
B. Treat with thrombolytic therapy
C. Continue the same management
D. Perform invasive thrombectomy

Answer: C
A 40-year-old man with paraplegia following road traffic accident one year ago is brought to the Emergency Department because of sudden onset dyspnea and chest pain. On examination, he is in severe distress and cyanosed. He has left parasternal heave and loud P2. Chest is clear to auscultation (see lab results and reports).
Blood pressure 80/55 mmHg Heart rate 110 /min Respiratory rate 30 /min Temperature 36.6 °C Oxygen saturation 86 %
Test Result Normal Values
ABG HCO3- 22 22-28 mmol/L

ABG PCO2 4.2 4.7-6.0 kPa
pH 7.4 7.36-7.45
ABG PO2 8.6 10.6-14.2 kPa
Echocardiogram: Dilated right ventricle with pulmonary artery pressure of 55 mmHg.
Venous Doppler: Confirmed DVT in the left leg extending to the thigh.
Which of the following is the best line of management?
A. Insertion of vena-caval filter
B. Alteplase infusion
C. Enoxaparin
D. Warfarin

Answer: B
A 50-year-old woman presented with fever, dyspnea and weight loss. Physical examination showed evidence of pleural effusion in the left hemithorax (see report).
Pleural fluid analysis: Shows exudative fluid with high gamma interferon level.
Which of the following is the most likely diagnosis?
A. Empyema
B. Mesothelioma
C. Parapneumonic effusion
D. Tuberculous pleural effusion

Answer:D
A 55-year-old man with COPD, presented with increasing dyspnea, cough and bilateral lower limb swelling. Physical examination showed elevated JVP, hepatomegaly and pitting edema in both lower limbs. Chest auscultation revealed scattered rhonchi in both sides.
Which of the following is the most appropriate next step in management?
A. Spirometry
B. Echocardiogram
C. Venous Doppler of lower limbs
D. High resolution CT scan of chest

Answer: B
In a patient with COPD and cor pulmonale, which of the following improves survival?
A. Montelukast
B. Inhaled steroids
C. Long term oxygen therapy
D. Long acting beta agonist inhaler

Answer: C
A 30-year-old woman is evaluated for a 6-month history of recurrent dyspnea, cough, and wheezing. She is awakened with asthma symptoms 6-8 nights a month. Her current medication is salbutamol inhaler. On examination, there are wheezes in both lung fields.
Which of the following is the most appropriate next best step in management?
A. Montelukast
B. Long-acting B-agonist
C. Inhaled budesonide/formoterol
D. Inhaled budesonide and inhaled albuterol as needed

Answer: C
A 30-year-old woman complains of recurrent attacks of sudden onset dyspnea, chest and throat tightness. Inhaled salbutamol is of little help; however, her symptoms improves suddenly. On examination, she is found to have inspiratory wheeze during dyspnea spells.
which of the following is the likely cause of symptoms?

A. Bronchial asthma
B. Endobronchial foreign body
C. Gastrosophageal reflux disease
D. Vocal cord dysfunction syndrome

Answer: D
A 55-year-old obese non-smoker man with hypertension presents with exertional dyspnea for 2 years. His wife stated that he is uncomfortable while sleeping at night and produces loud snores. Examination revealed central cyanosis, suffused conjunctivae and clear chest.
BMI 40 kg/m2 Which of the following has the highest diagnostic value?
A. Spirometry
B. Spiral CT chest
C. Polysomnography
D. Methacholine challenge test

Answer: C
A 55-year-old obese non-smoker man presents with loud snoring, daytime somnolence and exertional dyspnea for 4 years. Examination revealed central cyanosis: suffused conjunctivae and clear chest (see lab results).
BMI 40 kg/m2
Test Result Normal Values
ABG HCO3- 38 22-28 mmol/L
ABG PCO2 6.9 4.7-6.0 kPa
pH 7.38 7.36-7.45
ABG PO2 9 10.6-14.2 ka
Which of the following is the best treatment after polysomnography?
A. Home oxygen by venturi mask
B. Inhaled tiotropium
C. Home nasal CPAP
D. Oral prednisone

Answer:C
A 65-year-old man with idiopathic bronchiectasis presents with persistent cough of large amount of mucoid sputum and dyspnea on exertion. He is compliant with his medications, which include azithromycin. Physical examination showed finger clubbing and mild bilateral coarse crackles. Which of the following is the most appropriate treatment option?
A. Prednisone
B. N-Acetylcystine
C. Saline nebulizer
D. Postural sputum drainage

Answer: D
A 60-year-old man known to have COPD refused the advice to stop smoking.
What is the best statement should be given to him during a communication session?
A. "You better stop smoking otherwise you will die"
B. 'We will do everything to help you stop smoking"
C. "We have to talk to your family about this refusal"
D. "If you don't stop smoking we will stop seeing you in the clinic"

Answer: B
A 56-year-old man who has chronic productive cough underwent a CT chest, which was consistent with bronchiectasis. He came today to discuss his disease.
Which of the following would convey the diagnostic information to him?

A. Read to him the CT report
B. Show him the chest X-ray film
C. Inform him that he has bronchiectasis
D. Tell him that his air tubules are inflamed and damaged

Answer:D
A young patient has poorly controlled asthma and her doctor decided to start her on oral steroids. She is uncomfortable about that decision.
When counselling this patient, what is the most important communication aspect should be addressed?
A. Her living conditions
B. The patient's concerns
C. Dealing with her anger
D. The scientific basis of using steroids

Answer: B
A 74-year-old man, who smokes for 40 years, underwent a CT chest, which showed a large lung mass.Which of the following is the most appropriate way to break the bad news?
A. Avoid wasting time and tell him directly that he has lung cancer
B. Inform him that this is nothing to worry about
C. Inform him that he should stop smoking
D. Deliver the information in small steps

Answer: D
A patient with interstitial lung disease has repeated admissions for pneumonia. His son is worried about antibiotics induced diarrhoea.
Which of the following is the most appropriate way to address this situation?
A. Tell him that the diarrhea is not serious
B. Stress that the hospital infection control policy will be implemented
C. Agree with him that the antibiotics should be stopped
D. Apologize for repeatedly using antibiotics to treat his father

Answer: B
A 28-year-old man, who is a known drug addict, is found in deep coma and cyanosed. Blood pressure 100/55 mmH
Heart rate 55 /min
Respiratory rate 8 /min
Oxygen saturation 85 %
Which of the following is the most important initial management?
A. Obtain detailed history from the family
B. Mechanical ventilation
C. Do gastric lavage
D. Brain CT scan

Answer: B
A 53-year-old man with liver cirrhosis presents with several episodes of vomiting of blood and black tarry stool.
Blood pressure:
Supine 105/60 mmH
Sitting 85/50 mmHg
Heart rate: Supine 90 /min
Sitting 110 /min
Which of the following is the most appropriate next step in management?

A. Nasogastric tube placement
B. Urgent upper GI endoscopy
C. Resuscitation with adequate IV fluids
D. Intravenous infusion of H2 receptor antagonists

Answer:C
An elderly man, known to have COPD is brought to Emergency Department with severe respiratory distress, cyanosis and drowsiness. His upper airways were full of secretions (see lab results).
Blood pressure 90/60 mmH
Heart rate 110 /min
Respiratory rate 18 /min
Temperature 38.0 °C
Oxygen saturation 74 %
Test Result Normal Values
ABG HCO3 36 22-28 mmol/L
ABG PCO2 7.99 4.7-6.0 kPa
pH 7.24 7.36-7.45
ABG PO2 8.99 10.6-14.2 kPa
Which of the following is the best option in management?
A. Nasal CPAP
B. Mechanical ventilation
C. Aminophylline infusion
D. Oxygen administration via face mask

Answer: B
A 40-year-old woman is admitted with diagnosis of Guillain Barre Syndrome. Which of the following would be the most appropriate method for monitoring ventilatory impairment?
A. Rapid shallow breathing index
B. Serial arterial blood gas measurements
C. Bedside measurement of vital capacity
D. Continuous monitoring of oxygen saturation

Answer: C
A 60-year-old man with diabetes mellitus and ischemic heart disease is admitted to the Critical Care Unit in shock. The following parameters are obtained from pulmonary artery catheterization (see table).
Pulmonary Artery Catheter Test Result Normal Values
Right atrial pressure 2 2-8 mmHg
Pulmonary artery pressure 15/3 16-24/5-12 mmHg
Wedge pressure 4 5-12 mmHg
Cardiac output 2.5 4-6 L/min
Systemic vascular resistance 1800 800-1200 dyne/s/cm5
Which of the following is the most likely diagnosis?
A. Cardiogenic shock
B. Septicemic shock
C. Hypovolemic shock
D. Pulmonary embolism

Answer: C; Because wedge pressure and CO are both reduced.
A 45-year old man is just admitted to the Critical Care Unit with polytrauma following road traffic accident because of hypotension. He is sedated and mechanically ventilated. His heart sounds are distant with no added sounds. Lung auscultation showed equal air entry to both sides with no added sounds (see report).

Blood pressure 85/55 mmHg
Heart rate 110 /min
Temperature 36.0 °C
Oxygen saturation 100 %
Central venous pressure: Right atrial pressure 15 mmHg
Chest radiograph: Cardiomegaly and fractured sternum. Both lungs are normal. Which of the following is the most likely diagnosis?
A. Fat embolism
B. Septicemic shock
C. Hypovolemia due to internal bleeding
D. Cardiac tamponade due to hemopericardium

Answer: D
A 35-year-old man is evaluated in the Orthopedic Ward because of dyspnea, tachycardia and confusion. The patient had fractured left femur for which he undergone internal fixation 2-days ago. He is on enoxaparin as prophylaxis and pain-killer as needed. On examination, the heart sounds are normal and there are scattered crepitations in both lung fields. Petechial rash around the neck and axillae are noticed (see lab result and reports).
Blood pressure 120/70 mmH Heart rate 110 /min Respiratory rate 26 /min Temperature 37.8 °C
Oxygen saturation 85 %
Test Result Normal Values
ABG HCO3- 22 22-28 mmol/L
ABG PCO2 4.7 4.7-6.0 kPa
pH 7.26 7.36-7.45
ABG P02 9.5 10.6-14.2 kPa
Chest radiograph:Bilateral diffuse alveolar infiltrate.
Electrocardiogram: Sinus tachycardia.
Which of the following is the most likely diagnosis?
A. Pulmonary thromboembolism
B. Fat embolism syndrome
C. Aspiration pneumonia
D. Septicaemia

Answer: B
A 65-year-old woman with diabetes mellitus and old myocardial infarction is admitted to the Intensive Care Unit with urosepsis and septic shock. Her blood pressure did not respond to fluid challenge for which she is started on norepinephrine infusion. She has hypoxia despite optimizing the ventilator parameters and Fi02 of 100% (see table and report).
Pulmonary Artery Catheter Test Result Normal Values
Right atrial pressure 3 2-8 mm Hg
Pulmonary artery pressure 13/3 16-24/5-12 mm Hg
Wedge pressure 5 5-12 mm Hg
Cardiac output 4 4-6 L/min
Chest X-ray:
Bilateral lung infiltrate.
Which of the following is the most likely cause of the lung infiltrate?
A. Adult Respiratory Distress Syndrome
B. Cardiogenic pulmonary edema
C. Pulmonary embolism
D. Volume overload

Answer: A

A 55-year-old woman with a chronic obstructive pulmonary disease (COPD) on combination of long-acting B2-agonist and inhaled glucocorticoid and albuterol as needed presents to the clinic with worsening of her symptoms. On physical examination, there is no jugular venous distention. Pulmonary examination reveals decreased breath sounds. Bilateral lower extremity edema is noted to a level above the ankles. Echocardiogram is normal apart of high mean pulmonary artery pressure, 52 mmH (see lab results).
Blood pressure 125/75 mmH Heart rate 84 /min Respiratory rate 20 /min Temperature 36.6 °C
Oxygen saturation 86 % BMI 30 kg/m2
Test Result Normal Values ABG PCO2 7.5 4.7-6.0 kPa pH 7.6 7.36-7.45
ABG PO2 8.6 10.6-14.2 kPa
Which of the following is the most appropriate treatment at this time?
A. Start mucolytics
B. Start oral diuretics
C. Start oral prednisone
D. Start oxygen therapy

Answer: D
A 66-year-old woman is evaluated for a 6-week history of increasing dyspnea and right-sided pleuritic chest pain with occasional cough. She has lost significant body weight in the last 3 months. She has no other medical problems and takes no medications. Her physical examination reveals dullness to percussion and decreased breath sounds over both lower lung zones. A 4-cm right breast mass is palpated. Chest radiograph demonstrates bilateral pleural effusions but no evidence of infiltrate or pulmonary vascular congestion. Thoracentesis is performed. Pleural fluid Gram stain is negative (see lab results).
Test Result Normal Values
Total Protein 40 10-20 g/L
Fluid Protein/Serum Protein 0.67 <0.5
Glucose 3.5 >2.2 mmol/L
LDH 350 <200/UL
Total Proteins 60 60-84 g/L
Lactate dehydrogenase 125 60-160 IU/L
Leukocytes 3600 <1000/pL
Which of the following is the most appropriate statement about the diagnosis?
A. Exudative pleural effusion due to malignancy
B. Transudative pleural effusion due to malignancy
C. Exudative pleural effusion due to heart failure
D. Transudative pleural effusion due to empyema

Answer: A
A 28-year-old woman with bronchial asthma that controlled on budesonide and albuterol inhalers is evaluated in the clinic as she is 6 weeks pregnant. Her symptoms are well controlled on the treatment. However, symptoms usually flare in fall and she treats these symptoms with an add-on oral leukotriene inhibitor. She currently has no asthma symptoms. On examination, vital signs are normal. The lungs show good air entry and no wheezes. She is so concern about the safety of medications during pregnancy. What is the best information to be delivered to her with regard to medication safety?
A. Continue the current treatments as they are safe during pregnancy
B. Stop budesonide and continue albuterol inhaler
C. Stop albuterol and continue budesonide inhaler
D. Switch to as-needed albuterol therapy

Answer: A

A 26-year-old man presents to the clinic with shortness of breath and occasional wheeze that occurs several times per week. He develops significant shortness of breath when playing football. He notes that it takes several minutes to recover his breath when this occurs. He also notes coughing during exercise as well. He takes no medications. On physical examination, vital signs are normal. Mild nasal congestion is noted. His systemic examinations are unremarkable. A chest radiograph is normal. Spirometry findings are within normal parameters. Which of the following is the most appropriate next step in assessment?
A. Repeat spirometry
B. Clinical observation
C. Methacholine challenge test
D. Helical (spiral) computed tomography (CT) for the chest

Answer: C
A 31-year-old woman is evaluated because of daily wheezing and breathlessness that limit her activities several times a week. Medical history is otherwise unremarkable. She has never smoked. She is diagnosed with moderate persistent asthma. In addition to a short-acting 32-agonist, which of the following is the most appropriate treatment?
A. Add a low-dose inhaled corticosteroid and long-acting B2-agonist
B. Add a low-dose inhaled corticosteroid
C. Add a leukotriene antagonist
D. Add thiophylline

Answer: A
A 66-year-old woman with severe COPD is admitted with 4 days of dyspnoea and cough. She is alert, in moderate respiratory distress, and has an oxygen saturation of 93% on 3 L/min supplemental oxygen. A chest radiography demonstrates no consolidation. She receives intensive bronchodilator therapy and systemic steroids (see lab results).
Test Result Normal Values
ABG PCO2 8.8 4.7-6.0 kPa
pH 7.28 7.36-7.45
ABG PO2 9.9 10.6-14.2 kPa
Which of the following is the best next step in the management?
A. Non-invasive ventilation
B. Increase supplemental oxygen
C. Decrease supplemental oxygen
D. Intubation and mechanical ventilation

Answer: A
A 58-year-old woman with acute onset shortness of breath and pleuritic chest pain. She is diagnosed with pulmonary embolism. Which of the following is an indication to do thrombophilia workup?
A. Age above 55 years
B. Being on oral contraceptive pills
C. Has a history of connective tissue disease
D. Has negative family history of thrombosis

Answer: B
A 32-year-old man is evaluated in the clinic because of an abnormal chest computerised tomography (CT) scan that revealed a small lung lesion (7 mm) without any other abnormalities. He is non-smoker and he is asymptomatic. Physical examination shows no lymphadenopathy and normal lung examinations (see lab results).
Test Result Normal Values
Hb 140 130-170 g/L (Male)

120-160 g/L (Female)
Platelets count 300 150-400 x 109/L WBC 8 4.5-10.5 x 109/L
ESR 11 2-10 mm/h (Male)
3-15 mm/h (Female)
Calcium 2.25 2.15-2.62 mol/L
Which of the following is the most appropriate procedure to do next?
A. Repeat CT scan within 3-6 months
B. Pulmonary function test
C. Biopsy of this lesion
D. PET scan

Answer: A

Gastroenterology

A 30-year-old overweight man presents to the general physician with 6-month intermittent epigastric burning pain. He has also noted that for the past 2 months he has been experiencing episodes of nausea, an unpleasant taste in the mouth, hoarseness, and a nocturnal cough.

Which of the following is the most likely diagnosis?
A. Peptic Ulcer
B. Atypical angina
C. Chronic cholelithiasis
D. Gastroesophageal reflux disease

Answer: D

12. A 40-year-old man with Gastroesophageal Reflux Disease presents to the general physician with a 3-month history of intermittent retrosternal burning pain, worse on bending, nausea, regurgitation with an unpleasant taste in the mouth, and a nocturnal cough. The patient has been using over-the-counter antacids with no relief.

Which of the following is the most appropriate course of action?
A. Surgery
B. Antibiotic
C. Lifestyle modification
D. Proton pump inhibitors

Answer: D

Explanation: symptoms persist for 3 months >> indication for PPI initiation in mild to moderate GERD.

13. An individual who has received all 3 doses of hepatitis B vaccine and has never had hepatitis B virus infection.

Which of the following serologic markers is expected to be positive?
A. HBcAb
B. HBsAb
C. HBeAb
D. HBeAb and HBsAb

Answer: B

15. What is the immunological status of a patient with negative HBsAg, positive anti-HBc, and positive anti-HBs?
A. Immune due to hepatitis B vaccination
B. Immune due to natural infection
C. Chronically infected
D. Acutely infected

Answer: B

A 60-year-old man presents with epigastric pain and early satiety associated with 10% weight loss over 8 months (see lab results).

Test Result Normal Values
Hb 98 130-170 g/L (Male)
120-160 g/L (Female)
MCV 70 80-95 f1
Which of the following is the best initial action in management?
A. Gastroscopy
B. Ultrasound of abdomen
C. Empirical therapy for H. pylori
D. Trial of proton pump inhibitor for 6 weeks

Answer:A
A 30-year-old woman complains of epigastric pain, heartburn and bloating after meals for 3 months. No history of hematemesis or weight loss. A trial of esomeprazole 40mg for 8 weeks did not relief her symptoms (see lab result and reports).
Test Result Normal Values
Hb 140 130-170 g/L (Male)
120-160 g/L (Female)
Gastroscopy: Normal.
Urea breath test: Negative.
Ultrasound of abdomen: Normal.
Which of the following is the best next step in management?
A. Ranitidine
B. Hyoscine butylbromide
C. Tricyclic antidepressant
D. Increase the dose of esomeprazole to 80mg/day

Answer: C
In a patient with liver cirrhosis and bleeding esophageal varices, treated with esophageal banding, which of the following drugs is effective in preventing future bleeding?

A. Nadolol
B. Verapamil
C. Octreotide
D. Vasopressin

Answer: A
An 18-year-old girl presents with chronic diarrhea and intermittent abdominal pain. No history of upper or lower Gl bleeding. On examination, she is slim and has pallor. No jaundice (see lab results.
Hb 100 130-170 g/L (Male)
120-160 g/L (Female)
MCV 100 80-95 fl
Platelets count 540 150-400 x 109/L
WBC 6 4.5-10.5 x 109/L
Vitamin B12 (Women) 0.074 0.12-0.66 mol/L
Folic Acid (Women) 0.099 <0.1 mol/L
Which of the following is the most likely diagnosis?
A. Celiac disease
B. Crohns disease
C. Ulcerative colitis
D. Whipples disease

Answer: B

An 18-year-old girl brought by her parents to the Gastroenterology Clinic as they noticed her to have a fine tremor of both hands. Further examination showed jaundice and just palpable spleen (see lab results).
Test Result Normal Values
Copper 0.8 1.72-3.54 mmol/L (Male) 1.72-3.76 mmol/L (Female)
Direct bilirubin 22.32 1.5-6.5 mol/L
Total bilirubin 38.2 3.5-16.5 mol/L Aspartate aminotransferase 243 12-40 IU/L Alanine aminotransferase 165 5-40 IU/L Albumin 23 34-56 g/L
Total Proteins 46 60-84 cc
Serum ceruloplasmin Low
24 hours urine copper High
Which of the following is the most appropriate treatment?
A. Corticosteroid
B. D-penicillamine
C. Desferrioxamine
D. Ursodeoxycholic acid

Answer:B
A 70-year-old man is mechanically ventilated in the Intensive Care Unit following massive cerebral hemorrhage. 7 days later, he developed coffee ground vomitus. history is only significant for hypertension. Which of the following is the most likely the cause?
A. Stress gastritis
B. Functional dyspepsia
C. Helicobacter gastritis
D. Gastrosophageal reflux

Answer:A
A 45-year old with chronic hepatitis B is noted to develop increasing jaundice for the last 2 weeks assoclated with fatigue (see lab results).
Test Result Normal Values
Direct bilirubin 28.43 1.5-6.5 umol/L
Total bilirubin 39 3.5-16.5 mol/L
Aspartate aminotransferase 345 12-40 IU/L
Alanine aminotransferase 523 5-40 IU/L
Albumin 34 34-56 a/L
Total Proteins 60 60-84 cc
Hepatitis B DNA level Low
Which of the following is the most appropriate next step in management?
A. Start vitamin E
B. Perform liver biopsy
C. Start alpha interferon
D. Request anti-Delta antibody

Answer: D
A middle-aged woman with hepatomegaly underwent upper abdomen MRI and found to have a 3cm lesion in the liver that shows peripheral beaded enhancement in the early phase with progressive central filling in the delayed phase. Which of the following is the most likely diagnosis?
A. Hepatoma
B. Metastasis
C. Hydatid cyst
D. Hemangioma

Answer:D

A patient presents with chronic dyspepsia and recurrent upper Gl bleeds. Urea breath test is positive. Which of the following regimen is the best to treat the underlying cause?
A. Oral omeprazole for 6 weeks
B. Ranitidine plus bismuth for 6 weeks
C. Sucralfate plus metronidazole for 6 weeks
D. Esomeprazole, amoxicillin and clarithromycin for 14 days

Answer: D
A 40-year-old woman is under treatment for a recurrent helicobacter-associated duodenal ulcer. She noticed that her stool is turned non-tarry black (see lab result).
Blood pressure 110/70 mmH
Heart rate 70 /min
Test Result Normal Values
Hb 152 130-170 g/L (Male)
120-160 g/L (Female)
What drug is likely to be the cause of the black stools?
A. Ranitidine
B. Pantoprazole
C. Bismuth subsalicylate
D. Alumunium hydroxide

Answer: C
A 25-year-old patient presented with diarrhea for 5 months, which is mostly related to eating bread and wheat products. Which of the following skin lesions would suggest the likely diagnosis?
A. Psoriasis
B. Hyperpigmentation
C. Dermatitis herpitiformis
D. Pyoderma gangrenous

Answer: C
A 38-year old woman who was previously healthy came to the Emergency Department with recurrent episodes of passing black stools and sometimes fresh blood per-rectum over the last 2 months. Physical examination was unremarkable apart from pallor (see lab results and reports).
Test Result Normal Values
Hb 88 130-170 g/L (Male)
120-160 g/L (Female)
Upper and lower Gl endoscopy: Showed no significant pathology.
Which of the following is the best next investigation?
A. CT abdomen
B. Capsule endoscopy
C. Barium meal and follow
D. Radio-isotope scan of abdomen

Answer: B
A 23-year-old woman is brought to the Emergency Department after she ingested 15 tablets of paracetamol in a suicidal attempt 6 hours earlier. Activated charcoal is administered by the Emergency doctor. Which of the following is the best next option in management?
A. Hyperbaric oxygen
B. N-acetylcysteine
C. D-penicillamine
D. Hemodialysis

Answer: B

A 43-year-old man is known to have hepatitis B virus infection presented with fever and confusion. On examination, he is disoriented, has asterixis and positive shifting dullness (see report).
Ascetic Fluid analysis: Showed serum albumin ascitic fluid gradient (SAAG) of 1.2 and 350 neutrophils. Culture is bending. Which of the following is the best management?
A. IV diuretics and metronidazole
B. Lactulose enema and IV ceftriaxone
C. Oral Lactulose and sedation with diazepam
D. High volume ascetic fluid aspiration and propranolol

Answer:B

A 48-year-old man with a history of cirrhotic liver disease presents with abdominal distention. Clinical examination reveals tense ascites, which was drained. What is the appropriate type of diuretic to help prevent re-accumulation of ascites?
A. Aldosterone antagonist
B. Thiazide diuretic
C. Osmotic diuretic
D. Loop diuretic

Answer: A

A 55-year-old man known case of liver cirrhosis with ascites presented to the Emergency Department complaining of progressive non- specific abdominal pain for 6 days, which was associated with vomiting. Examination confirmed generalized abdominal tendemess (see lab result and report).
Blood pressure 100/55 mmH
Test Result Normal Values
WBC 23.3 4.5-10.5 x 109/L.
CT scan abdomen: Moderate ascites with cirrhotic liver.
Which of the following is the most appropriate step in treatment?
A. Antibiotics
B. Observation
C. Diagnostic laparoscopy
D. Exploratory Laparotomy

Answer: A

A 46-year-old man presents with a 1-day history of abdominal pain, vomiting and **hematemesis**. History reveals he is a heavy alcohol drinker. He has previously been admitted with peptic ulceration and is receiving proton pump inhibitor therapy. On examination, he has tenderness over the epigastrium. Blood pressure 110/69 mmH
Heart rate 112 /min
Which of the following is the most likely diagnosis?
A. Acute pancreatitis
B. Chronic pancreatitis
C. Oesophageal varices
D. Pancreatic pseudocyst

Answer: C

A 30-year-old man comes to the clinic to discuss his recent blood test results. A few months ago, he had a health visit that showed an abnormality on his liver function tests when the bilirubin level of

41 mol/L (normal 3-20) with the remainder of the liver function profile being within normal limits. He has no significant medical history. On examination, no significant findings. The physician orders second blood test, results show a normal full blood count, renal function and thyroid function. The liver function test is done (see lab results).

Test	Result	Normal Values
Alkaline phosphatase	115	39-117 IU/L
Alanine aminotransferase	38	5-40 IU/L
Direct bilirubin	7	1.5-6.5 mol/L
Total bilirubin	42	3.5-16.5 mol/L

Which of the following is the most likely diagnosis?
A. Cholestasis
B. Viral hepatitis
C. Gilberts syndrome
D. Haemochromatosis

Answer:C

A 61-year-old man complains of indigestion that has been bothering him for the past 2 months. He has never had these symptoms before and he is eating and drinking well. He says he has lost weight over this period. His bowel habits are normal and he does not complain of any abdominal pains. Abdominal examination is normal. Blood tests including full blood count, renal function, and liver function as well as C-reactive protein are normal. Which of the following is the best management plan?
A. Urgent endoscopy
B. Routine abdominal ultrasound
C. Abdominal and erect X-ray today
D. Routine referral to gastroenterology

Answer: A

A 29-year-old man diagnosed with celiac disease 2 years ago. He complains of ongoing diarrhoea, feeling tired and bloating. Poor compliance with a gluten-free diet is suspected. Which of the following is the most appropriate method to ascertain this?
A. Food diary
B. Refer for a duodenal biopsy
C. Anti-endomysial antibodies
D. Tissue transglutaminase antibodies

Answer: D

Which of the following extra-intestinal manifestations of Cohn's disease activity?
A. Clubbing
B. Erythema nodosum
C. Pyoderma gangrenosum
D. Primary sclerosing cholangitis

Answer: B

A 28-year-old woman comes to the clinic for a check-up. She believes that she is allergic to wheat; her symptoms of abdominal bloating and loose stool have improved over the past few months by eating a gluten-free diet.
Which of the following is the most appropriate advice to her?
A. Dietician referral

B. Gastroenterology referral
C. No need to Continue on gluten-free diet
D. Resume eating gluten, bloods for celiac screen

Answer: D

A nurse is bitten by a patient who is known to have hepatitis B. She has a documented full history of hepatitis B vaccination and was known to be a responder. What is the most appropriate action to reduce the chance of contracting hepatitis B?
A. Give hepatitis B immune globulin + hepatitis B vaccine booster
B. Give hepatitis B immune globulin
C. Give hepatitis B vaccine booster
D. Give oral ribavinn for 4 weeks

Answer: C

A 40-year-old woman is noticed to be jaundiced (see lab results). Test Result
Anti-HBs Negative
Anti-HBc Positive
HBs antigen Positive
IgM anti-HBc Negative
What is the patient's hepatitis B status?
A. Chronic hepatitis B
B. Acute hepatitis B infection
C. Previous hepatitis B infection
D. Previous immunisation to hepatitis B

Answer: A

A 36-year-old man has just completed a 2-week course of Helicobacter pylori therapy for dyspepsia. He has stopped his eradication therapy today and he wants to repeat the test. How soon can the test be re- performed to test for eradication?
A. In 4 weeks
B. In 2 weeks
C. In 1 weeks
D. Straight away

Answer: A
A 28-year-old woman with irritable bowel syndrome comes to the clinic complaining that loperamide and antispasmodic medication have not eased her symptoms. She has no new symptoms suggestive of any serious condition. Which of the following is the most appropriate treatment?
A. Cognitive behaviour the rap psychological therapy
B. Tricyclic antidepressant
C. Psychological therapy
D. Acupuncture

Answer: B
A 44-year-old woman is diagnosed with non-alcoholic steatohepatitis following a liver biopsy. What is the single most important step can help to prevent the progression of her disease?
A. Weight loss
B. Stop smoking
C. Start statin therapy

D. Eat more omega-3 fatty acids

Answer: A

What is the most common location of Crohn's disease?
A. Ileal
B. Colonic
C. Duodenum
D. Ileocolonic

Answer: D

A 30-year-old woman presented with 1-week history of fatigue malaise and nausea. 2 days ago, she noticed yellowish discoloration of the sclera. She denied any history of abdominal pain or drugs intake. She has no significant medical history. On examination, she is jaundiced with no stigmata of chronic liver disease, conscious with no flapping tremors (see lab results)
Blood pressure 110/70 mmHg
Heart rate 76 /min
Temperature 36.6 °C
Test Result Normal Values
Direct bilirubin 108 1.5-6.5 mol/L
Total bilirubin 129 3.5-16.5 mol/L
Aspartate aminotransferase 1235 12-40 IU/L
Alanine aminotransferase 1987 5-40 IU/L
Alkaline phosphatase 156 39-117 IU/L
Gamma glutamyltransferase 60 6 to 37 IU/L
Albumin 41 34-56 g/L
Which of the following is the most appropriate management?
A. Order PT, INR
B. Discharge her home
C. Admit and start IV dextrose
D. Refer her to liver transplant center

Answer: A
A 40-year-old man comes to the clinic complaining of diarrhea and abdominal cramps. The symptoms started a day after he returned home from trip to India. The stool are watery with no mucus or blood. Blood pressure 100/70 mmHg
Heart rate 98 /min
Temperature 37.3 °C
Which of the following is the most likely pathogen?
A. Rotavirus
B. Escherichia coll
C. Bacillus cereus
D. Clostridum perfringens

Answer: B
A 60-year-old man with chronic hepatitis C and liver cirrhosis presented to Emergency Room with perforated peptic ulcer that requires urgent laparotomy (see lab results).
Test Result Normal Values
Hb 90 130-170 g/L (Male) - 120-160 g/L (Female)
Platelets count 90 150-400 × 109/L
INR 2 0.8-1.2

Prothrombin time 17 10-13 sec
which of the following is the most appropriate next step in the management prior to the procedure?
A. Platelets
B. Cryoprecipitate
C. Fresh frozen plasma
D. Packed red blood cells

Answer: C
What is the recommended age for screening colonoscopy for average risk population?
A. 40
B. 50
C. 55
D. 60

Answer: B
A 45-year-old obese woman presented with 1-week history of jaundice, anorexia and abdominal pain. She has no chronic illnesses and not on any medications. On examination, she is jaundiced with stable vital signs and no fever. Abdomen exam showed mild tender liver with span of 13 cm, no splenomegaly (see lab results).
Test Result Normal Values
RBC 4 4.7-6.1 x 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 110 130-170 g/L. (Male)
120-160 g/L (Female)
Platelets count 212 150-400 × 109/L
WBC 18 4.5-10.5 x 109/L
Direct bilirubin 79 1.5-6.5 mol/L
Total bilirubin 98 3.5-16.5 mol/L
Aspartate aminotransferase 70 12-40 IU/L
Alanine aminotransferase 150 5-40 IU/L
Alkaline phosphatase 1000 39-117 IU/L
Which of the following is the next step in the management?
A. MRCP
B. Liver biopsy
C. Abdominal ultrasound
D. Abdominal CT scan of the liver

Answer: C
Charcot triad of fever, upper abdominal pain and jaundice is diagnostic of which of the following conditions?
A. Hepatitis
B. Cholangitis
C. Pancreatitis
D. Cholecystitis

Answer: B
A 60-year-old woman with recent history of feeling tiredness, fatigue and weight loss. She noticed as well worsening of her chronic constipation. Stool for occult blood was positive (see lab resuits).
Test Result Normal Values
RBC 3 4.7-6.1 x 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 89 130-170 g/L (Male)
120-160 g/L (Female)

HCT 0.25 0.42-0.52 (Male)
0.37-0.48 (Female)
Platelets count 234 150-400 × 109/L
Which of the following is the best next step in diagnosis?
A. CT enterography
B. Barium enema
C. Sigmoidoscopy
D. Colonoscopy

Answer: D
A 48-year-old woman with 4-months history progressive dysphagia for solid and liquid and she noticed as well weight loss. Lab investigation was normal apart from mild anemia. Upper GI endoscopy was recommended but she refused as her relative died post colonoscopy. Which of the following is the best next action?
A. Explore her concern
B. Discharge her from the clinic
C. Ask her son to sign the consent
D. Call for a meeting with the family

Answer: A
A 20-year-old man present to the clinic for advice. He has a positive family history of familial polyposis syndrome. He is asymptomatic with normal Lab tests. Which of the following is the best advice at this stage?
A. Close follow-up
B. Order colonoscopy
C. Tumour markers panel
D. His risk is very low and no further action

Answer: B
A 55-year-old man presented to the Colonoscopy Unit to perform screening colonoscopy. The consent was taken and procedure started. During colonoscopy, a large polyp 4 cm was seen which need to be removed by snare polypectomy which carries a high risk of bleeding and perforation than the screening colonoscopy. Which of the following is the best action?
A. Perform the polypectomy
B. Ask the relative to sign the new consent
C. Let the patient sign after the procedure
D. Repeat the procedure with high risk consent

Answer: D
A 32-year-old man with 1-year history of intermittent dysphagia for solid with no other alarm symptoms. He has no chronic medical illness. Upper GI endoscopy was recommended but he refused the procedure. Which of the following is the best next action?
A. Discharge from the clinic
B. Ask the brother to sign the consent
C. Show him that you are upset with his decision
D. Discuss with him other alternative like barium swallowing

Answer: D
A 30-year-old woman presented with hematemesis after a bout of prolonged vomiting. She has no significant medical history and not on any medications. Which of the following is the most likely diagnosis?
A. Gastritis

B. Duodenal ulcer
C. Mallory-Weiss tear
D. Barrett's esophagus

Answer: C
A 32-year-old man diagnosed with fibrostenotic ileal crohn's disease underwent ileocecal resection of 70 cm from the ileum. 1-month post-operative, started to complain of diarrhoea watery 5 times/ day with no mucus or blood. No history of abdominal pain. On examination, he is unremarkable. All basic work up was normal including stool culture (see lab result and report).
Test Result Normal Values
ESR 7 2-10 mm/h (Male)
3-15 mm/h (Female)
CT scan of abdomen:
Showed normal anastmosis site.
Which of the following is the most appropriate treatment?
A. Mesalamine
B. Prednisolone
C. Metronidazole
D. Cholestyramine

Answer: D

Which of the following diagnostic tools is the best option for screening for hepatocellular carcinoma in cirrhotic patients?
A. Liver function test
B. Alpha fetoprotein
C. Ultrasound abdomen
D. Triphasic CT scan of liver

Answer: C

A 36-year-old man presented to the clinic for evaluation of elevated liver enzyme. He is asymptomatic with no significant past medical history. He is a smoker and drinks alcohol every weekend. Examination was unremarkable (see lab results and reports).
Test Result Normal Values
Total bilirubin 12 3.5-16.5 pmol/L. Aspartate aminotransferase 147 12-40 IU/L Alanine aminotransferase 76 5-40 IU/L Alkaline phosphatase 119 39-117 IU/L Albumin 33 34-56 g/L
Iron 35 11.7-31.8 mol/L (Male)
9-30.4 pmol/L (Female)
Total iron binding capacity 34 11.6-31.3 mol/L (Male)
9-30.4 pmol/L (Female)
Ferritin 423 20-300 pg/L (Male)
20-120 g/L (Female)
Liver function tests: Hepatitis screen; Negative HCV, HBAg was negative with positive HBSAD. US Abdomen: Enlarged liver with no focal lesion.
Which of the following is the most likely cause for his underlying liver disease?
A. Hemochromatosis
B. Alcoholic hepatitis
C. Hepatocellular carcinoma
D. Chronic hepatitis B infection

Answer: B; AST:ALT Ratio is 2:1.

A 67-year-old man a known case of HCV cirrhosis admitted to Emergency Room with massive hematemesis. Variceal bleeding is suspected. On examination, he is drowsy with tense ascites (see lab results).
Blood pressure 110/70 mmH, Heart rate 86 /min
Oxygen saturation 95 % Test Result Normal Values Hb 8 130-170 g/L (Male) 120-160 g/L (Female)
INR 1.3 0.8-1.2
Which of the following management options has a great benefit in reducing this patient mortality?
A. Octreotid
B. IV ceftriaxone
C. Blood transfusion
D. Panataprazole infusion

Answer: B

A 42-year-old man with hepatitis B cirrhosis evaluated for focal lesion at US abdomen. He is asymptomatic. Currently on diuretics with well-controlled ascites (see lab results and report). Test Result Normal Values
Hb 110 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 123 150-400 x 109/L
INR 1.5 0.8-1.2
Total bilirubin 16 3.5-16.5 mol/L
Aspartate aminotransferase 98 12-40 IU/L
Alanine aminotransferase 65 5-40 IU/L
Alkaline phosphatase 120 39-117 IU/L
Albumin 30 34-56 g/L
CT scan of liver: were patent.
Demonstrated a 6 cm lesion with arterial enhancement at RT lobe, portal vein and hepatic veins
Which of the following is the best treatment option?
A. Sarafenib
B. Liver transplant
C. Surgical resection
D. Transarterial chemoembolisation

Answer: D; Child-Pugh Score is B so not a candidate for Resection

A 22-year-old pregnant woman complains of heartburn, which is mainly at night after the dinner when she is lying on the bed. She was started on antacid syrup with somewhat response to that. She denies history of abdominal pain or nausea. Which of the following is the best next step in the management?
A. Start panotprazole
B. Ambulatory pH testing
C. Lifestyle modifications
D. Perform upper endoscopy

Answer: C

A 24-year-old woman presents with 9-months of intermittent abdominal pain and anxiety. Occasionally, the pain associated with loose motions. She denies any alarm symptoms. All basic work-up (CBC, LFT, TSH, CRP and Celiac serology) were normal, Stool C/S and fecal calprotein was negative.
Which of the following is the most appropriate treatment?
A. Metronidazole

B. Increase fibre intake
C. Trial of gluten free diet
D. Tricyclic antidepressants

Answer: D

A 62-year-old man with no significant medical history presented to Emergency Room with 1-day history of 4 episodes of hematochezia. He denies history of abdominal pain, constipation or weight loss.
Blood pressure 90/50 mmH
Heart rate 112 /min
Which of the following is the most likely diagnosis?
A. Cecal cancer
B. Diverticulosis
C. Internal piles
D. Ischaemic colitis

Answer: B

A 45-year-old woman was diagnosed with GERD for 4 years on daily dose of PPI with a good control of her symptoms. However, she reported occasionally heartburn. She underwent endoscopy. which was normal. Gastric biopsy revealed Helicobacter pylori. She gave history of smoking 20 pack-year but she quit 3 years ago, but she gave history of social alcohol drink. On examination, she is obese otherwise normal. Which of the following risk factors is mostly contribute to worsening of her GERD and she should control it?
A. Obesity
B. Smoking
C. Alcohol intake
D. Helicobacter pylori

Answer: A
A 46-year-old man came to the clinic asking for advice about eradication of H.pylori. He is known to have GERD on daily dose of PPI for 8 years with good control. He developed new onset of epigastric pain and was found to have helicobacter pylori By ELISA testing. Which of the following can happen post H.pylori eradication in such patient?
A. No sequel post eradication
B. Need to increase the PPI dose
C. Reducing risk of barrett's esophagitis
D. Increase the risk of atrophic gastritis

Answer: B
A 60-year-old man with a history of diabetic mellitus, hypertension and osteoarthritis presents with melena. He gave a history of NSAID use for knee pain. He underwent upper GI endoscopy, which showed 1 cm duodenal ulcer with bleeding vessel, hemoclip was applied with good hemostasis (see lab result).
Blood pressure 100/60 mmHg Heart rate 110 /min
Oxygen saturation 95 %
Test Result Normal Values
Hb 8.4 130-170 g/L (Male)
120-160 g/L (Female)
Which of the following is the most appropriate next step in the management?
A. High dose of oral PPI BID
B. High dose of oral H2 blocker

C. IV PPI infusion for 24 hours then oral PPI
D. IV PPI infusion for 72 hours then oral PPI

Answer:D
A 44-year-old woman known to have gall bladder stones presented with biliary pain and jaundice. US abdomen confirmed CBD stones. She underwent ERCP for stone extraction. Few hours later, she developed sever epigastric pain, nausea and vomiting (see lab results).
Test Result Normal Values
Hb 11 130-170 g/L (Male)
120-160 g/L (Female)
WBC 18 4.5-10.5 x 109/L
Urea 11 2.75-7.4 mmol/L
Creatinine 110 44-115 mol/L
Total bilirubin 23 3.5-16.5 mol/L
Lipase 3000 0-160 IU/L
Which of the following management options has proven a survival benefit in such patient?
A. Urgent repeat of ERCP
B. Administration of IV antibiotics
C. Rectal indomethacin and IV steroid
D. IV ringer lactate at rate of 250 ml/ours

Answer:D
A 60-year-old man with a history of chronic alcohol intake presents with chronic dull epigastric pain and diarrhoea for 6 months. He reported 5 bulky stool a day (see reports).
72 hours stool collection:
>42 gram of fat/24 hours.
Plain abdomen X-ray: Showed pancreatic calcification.
Which of the following is the most effective strategy to control his symptoms?
A. High dose of PPI BID
B. Long course of metrodidazole
C. Lipase 30.000 IU with each meal
D. Diet modification with small frequent low fat meals

Answer: C; Chronic Pancreatitis causing Exocrine Pancreatic insufficiency
A 60-year-old man presented to Emergency Room with 3-days history of epigastric and RT upper quadrant pain and nausea. He gave a history of fever and rigors for 1-day duration. On Examination, Jaundice and tender RT upper abdomen with no rebound (see lab results and report).
Blood pressure 100/70 mmH Heart rate 120 /min Temperature 38.6 °C
Test Result Normal Values Hb 11 130-170 g/L. (Male) 120-160 g/L (Female)
WBC 16.4 4.5-10.5 x 109/L
Direct bilirubin 49 1.5-6.5 umol/L.
Total bilirubin 57 3.5-16.5 mol/L
Aspartate aminotransferase 150 12-40 IU/L
Alanine aminotransferase 200 5-40 IU/L
Alkaline phosphatase 250 39-117 IU/L
US abdomen: Reveals multiple small galistones with mild thickening of the wall. CBD dilated 9 mm.
Which of the following is the most likely diagnosis?
A. Acute Hepatitis
B. Liver abscess
C. Acute cholecystitis
D. Ascending Cholangitis

Answer: D

Which of the following is the least risk factors for cholesterol gallstones?
A. Obesity
B. Nulliparity
C. Rapid weight loss
D. Hypertriglyceridemia

Answer: B
A 34-year-old woman presented to the clinic for evaluation of persistent high liver enzymes for 6 months. She is asymptomatic. She has no significant medical or drugs history (see lab results and reports).
Test Result Normal Values
Direct bilirubin 1.1 1.5-6.5 mol/L
Total bilirubin 1.9 3.5-16.5 mol/L
Aspartate aminotransferase 53 12-40 IU/L
Alanine aminotransferase 43 5-40 IU/L
Alkaline phosphatase 546 39-117 IU/L
Albumin 35 34-56 g/L
US Abdomen: Unremarkable.
MRCP: Showed multi focal segmental biliary strictures and dilation. Which of the following is the best next step in the diagnosis?
A. Liver biopsy
B. Perform colonoscopy
C. Repeat abdomen ultrasound
D. Check anti-nuclear antibodies

Answer: B
What is the most common HCV genotype in Saudi Arabia?
A. 1
B. 2
C. 3
D. 4

Answer: D
A 50-year-old man referred to the clinic with positive CV. He is asymptomatic with no history of drug use or previous blood transfusion. Physical examination was unremarkable, Reviewing of his lab test showed normal LFT with positive anti-HCV (ELISA), Hepatitis C RNA was negative.
US abdomen showed normal liver.
Which of the following is the most appropriate next step in management?
A. Liver fibroscan
B. No further tests
C. Triphasic CT scan of liver
D. Repeat HCV RNA in 6 months

Answer: B
A 43-year-old man who is known to have chronic hepatitis B presents to the clinic for evaluation. He is asymptomatic with normal physical examination (see lab results and report).
Test Result Normal Values
RBC 5
4.7-6.1 x 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 11.9 130-170 g/L. (Male)
120-160 g/L (Female)
Platelets count 145 150-400 x 109/L

WBC 3.9 4.5-10.5 x 109/L
Direct bilirubin 17 1.5-6.5 mol/L
Total bilirubin 21 3.5-16.5 mol/L
Aspartate aminotransferase 89 12-40 IU/L
Alanine aminotransferase 61 5-40 IU/L
Alkaline phosphatase 160 39-117 IU/L
US Abdomen: Showed mild coarse liver echotexture.
Which of the following is the most appropriate treatment option?
A. Entecavir
B. Observation
C. Lamivudine
D. Pegylated Interferon

Answer: A

A 62-year-old man presents with a 3-month history of epigastric pain after eating and intermittent heartburn between meals. He has lost weight but does not have any nausea or vomiting. He has not noticed any change in bowel habit and has not passed any blood or melena stools. He has no past medical history. He quit smoking 2 years ago. What is the most appropriate management strategy?
A. Routine ultrasound of the abdomen
B. Urgent referral for upper Gl endoscopy
C. An oral proton pump inhibitor for 4 weeks
D. Blood testing for helicobacter pylori infection

Answer: B
A 38-year-old man presents to the clinic complaining of severe epigastric pain that is worse after eating. He describes the pain as burning in nature with mild improvement with antacid therapy. Physical examination is unremarkable (see lab results).
Test Result Normal Values
Hb 120 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 450 150-400 x 109/L WBC 10 4.5-10.5 x 109/L
Urea 8.5 2.75-7.4 mmol/L
Creatinine 65 44-115 mol/L
What is the most appropriate test to establish the diagnosis?
A. CT scan of the abdomen
B. upper endoscopy
C. Barium swallow
D. H. pylori antibody

Answer: B
A 29-year-old man who is smoker for 5 years had appendectomy 2 years ago. He came to the clinic to discuss with you his risk to develop Crohn's disease because his brother has Crohn's disease that difficult to be controlled. Which of the following increase the risk for the development of Crohn's disease?
A. Being male
B. Cigarette smoking
C. No known risk factors
D. History of appendectomy

Answer: B
A 30-year-old woman presents to the Emergency Department with 2-days history of severe abdominal pain that radiated to the back, which is associated with nausea and vomiting. On

examination, she has normal vital signs with tender epigastrium but without rebound (see lab results).
Test Result Normal Values
Platelets count 450 150-400 x 109/L WBC 14 4.5-10.5 x 109/L
Alanine aminotransferase 80 5-40 IU/L Aspartate aminotransferase 65 12-40 IU/L Amylase 800 24-151 IU/L
Lipase 750 0-160 IU/L
HCT 49 0.42-0.52 (Male)
0.37-0.48 (Female)
Which of the following predict a poor prognosis for this patient?
A. Hematocrit > 44%
B. Lipase > 500 IU/L
C. Amylase > 600 IU/L
D. Albumin > 3.5 g/di

Answer: A
A 30-year-old man with 3 years history of intermittent abdominal pain, bloating and diarrhoea is seen in the clinic. Which of the following complains is highly suggestive of irritable bowel syndrome?
A. Diarrhoea that alternate with constipation
B. Undiagnosed food particles in the stool
C. Blood streak mixed with stool
D. Diarrhoea at night

Answer: A
A 55-year-old man with chronic hepatitis C present to the clinic for regular follow-up. His examination reveals a palpable right upper quadrant mass. CT scan of his abdomen shows a large 6*6 cm mass in the right lobe of the liver. A mass biopsy confirms the diagnosis of hepatocellular carcinoma. What is the most appropriate management steps at this stage?
A. Start chemotherapy
B. Start radiotherapy
C. Referral for palliative medicine
D. Referral for surgical resection

Answer: D
A 30-year-old asymptomatic woman presents to the clinic as pre-employment screening. Her examinations are unremarkable. Her laboratory tests revealed, positive HbsAg, anti-HBc (IgG) and anti- HBe but negative anti-HBs and HBeAg. Which of the following statements is the most accurate?
A. She has acute hepatitis B
B. She has chronic hepatitis B
C. She has received hepatitis B vaccine
D. She has been exposed to hepatitis B but now in recovery period

Answer: B
A 28-year-old woman has 2-week history of jaundice and an 11-month history of progressive fatigue and arthralgia. She is not on any regular medication and does not drink alcohol. Her examination reveals hepatosplenomegaly. Viral hepatitis screening tests are negative (see lab results).
Test Result Normal Values
Hb 10 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 120 150-400 x 109/L
WBC 7 4.5-10.5 x 109/L
INR 1.5 0.8-1.2

Total bilirubin 26 3.5-16.5 mol/L
Aspartate aminotransferase 920 12-40 IU/L
Alanine aminotransferase 890 5-40 IU/L
Alkaline phosphatase 160 39-117 1U/L
Albumin 30 34-56 g/L
Which of the following is the most likely diagnosis?
A. Hepatocellular carcinoma
B. Autoimmune hepatitis
C. Primary biliary cirrhosis
D. Acute cholangitis

Answer: B
A 34-year-old man presents to the clinic with symptoms of gastrosophageal reflux disease that not responding well to esomeprazole 20 mg once a day. Despite mild improvement in his symptoms, he continues to wake up with an acid taste in the middle of the night. He has not had difficulty swallowing. He confirms that he takes the medication properly. Physical examination is unremarkable. His laboratory investigation are within normal range. Which of the following is the most appropriate management for this patient?
A. Switch to pantoprazole
B. Perform upper endoscopy
C. Add amoxicillin and clarithromycin
D. Increase the dose of esomeprazole

Answer: D
A 35-year-old nurse is seen in the clinic after a recent routine screening for viral hepatitis infection. She was found to have positive antibody to hepatitis C virus (HCV). Her medical history is unremarkable; she has not used illicit drugs or had any history of blood transfusions She currently feels well and takes no medications. Vital signs and physical examination are normal. Laboratory studies reveal a positive CV antibody test, but HCV RNA testing is negative (see lab results).
Test Result Normal Values
Hb 120 130-170 q/L (Male)
120-160 g/L (Female)
Platelets count 300 150-400 x 109/L WBC 8 4.5-10.5 x 109/L
Aspartate aminotransferase 26 12-40 IU/L
Alanine aminotransferase 30 5-40 IU/L
Alkaline phosphatase 80 39-117 IU/L
Gamma glutamy/transferase 25 6 to 37 IU/L
Which of the following is the most appropriate action to be done?
A. No further testing
B. Perform liver ultrasound
C. Repeat HCV antibody testing
D. Perform serial alanine aminotransferase monitoring

Answer: A
A 40-year-old man is evaluated in the Emergency Department for several episodes of hematemesis. He has no previous similar episodes. His history is unremarkable. He takes no medications. There is no jaundice or other signs of chronic liver disease. Abdominal examination reveals no tenderness, guarding, or rebound (see lab results).
Blood pressure 105/65 mmHg Heart rate 110 /min Temperature 36.8 °C
Test Result Normal Values
Hb 95 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 250 150-400 x 109/L

WBC 12 4.5-10.5 x 109/L
Prothrombin time 11 10-13 sec
Alanine aminotransferase 30 5-40 IU/L
Alkaline phosphatase 26 39-117 IU/L
Blood urea nitrogen 14 2.8 to 8.9 mmol/L
Creatinine 88 44-115 mol/L
Which of the following is the most likely cause of his gastrointestinal bleeding?
A. Erosive gastritis
B. Peptic ulcer disease
C. Oesophageal varices
D. Mallory-Weiss tear

Answer: B
A 45-year-old man with a diagnosis of gastrosophageal reflux disease who is taking a maximum dose of pantoprazole for several weeks presents to the clinic with heartburn that is minimally improved with treatment. The patient verified that he is taking the medication 30 minutes before meals. He takes no other medications. His physical examination is unremarkable. Upper endoscopy findings are normal. Which of the following is the most appropriate next step in the management?
A. Repeat endoscopy
B. Barium esophagogram
C. Esophageal manometry
D. Ambulatory pH impedance monitoring

Answer: D
A 50-year-old man presents to the Emergency Department with acute onset of abdominal pain, nausea and vomiting. Abdominal examination shows tenderness and guarding at the epigastrium. CT scan of the abdomen reveals fluid around the pancreas. He was admitted and started on intravenous fluids; analgesics and nasogastric suction (see lab results).
Test Result Normal Values
Hb 120 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 190 150-400 x 109/L WBC 18 4.5-10.5 x 109/L
Aspartate aminotransferase 55 12-40 IU/L Alanine aminotransferase 65 5-40 IU/L Lactate dehydrogenase 6 1.4-3.56 mol/L Amylase 850 24-151 IU/L
Lipase 900 0-160 IU/L
Which of the following is the most appropriate next step in the management?
A. Add imipenem
B. Add pantoprazole
C. Add methyiprednisolone
D. Urgent surgical consultation

Answer: A
A 32-year-old man is evaluated for a 10-month history of fatigue and diarrhoea. He has a good appetite but has unintentionally significant weight loss over the past 10 weeks. He has not had blood in the stool, melena, or other obvious source of blood loss. His systemic examinations are unremarkable. Stool is negative for occult blood (see lab results).
Blood pressure 110/70 mmHg Heart rate 80 /min Respiratory rate 16 /min Temperature 36.6 °C
BMI 18 kg/m2
Test Result Normal Values
Hb 109 130-170 g/ (Male)
120-160 g/L (Female)
MCH 20 28-33 pg/cell
MCV 70 80-95 f1

Platelets count 455 150-400 x 109/L
WBC 8 4.5-10.5 x 109/L
Aspartate aminotransferase 30 12-40 IU/L Alanine aminotransferase 26 5-40 IU/L Albumin 32 34-56 g/L
Total bilirubin 10 3.5-16.5 pmol/L
Which of the following is the best treatment?
A. Loperamide (anti-diarrhoea agent)
B. Metronidazole (antibiotic drug)
C. Mebendazole (anti-parasitic drug)
D. Gluten-free diet

Answer: D

A 34-year-old woman presents with a 4-week history of retrosternal burning pain. The pain is often worse following eating. Her past medical history includes depression and she is on escitalopram 10 mg OD. Clinical examination is unremarkable. Which of the following is the most likely diagnosis?
A. Pericarditis
B. Pneumothorax
C. Myocardial infarction
D. Gastro-oesophageal reflux disease

Answer: D
A 38-year-old woman with a history of chronic abdominal pain and diarrhea develops tender red nodules on her shins. The findings are most consistent with which of the following?
A. Celiac disease
B. Crohn's disease
C. Diverticular disease
D. Irritable bowel syndrome

Answer: B
A 38-year-old woman presented with a 7-weeks of frequent, loose and bloody bowel movements that associated with campy abdominal and rectal pain. She has family history of crohn's disease. Stool analysis is negative (see lab results).
Blood pressure 110/70 mmH, Heart rate 76 /min, Temperature 36.6 °C
Test Result Normal Values
RBC 9.2 4.7-6.1 × 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 111 130-170 g/L (Male)
120-160 g/L (Female)
HCT 0.41 0.42-0.52 (Male)
0.37-0.48 (Female)
MCH 27 28-33 pg/cell
MCHC 330 320-360 g/L
MCV 77 80-95 f1
Platelets count 220 150-400 x 109/L
C-reactive peptide 22 <8.2 mg/L
Which of the following is the most appropriate next step in establishing diagnosis?
A. Anorectal manometry
B. Upper GI endoscopy
C. Celiac disease testing
D. Measurement of fecal calprotectin

Answer: B
A 28-year-old man with 4-months history of non-specific crampy abdominal pain and diarrhea alternate with constipation that is not related to any specific food. He has no history of food allergy. No history of weight loss. Blood pressure 110/70 mmH, Heart rate 76 /min, Temperature 36.6 °C, BMI 23 kg/m2
Which of the following is the most likely diagnosis?
A. Irritable bowel syndrome
B. Lactose intolerance
C. Colonic cancer
D. Crohn's disease

Answer: A
A 34-year-old man previously healthy brought a laboratory results for hepatitis B virus from a private clinic (see lab results).
Test Result
HBAg Negative
HBeAg Negative
Anti- HBsAg Positive
Anti-HBc IgG Negative
What is the clinical state the laboratory results indicate?
A. Previous Hepatitis B infection
B. Acute Hepatitis B infection
C. Chronic Hepatitis B infection
D. Previous hepatitis B immunization

Answer: D
A 31-year-old woman presented to the Emergency Room with mild acute cholecystitis. She was started on intravenous fluids and empiric antibiotics. Which of the following is the most appropriate next step?
A. Endoscopic retrograde cholangiopancreatography
B. Laparoscopic cholecystectomy within 72 hours
C. Cholecystectomy after 3 months
D. No more intervention needed

Answer: B
A 34-year woman presented with recurrent biliary colic for 5 months. Liver function test is within normal limits. Liver ultrasound shows cholelithiasis. Which of the following procedures is the most appropriate?
A. Cholecystectomy
B. Upper Gi endoscopy
C. Gall bladder lithotripsy
D. Endoscopic retrograde cholangiopancreatography

Answer: A
A 29-year-old man presented with painless jaundice after resolution of upper respiratory tract viral infection, otherwise he is asymptomatic. Clinical abdominal examination is unremarkable. Peripheral blood smear is normal (see lab results).
Test Result Normal Values
RBC 5.1 4.7-6.1 x 1012/L (Male) 4.2-5.4 x 1012/L. (Female)
Hb 133 130-170 g/L. (Male) 120-160 g/L (Female)
MCV 88 80-95 fl
Reticulocyte 0.21 0.2-1.2 %
Platelets count 220 150-400 x 109/L

WBC 5.2 4.5-10.5 x 109/L
Alkaline phosphatase
41 39-117 1U/L
Alanine aminotransferase 12 5-40 IU/L
Aspartate aminotransferase 20 12-40 IU/L
Indirect bilirubin 18 3.2-12.1 mol/L
Direct bilirubin 1.51 1.5-6.5 mol/L
Total bilirubin 19.51 3.5-16.5 mol/L
HCT 0.44 0.42-0.52 (Male)
0.37-0.48 (Female)
Which of the following is the most likely cause of his jaundice?
A. Hemolytic anemia
B. Fatty liver disease
C. Gilbert syndrome
D. Common bile duct obstruction

Answer: C

A 62-year-old woman presents with dysphagia. Her symptoms were first noticed a few months ago and have steadily worsened. Initially she had difficulty swallowing solids but over recent weeks has also been having difficulty swallowing liquids. She lost 5 Kg over 5 weeks. On examination, she looks thin and slightly pale. Which of the following is the most likely diagnosis?
A. Achalasia
B. Pharyngeal pouch
C. Oesophageal cancer
D. Gastro-oesophageal reflux disease

Answer:C
A 24-year-old woman is diagnosed with celiac disease Which of the following foods should she avoid?
A. Rice
B. Barley
C. Maize
D. Potatoes

Answer: B
A 40-year-old woman brought to Emergency Room with severe colicky right upper quadrant pain. The pain started after eating fish and chips for lunch. By the time she is seen, an hour later, the pain has settled and she is otherwise asymptomatic. Clinical examination is unremarkable. Which of the following is the most likely cause of her symptoms?
A. Acute cholecystitis
B. Gastroenteritis
C. Renal colic
D. Biliary colic

Answer: D
A 21-year-old woman presents with a 3-months history of upper abdominal pain. The discomfort and bloating becomes worse by eating. There is no diarrhea or constipation and her appetite is good. Clinical examination of her abdomen is normal. Helicobacter test is negative and other blood tests are all normal, Which of the following is the most likely diagnosis?
A. Irritable bowel syndrome
B. Functional dyspepsia
C. Celiac disease

D. Crohn's disease

Answer: B
A 25-year-old woman presented with episodic abdominal pain and altered bowel habits that have been present intermittently for the past year. The pain is "crampy." and moderate to severe. Defecation brings some relief, but she sometimes must pass several loose, watery stools before the pain resolves. She occasionally notes mucus in her stool and sometimes feels that evacuation is incomplete even though she is unable to pass more stool. Her body weight is stable. Abdominal examination are normal.
Which of the following is the most appropriate test to establish diagnosis?
A. Helicobacter pylori test
B. Celiac disease testing
C. Thyroid function test
D. CT abdomen

Answer: B

A 60-year-old man presents with low-grade fever, insomnia and progressive abdominal distention with vague abdominal pain. Physical examination reveals flapping tremor and moderate ascites. Ascitic fluid analysis showed high SAAG and neutrophil count of 350 cells/L. Which of the following is the most appropriate management?
A. Intravenous antibiotic
B. Total paracentesis
C. Syrup lactulose
D. Oral diuretic

Answer: A
A 55-year-old man with Liver cirrhosis is admitted to the hospital with a massive upper gastrointestinal (GI) bleeding. In addition to volume restoration, which intravenous medication is recommended before upper GI endoscopy?
A. Vitamin K
B. Octreotide
C. Propranolol
D. Vasopressin

Answer: B
A 28-year-old woman is evaluated because of recurrent lower abdominal pain and diarrhea for the last 6 months. She also reported having occasional joint pains and a 3 kg weight loss. She received treatment with antibiotics, dietary restrictions, and antispasmodics without improvement. Barium series showed thickening of the terminal ileum and few strictures in the jejunum (see lab results).
Test Result Normal Values
Hb 94 130-170 g/. (Male)
120-160 g/L (Female)
ESR 85 2-10 mm/h (Male)
3-15 mm/h (Female)
Which of the following is expected feature on physical examination?
A. Nail pitting
B. Splenomegaly
C. Perianal diseases
D. Erythema marginatum

Answer: C

A 56-year-old man presents to the Gastroenterology Clinic with frequent heartburn. He had upper GI endoscopy and biopsy, which revealed Barret's esophagus with low dysplasia. Which of the following is the best initial option in management?
A. Esophageal resection
B. Pantoprazole
C. Ranitidine
D. Sucralfate

Answer: B
A 60-year-old man with diabetes presented with progressive ascites, fever and weight loss for 2-months. Paracentesis revealed low serum-ascites albumin gradient (SAAG).
Which of the following is the most likely diagnosis?
A. Liver cirrhosis
B. Peritoneal tuberculosis
C. Veno-occlusive disease
D. Restrictive cardiomyopathy

Answer:B
A 44-year-old man with HBV cirrhosis presents with increasing abdominal girth, lower limb edema, and 10 kg weight gain over the last 3 months. On examination, he has jaundice and moderate ascites (see lab resuits and report).
Test Result Normal Values
Sodium 132 134-146 mmol/L Potassium 3.8 3.5-5.1 mmol/L Creatinine 65 44-115 mol/L
Albumin 30 34-56 g/L
Total bilirubin 25 3.5-16.5 mol/L
Ascetic fluid analysis: showed an albumin level of 12 g/L and neutrophil count of 180/ul.
In addition to salt-restricted diet, what is the best next step in management?
A. Trans-jugular Intrahepatic Portosystemic shunt
B. Furosemide and spironolactone
C. Large-volume paracentesis
D. Terlipressin and albumin

Answer: B
In a patient with ulcerative colitis, which of the following is associated with increased risk of colorectal cancer?
A. Colitis limited to the most distal 15 cm of colon
B. Disease of three-year duration
C. Primary sclerosing cholangitis
D. Mild inflammation

Answer: C

A previously healthy 22-year-old woman reports 3-month history of frequent bloody bowel movements with mucus and occasional right hip pain. She has had no history of contact with sick patient and no recent intake of drugs. Physical examination findings are within normal except for gross blood on rectal examination (see lab results and report).
Test Result Normal Values
Hb 110 130-170 g/L. (Male) 120-160 q/L (Female)
MCV 72 80-95 fl
Stool analysis: Shows many RBCs and few WBCs.
Which of the following is the most likely diagnosis?
A. Internal haemorrhoids
B. Ulcerative colitis

C. Infectious colitis
D. Celiac disease

Answer: B
A 20-year-old woman with no previous medical illnesses, presented with 7-day history of increasing jaundice, nausea, vomiting and right sided upper abdominal pain (see lab results).
Test Result Normal Values
Total bilirubin 26 3.5-16.5 umol/L
Aspartate aminotransferase 700 12-40 IU/L
Alanine aminotransferase 1650 5-40 IU/L
Alkaline phosphatase 126 39-117 IU/L
Which of the following tests has the greatest prognostic value?
A. Bilirubin
B. Albumin
C. Prothrombin time
D. Alanine aminotransferase

Answer: C
A 28-year-old woman presents with right lower quadrant abdominal pain, loose bowel motions, and weight loss for 3 months. Colonoscopy and biopsy of the terminal ileum demonstrated non-caseating granulomas. Which of the following is the most likely diagnosis?
A. Pseudomembranous colitis
B. Intestinal tuberculosis
C. Crohn's disease
D. Celiac disease

Answer: C
A 25-year-old woman presents with abdominal pain, bloating and weight loss. Physical examination is unremarkable except for BMI 18% (see lab results and report),
Test Result Normal Values
Hb 90 130-170 g/L. (Male)
120-160 g/L (Female)
MCV 70 80-95 fl
Calcium 2.05 2.15-2.62 mol/L
Phosphate, inorganic 0.79 0.82-1.51 mmol/L
Tissue transglutaminase antibody: Positive.
Which of the following is the best management?
A. Glucocorticoids
B. Gluten free diet
C. Metronidazole
D. Ciprofioxacin

Answer: B
A 56-year-old man presents with epigastric pain, bloating and weight loss. Gastroscopy showed antral mass and biopsy is taken (see reports).
Helicobacter pylori breath test: Positive.
Biopsy report: Mucosa-Associated Lymphoid Tissue lymphoma (MALToma).
Which of the following is the next step in management?
A. Eradication therapy for H. Pylori
B. Referral for gastrectomy
C. Chemotherapy
D. Radiotherapy

Answer: A
A young woman who is known to have pan ulcerative colitis is admitted for a severe relapse. Since there was no response to medical treatment, surgery is decided. The patient is very sceptical about surgery and requested more information.
What is the appropriate way to address her concerns?
A. Tell her that she will die if surgery is not performed
B. Discard her concerns as unfounded and push her for surgery
C. Inform her that surgery will leave her with a colostomy bag
D. Agree on another meeting with the surgeon and the gastroenterologist

Answer: D
A 52-year-old man, who has been medically treated as peptic ulcer for a long time, underwent upper endoscopy, which showed a malignant ulcer. During the visit, he started blaming his primary physician and wanted to file a complaint.
Which of the following is the most appropriate response?
A. Agree that it is his right to do so
B. Take a neutral stand in this issue
C. Inform him that his doctor did what he could do
D. Tell him that he should blame himself not his doctor

Answer: A
A married man, who had a trip to Thailand 2 months ago, is complaining of recurrent diarrhoea and weight loss. He underwent HIV testing which came reactive. However, he requested not to tell anyone including his wife. Which of the following is the best response?
A. Disclose the result to his emplover
B. Request the social worker to inform his wife
C. Report to the infection control unit in the hospital
D. Respect his request not to disclose the result to anyone

Answer: C
A 60-year-old man presented with high-grade fever and left pleuritic chest pain. On examination, he has early clubbing and evidence of moderate pleural effusion in the left side of the chest. Pleural fluid analysis revealed pH of 7.0. Which of the following is the most likely diagnosis?
A. Empyema
B. Chylothorax
C. Parapneumonic effusion
D. Tuberculous pleural effusion

Answer: A

A 59-year-old woman comes to the clinic with headache, dyspnea on exertion and occasional palpitations (see lab reults).
Test Result Normal Values
Hb 95 130-170 g/L (Male)
120-160 g/L (Female)
MCV 68 80-95 fl
Platelets count 530 150-400 x 109/L
Total iron binding capacity 28 11.6-31.3 mol/. (Male)
9-30.4 umol/L (Female)
Ferritin 3 20-300 pg/L (Male)
20-120 pg/L (Female)

Apart from starting iron supplement, which of the following is the most appropriate next step in management?
A. Urea breath test
B. Stool occult blood
C. CT-scan abdomen and pelvis
D. Endoscopy and colonoscopy

Answer: D

A 46-year-old man comes to the clinic with dyspnea on exertion and palpitations. On examination, he has patches of vitiligo (see lab results).
Test Result Normal Values
Hb 89 130-170 g/L (Male)
120-160 g/L (Female)
MCV 103 80-95 fl
Platelets count 128 150-400 x 109/L
Ferritin 105 20-300 g/L (Male)
20-120 g/L (Female)
Vitamin B12 (Men) 122 160-900 ng/L
Folic Acid (Men) 8 <4 ug/L
Which of the following is the most appropriate test to be ordered next?
A. Gastroscopy and antral biopsy
B. Intrinsic factor antibodies
C. Anti-gliadin antibodies
D. Colonoscopy

Answer: B
A 60-year-old man admitted to the Emergency Room with 1-day history of hematemesis. He has multiple spider angiomata on his chest with bilateral gynecomastia. Abdominal examination showed splenomegaly (see lab results).
Blood pressure 100/60 mmHg, Heart rate 100 /min, Respiratory rate 18 /min, Temperature 36.6 °C, Oxygen saturation 95 %
Test Result Normal Values
Hb 8.5 130-170 g/L (Male)
120-160 g/L (Female)
HCT 0.23 0.42-0.52 (Male)
0.37-0.48 (Female)
Which of the following is the most appropriate next step in management after the initial fluid resuscitation?
A. Blood transfusion
B. Upper GI endoscopy
C. Selective angiography
D. Nasogastric tube insertion

Answer:B
A 22-year-old man presented with 6-month history of non-bloody diarrhea, recurrent abdominal cramps and weight loss. On examination, he looked pale with oral ulcers. Abdominal examination showed tender right lower quadrant.
Blood pressure 110/70 mmH
Heart rate 100 /min
Temperature 38.1 °C
Which of the following is the most likely diagnosis?
A. Celiac disease

B. Crohn's disease
C. Ulcerative colitis
D. Chronic appendicitis

Answer: B
A 35-year-old wornan presented with chronic vague gastric pain for several years. The pain is revealed by food. Serum immunoglobulin studies for igG against helicobacter pylori was positive. Upper Gl endoscopy showed antral gastritis with small clean bases ulcer at duodenum. The patient was treated for Hi.pylori.
Which of the following is the most appropriate test to ensure h pylori eradication?
A. Rapid urease test
B. Urea breath test
C. Culture of gastric mucosa
D. Repeat lgA and IgG against H.pylori

Answer:B
A 60-year-old man presented with jaundice for 2 months. Physical examination revealed a nodular enlarged liver (see report).
CT Scan: Showed cirrhotic liver with 2 lesions at right lobe measuring 4 x 5 cm.
Which of the following risk factors is strongly associated with this condition?
A. Aflatoxin
B. Lead toxin
C. Wilson disease
D. Hepatitis B infection

Answer: D
A 35-year-old man with a 7-year history of ulcerative colitis. presented with insidious onset of progressive fatigue, pruritis and jaundice. He underwent MRCP, which demonstrates multiple short strictures of biliary tree at both intrahepatic and extrahepatic ducts (see lab results). Test Result Normal Values
Alkaline phosphatase 785 39-117 IU/L Alanine aminotransferase 60 5-40 IU/L Aspartate aminotransferase 40 12-40 IU/L
Creatinine kinase 34 24-170 IU/L (Female)
24-195 IU/L (Male)
Which of the following is the most likely diagnosis?
A. Liver cirrhosis
B. Bile duct atersia
C. Primary biliary cholangitis
D. Primary sclerosing cholangitis

Answer: D
A 40-year-old man with a history of hypertriglyceridemia presents to Emergency Room with 4- hour of sever epigastric pain, nausea, and vomiting. The pain was steady and radiate to the back. The patient was in pain and agitated (see lab results and report).
Blood pressure 100/70 mmHg
Heart rate 110 /min Respiratory rate 23 /min Temperature 38.3 °C Oxygen saturation 94 %
Test Result Normal Values
RBC 5 4.7-6.1 x 1012/L (Male) 4.2-5.4 x 1012/L (Female) Hb 12 130-170 q/L. (Male) 120-160 g/L (Female)
HCT 39 0.42-0.52 (Male)
0.37-0.48 (Female)
Platelets count 344 150-400 x 109/L
WBC 16 4.5-10.5 x 109/L

Alanine aminotransferase 97 5-40 IU/L
Aspartate aminotransferase 87 12-40 IU/L
Amylase 150 24-151 IU/L
Lactate dehydrogenase 169 60-160 IU/L
Lipase 30 0-160 IU/L
Plain Abdominal X-ray:Showed an air-filled intestinal loop at left upper quadrant. Which of the following is the most likely diagnosis?
A. Acute pancreatitis
B. Bowel perforation
C. Mesentric ischemia
D. Diverticular disease

Answer: A
A 62-year-old patient known case of chronic hepatitis C presented with abdominal pain and swelling. Examination showed ascites, abdominal tenderness and peripheral edema. Diagnostic ascitic tap done and revealed a neutrophil count of 400/mm (<250 mm)
Blood pressure 110/70 mmHg
Heart rate 90 /min
Temperature 37.6 °C
Which of the following is the appropriate next step in the management?
A. Fluid restriction
B. Oral spironolactone
C. Intravenous antibiotics
D. Therapeutic paracentesis

Answer: C
A 69-year-old man seen in Outpatient Clinic for a routine check-up. No relevant history, apart from occasionally low back pain on non-steroidal anti-inflammatory drugs as needed. No chronic medical history. On examination, his abdomen was soft with no masses. Rectal examination was normal (see lab results).
Test Result Normal Values
RBC 4 4.7-6.1 x 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 8 130-170 g/1. (Male)
120-160 q/L (Female)
HCT 38 0.42-0.52 (Male)
0.37-0.48 (Female)
MCH 23 28-33 pg/cell
MCV 70 80-95 fl
Platelets count 456 150-400 x 109/L
WBC 5 4.5-10.5 x 109/L
Which of the following is the most important next step?
A. Sigmoidoscopy
B. Ultrasound abdomen
C. Stool for occult blood
D. Upper and lower Gl endoscopy

Answer: D
A 34-year-old pregnant woman, known case of chronic hepatitis B, comes to the clinic for counselling regarding the risk of hepatitis B transmission to her baby (see lab results and report).
Test Result Normal Values
Alkaline phosphatase 198 39-117 IU/L
Alanine aminotransferase 120 5-40 IU/L

Aspartate aminotransferase 67 12-40 IU/L
Hepatitis B viral load: 1 million copies/mi.
Which one of the following the baby should receive to reduce the vertical transmission risk?
A. Hepatitis b vaccine
B. Hepatitis b immunoglobulin
C. Hepatitis b vaccine and immunoglobulin
D. Hepatitis b vaccine, immunoglobulin and entecavir

Answer: C
A 32-year-old woman 30-week pregnant G1+0+0, presented to the clinic with pruritus and jaundice. There is no fever, abdominal pain, or vomiting. She is not on any medications and no chronic medical illness (see lab results).
Test Result Normal Values
Hb 11 130-170 g/L. (Male)
120-160 g/L (Female)
Platelets count 160 150-400 x 109/L
WBC 7 4.5-10.5 x 109/L
Alkaline phosphatase 700 39-117 IU/L
Alanine aminotransferase 79 5-40 IU/L
Aspartate aminotransferase 100 12-40 IU/L
Which of the following is the most likely diagnosis?
A. Viral hepatitis
B. Hellp syndrome
C. Budd chiari syndrome
D. Cholestasis of pregnancy

Answer: D
A 47-year-old alcoholic patient presented with 6-week history of weight loss, fever, and right hypochondrum pain (see lab resuit and report).
Test Result Normal Values
Alpha fetoprotein 120 0-40 ng/mL
Abdominal Ultrasound: Showed ascites and focal liver lesion in a cirrhotic liver. Which of the following is the most likely diagnosis?
A. Liver abscess
B. Pancreatic cancer
C. Hydatid liver disease
D. Hepatocelluar carcinoma

Answer: D
A 23-year-old woman was informed that her husband has chronic hepatitis B infection. She was worried that he had transmitted the virus to her. She is currently asymptomatic apart from fatique. What is the first evidence of acute infection in her case?
A. HBEAg
B. HBsAg
C. Anti-HBc
D. Anti-HBs

Answer: B
A surgeon was performing a procedure on a patient with chronic hepatitis C infection. He accidentally had a needie stick injury. What is the likelihood of hepatitis C transmission?
A. 0.03 %
B. 0.3 %
C. 3 %

D. 30 %

Answer: C
A 16-year-old man presented to the Outpatient Department complaining of diarrhea for 3 years. He was passing 3-4 large volume stool per day, which was foul smelling and contained undigested food but no blood or mucus. He was also complaining of postprandial bloating and easy fatigability. On examination, he looked pale and has no secondary sexual characteristics
BMI 14 kq/m2
Test Result Normal Values
Hb 10 130-170 g/L (Male)
120-160 g/L (Female)
MCV 70 80-95 fl
Platelets count 433 150-400 x 109/L
Alanine aminotransferase 90 5-40 IU/L
Aspartate aminotransferase 60 12-40 IU/L
ESR 12 2-10 mm/h (Male)
3-15 mm/h (Female)
Which of the following is the most appropriate diagnostic test?
A. Stool for fat content
B. Barium follow through
C. Lactose hydrogen breath test
D. IgA antiendomysial antibodies

Answer: D
A 24-year-old man presented with 2-month of bloody diarrhea. He passes 3-4 loose motions day associated with blood and mucus. He gave history of left lower quadrant pain. No fever or weight loss. He underwent sigmoidoscopy, which showed loss of normal vascular pattern with erosions up to splenic flexure (see lab results).
Test Result Normal Values
RBC 5 4.7-6.1 x 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 10.9 130-170 g/L (Male) 120-160 g/L (Female)
Platelets count 488 150-400 x 109/L WBC 6 4.5-10.5 x 109/L
ESR 32 2-10 mm/h (Male)
3-15 mm/h (Female)
Consistency LIQUID (soft or tight)
Color YELLOW Brown
Mucus and blood ++ absent
Leukocytes +++ solitary
Which of the following is the best treatment option?
A. Mesalamine enema
B. Mesalamine suppository
C. Mesalamine oral and enema
D. Mesalamine oral and suppository

Answer: C
A 60-year-old man with chronic history of heartbum for several years, presented with occasional hematemesis. Upper GI endoscopy revealed intestinal-type metaplasia at the distal esophagus.Which of the following is the most likely diagnosis?
A. Oesophagitis
B. Mallory-weiss tear
C. Barretts oesophagus
D. Plummer vinson syndrome

Answer: C
A 30-year-old woman with left-sided ulcerative colitis presented to Emergency Room with 7-8 bloody diarrhea /day associated with low-grade fever, abdominal pain and joint pain at her knees. On examination, she looks ill and in pain. All stool cultures were negative.
Blood pressure 90/70 mmHg
Heart rate 120 /min
Respiratory rate 18/min Temperature 37.3 °C
Oxygen saturation 95 %
Test Result Normal Values
Hb 9 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 456 150-400 x 109/L. WBC 10 4.5-10.5 x 109/L.
ESR 122 2-10 mm/h (Male)
3-15 mm/h (Female)
After fluid resuscitation what is the best next step in management?
A. Azathioprine
B. Sulphasalazine
C. Oral budesonide
D. Iv methylprednisolone

Answer: D
A 28-year-old man who was previously healthy, presented to the Emergency Department with 5-days history of right upper quadrant abdominal pain, which was associated with nausea and 2 episodes of vomiting. 2 days later, his family noticed yellowish discolouration of his eyes and skin. No history of alcohol or drug use. On examination, he was jaundiced. Abdominal exam showed tender right upper abdomen (see lab results),
Test Result Normal Values
Alkaline phosphatase
109 39-117 IU/L
Alanine aminotransferase 990 5-40 IU/L
Aspartate aminotransferase 789 12-40 IU/L
Which of the following is the most appropriate next step in diagnosis?
A. Hepatitis A IgG
B. Hepatitis A IgM
C. Hepatitis B antigen
D. Hepatitis C antibodies

Answer: B
A 17-year-old man with no previous medical history, presented with 3-weeks history of bloody diarrhoea. He reported frequent urges to defecate, associated with mild lower abdominal cramps. He had experienced similar attacks in the past, but milder associated with bloating and loose stool, which was mucus with few blood drops. Abdominal examination was soft with mild tenderness at left lower quadrant but no guarding.
Blood pressure 110/70 mmHg
Heart rate 76 /min
Temperature 36.6 °C
Oxygen saturation 95 %
Which of the following is the most likely diagnosis?
A. Diverticulitis
B. Ischaemic colitis
C. Crohn's disease
D. Ulcerative colitis

Answer: D
A 20-year-old man with a history of high liver function test in the past presents, with tremors and dysarthria and dyskinesia. Slit-lamp examination reveals a greenish-brown ring at comeoscleral junction. Which of the following is the most likely diagnosis?
A. Antitrypsin deficiency
B. Hepatic hemangioma
C. Hemochromatosis
D. Wilsons disease

Answer: D
A 22-year-old woman with chronic diarrhea and malabsorption is suspected of having celiac disease, a jejunal biopsy taken. Which of the following histological finding is expected?
A. Flattened crypts
B. Subtotal villous atrophy
C. Epithelial cells with fat globules
D. Fissures and ulcers penetrating into submucosa

Answer: B
A 50-year-old man presents with 3-months history of jaundice, arthralgia, increase in skin pigmentation, and upper abdominal discomfort. He is on oral hypoglycemic medications for recently diagnosed diabetes mellitus. His viral hepatitis screening was negative. Examination revealed a bronze coloured skin, enlarged liver with span of 15 cm, no splenomegaly, and testicular atrophy. Cardiac examination revealed a 3rd heart sound (see lab results).
Test Result Normal Values
Direct bilirubin 24 1.5-6.5 mol/L
Total bilirubin 31 3.5-16.5 umol/L
Aspartate aminotransferase 79 12-40 IU/L
Alanine aminotransferase 112 5-40 IU/L
Alkaline phosphatase 109 39-117 IU/L
Random Glucose 7 3.9-5.5 mmol/L
Ferritin 400 20-300 g/L. (Male)
20-120 pg/L (Female)
Which of the following investigations is the most appropriate next step?
A. Alpha 1-antitrypsin
B. 24 hours urinary copper
C. Antimitochondrial antibodies
D. Hepatic iron load in liver biopsy

Answer: D
A 31-year-old patient came to Emergency Room with 1-day history of melena and dizziness. He has no significant medical history and takes no medications apart from ibuprofen for dental pain for 5 days. Examination revealed cold extremities, and faint peripheral pulses. Per-rectal examination revealed maroon coloured stool (see lab results).
Blood pressure 90/60 mmHg Heart rate 120 /min Respiratory rate 18 /min Temperature 36.6 °C Oxygen saturation 95 %
Test Result Normal Values
RBC 4.3 4.7-6.1 x 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 8.7 130-170 g/L. (Male)
120-160 g/L (Female)
Platelets count 343 150-400 × 109/L

WBC 8 4.5-10.5 x 109/L
Which of the following is the most appropriate next step in the management?
A. Normal saline
B. Colloid solution
C. Blood transfusion
D. Albumin and ringer lactate

Answer: A
A 25-year-old woman came to the clinic for an advice. She is planning to get pregnant; she is a known case of distal ulcerative colitis diagnosed 3 years ago on 3 grams of mesalamine granules. Her last exacerbation was a year ago required a short course of steroid. Currently she has a 2-3 bowel motions a day, occasionally seen a streaks of blood once every 4-5 weeks. Her latest fecal calprotectin is 90 (normal 0-50). Which one of the following is the most appropriate advice?
A. It is difficult to get pregnant now
B. She can get pregnant as her disease is controlled
C. Her risk of abortion is high and should be counselled
D. It is advisable to postpone the pregnancy for 3 years

Answer: B
A 65-year-old man known case of diabetes mellitus and hypertension presents to Emergency Room with 1-day history of vomiting fresh blood and melena. He is taking aspirin 81 mg daily. After fluid resuscitation, he was started on IV PPI. He underwent upper Gl endoscopy, which revealed a 2 cm duodenal ulcer at the bulb with visible vessel seen, endoscopic hemostasis was done using endo clip
Which of the following is the recommended duration for IV PPI post endoscopic management?
A. 24 hours
B. 36 hours
C. 48 hours
D. 72 hours

Answer: D
A 30-year-old woman a known case of fistulizing crohn's disease on biological therapy and azathioprine. She is pregnant on her third trimester. Her physician is planning to communicate with her obstetrician prior to her delivery about her delivery method and the vaccination of the child.
Which of the following is the recommended advice for the child vaccinations?
A. All vaccines can be given safely
B. Live attenuated vaccines are contraindicated
C. All vaccines should be postponed till 6 months
D. Live attenuated vaccines should be postponed for 12 months

Answer: D
A 65-year-old man presented with melena and hematemesis. He is a known case of diabetes mellitus, hyperlipidaemia, and he is on aspirin 81 mg, lipitor 20 mg, diamicron 80 mg and warfarin 3 mg a day. Underwent upper Gl endoscopy and showed multiple superficial ulcers at the antrum and duodenal bulb.
Which of the following medications is most strongly associated with his presentation?
A. Aspirin
B. Lipitor
C. Warfarin
D. Diamicron

Answer: A
A 21-year-old patient presented with 4-months history of itchy skin rash over his elbows with abdominal bloating and diarrhoea. No other medical problems. He has a cousin with similar

condition. He underwent upper and lower I endoscopy. Multiple duodenal erosions seen at the 1st and 2nd part with scalloping (see lab results and reports).
Test Result Normal Values RBC 4 4.7-6.1 x 1012/L (Male) 4.2-5.4 x 1012/L (Female)
Hb 10.9 130-170 g/L. (Male)
120-160 g/L (Female)
Alkaline phosphatase
112 39-117 IU/L
Alanine aminotransferase 123 5-40 IU/L Aspartate aminotransferase 79 12-40 IU/L Calcium 1.9 2.15-2.62 mmol/L
Random Glucose 6 3.9-5.5 mmol/L
ileocolonoscopy: Normal.
Duodenal biopsy: Showed subtotal villous atrophy. Which of the following is the best treatment option?
A. Oral steroid
B. Iv pantoprazole
C. Start gluten free diet
D. Oral iron tablets and repeat endoscopy in 3 months

Answer: C
Osteoporosis is one of common extra-intestinal manifestations of crohn's disease, which carries a concomitant fracture risk. Which of the following risk factors are more likely to be strongly associated with increase osteoporosis risk in crohn's disease?
A. Age 20-40
B. Premenopausal
C. Long steroid use
D. Extensive colonic involvement

Answer: C
A 35-year-old woman with heartburn and regurgitation for 6 months is referred for evaluation of chronic GERD symptoms. The patient states that she notes burning in her chest primarily after meals and occasionally experiences episodes of chest pain. She denies change in weight or lifestyle. She often consumes dinner 2 hours before going to bed. Her heartbum occurs mainly during the night. Physical examination is normal except for a body mass index (BMI) of 28 kg/m2. A complete blood count and metabolic panel are normal. The patient has been taking omeprazole 20 mg daily. An upper endoscopic examination is performed, and the appearance
of distal esophagitis. Which of the following is the best next step?
A. Add antacid twice/day
B. Ranitidine at bed time
C. Advise her to lose weight
D. Refer her for Nilsson Fundoplication

Answer: B
A 32-year-old man came to the clinic for follow-up visit. He is diagnosed of ileocolonic crohn's disease started on steroid and pentasa for 3 months. He noticed perineal pain and discharges for 3 weeks duration. His abdominal symptoms are well controlled with no abdominal pain or diarrhoea. Examination was normal apart from 2 perianal fistula were seen (see lab results and report),
Test Result Normal Values
RBC 4 4.7-6.1 x 1012/L (Male) 4.2-5.4 x 1012/L. (Female)
Hb 11 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 421 150-400 x 109/L WBC 7 4.5-10.5 x 109/L
ESR 36 2-10 mm/h (Male)

3-15 mm/h (Female)
MRI Pelvis: Confirmed the presence of perianal fistulae. Which of the following is the best next step in management?
A. Methotrexate
B. Azathiporine
C. Budesonide
D. Infliximab

Answer: D
A 40-year-old healthy man came to the clinic requesting a screening colonoscopy. He is asymptomatic with no family history of colon cancer. Clinical examination is unremarkable (see lab results).
Test Result Normal Values
RBC 4 4.7-6.1 x 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 13.3 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 321 150-400 x 109/L
WBC 4 4.5-10.5 x 109/L
Based on the latest recommendation from local and international gastroenterology associations, what is the recommended age of screening colonoscopy for this average risk patients?
A. 45
B. 55
C. 65
D. Order for him now

Answer: A
A 40-year-old man presented with several months history of dyspepsia. He denies any history of dysphagia, vomiting, or weight lost. He has no significant past medical history and not on any medications. He received helicopter pylori eradication (PPI standard dose, clarithromycin 500mg, and amoxil 1g bid for 10 days) with partial response. Basic routine lab was normal (see reports).
Endoscopy: Showed gastric erosions.
Biopsy: Showed h pylori organism.
Which of the following is the best treatment regimen?
A. Repeat the same regimen for a longer duration 14 days
B. PPI, nitromidazole 500mg bid, doxacyclin 100mg qid for 14 days
C. PPI, levofloxacin 500mg gid, amoxil 1 g bid, nitromidazole 500mg bid for 14 days
D. PPI, bismuthsubcitrate 300mg qid, tertacycline 500mg id , metronidazole 500mg tid for 14 days

Answer: D
A 45-year-old woman presented with 9-months history of worsening dysphagia for both solid and liquid. Recently she had episodes of ill defined central chest discomfort and nocturnal cough. Clinical examination was unremarkable (see lab results).
Test Result Normal Values
RBC 5 4.7-6.1 x 1012/L (Male) 4.2-5.4 x 1012/L (Female)
Hb 11.8 130-170 g/L. (Male) 120-160 g/L (Female)
Platelets count 300 150-400 x 109/L WBC 7 4.5-10.5 x 109/L
ESR 13 2-10 mm/h (Male)
3-15 mm/h (Female)
Which of the following is the most likely diagnosis?
A. Achalasia
B. Esophageal web
C. Pharyngeal pouch

D. Squamous cancer of esophagus

Answer: A
A 45-year-old man with several months of worsening dyspepsia. He denies any history of diarrhoea, fever, or weight loss. He has no medical problems. He was prescribed pantoprazole 40 mg daily with no relieve of his symptom (see reports).
Upper GI endoscopy: Showed multiple antral superficial ulcers.
Biopsy: Revealed mucosa-associated lymphoid tissue (malt) lymphoma with numerous H.pylori organism
CT scan of abdomen and chest: Normal.
Which of the following is the best management option?
A. Chemotherapy
B. Radiotherapy
C. Gastrectomy
D. H.pylori eradication

Answer: D
A 45-year-old healthy man recently diagnosed with hepatitis C virus (HCV) infection and was referred to the clinic for initiation of HCV therapy. HCV RNA viral load quantitation was 2 million I/ml and a Genotype 4, a Liver ultrasound that was normal with no cirrhosis or focal liver lesions (see lab results and report).
Test Result Normal Values
RBC 5 4.7-6.1 × 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 13 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 233 150-400 x 109/L
WBC 5 4.5-10.5 x 109/L
Alanine aminotransferase 89 5-40 IU/L
Aspartate aminotransferase 56 12-40 IU/L
Creatinine 100 44-115 umol/L
Fibroscan: Showed fibrosis score of F3.
Which of the following is the recommended treatment?
A. Entacavir
B. Peg-interferon
C. Ledipasvir/sofosbuvir
D. Observe and repeat lab in 3 months

Answer: C
A 50-year-old woman presented with few months history of feeling dizziness when getting out of bed she also feels very warm and flushed. She visited the Emergency Room several times with frequent episodes of abdominal cramps and watery diarrhea over the past year. She gave a history of itchy skin. On examination, she has a 2/6 systolic murmur at left lower sternal border, chest examination reveals wheezing, abdomen was unremarkable. Laboratory results and stool cultures were normal.
Which of the following is the most appropriate next step in diagnosis?
A. Ultrasound abdomen and pelvis
B. Double contrast barium enema
C. Serum lipase and amylase level
D. Urine for 5-hydroxindoleacetic acid

Answer: D

A 30-year-old man presented with 1-day of sever epigastric pain radiating to the back with vomiting. He gave a history of alcohol drinking occasionally. No other medical problems and not on any medications. On examination, tender upper abdomen without guarding (see lab results).
Blood pressure 98/70 mmHg
Heart rate 110 /min
Respiratory rate 18 /min
Temperature 37.2 °C
Test Result Normal Values
Hb 11.6 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 311 150-400 × 109/L
WBC 13 4.5-10.5 x 109/L
Alkaline phosphatase
120 39-117 IU/L
Alanine aminotransferase 65 5-40 IU/L
Aspartate aminotransferase 89 12-40 IU/L
Lipase 770 0-160 IU/L
Which of the following is the most appropriate next step?
A. Normal saline
B. Order urgent ercp
C. Broad spectrum antibiotic
D. CT scan of the abdomen with contrast

Answer: A
A 34-year-old man diagnosed of ileocolonic crohn's disease on infliximab and azathioprine came to the clinic complaining of perianal pain and discharge with low-grade fever for 3 weeks durations. He has a 3-4 non-bloody bowel motions /day. He has a perianal induration and purulent discharge on pressure.
Blood pressure 110/70 mmHg
Heart rate 76 /min
Temperature 37.9 °C
Which of the following is the appropriate next step in the patient management?
A. Order MRI pelvis
B. Broad spectrum antibiotics
C. Swab from the perianal discharge
D. Increase infliximab dose and reduce the frequency

Answer: A
A 23-year-old woman complaining of 4-months history of abdominal discomfort with alternating bowel habit. Recently she gave a history of bloating and passing mucus with stool and noticed increase in bowel motions 3-4/day. The abdominal pain was relieved by defecation with no nocturnal symptoms. She has no weight loss or fever. She is worried about serous diagnosis as her uncle was diagnosed as colon cancer at the age of 60. Clinically is unremarkable (see lab results).
Test Result Normal Values
RBC 3.7 4.7-6.1 x 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 11 130-170 g/L. (Male)
120-160 g/L. (Female)
Platelets count 332 150-400 x 109/L. WBC 5 4.5-10.5 x 109/L
ESR 10 2-10 mm/h (Male)
3-15 mm/h (Female)
Alanine aminotransferase 34 5-40 IU/L
Aspartate aminotransferase 23 12-40 IU/L

Amylase 122 24-151 IU/L
Which of the following is the most likely diagnosis?
A. Gastroenteritis
B. Ulcerative colitis
C. Colo-rectal cancer
D. Irritable bowel disease

Answer: D
A 62-year-old woman with long standing history of pruritis and fatigue presented to the clinic with a 2-months history of jaundice and abdominal swelling. On examination, scratch marks seen. She was jaundiced with clubbing of her fingers. Abdominal examination revealed splenomegaly with ascites (see lab results and report).
Test Result Normal Values
RBC 4 4.7-6.1 x 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 11 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 102 150-400 x 109/L
WBC 3 4.5-10.5 x 109/L
INR 1.5 0.8-1.2
Alkaline phosphatase 170 39-117 IU/L
Alanine aminotransferase 78 5-40 IU/L
Aspartate aminotransferase 112 12-40 IU/L
US Abdomen: Showed coarse liver with 3 cm focal lesion at rt lobe. Which of the following is the most appropriate next step?
A. Alpha- fetoprotein
B. Biopsy for the lesion
C. Doppler ultrasound of liver
D. Triphasic CT scan of the abdomen

Answer: D
A 34-year-old woman presented with 2-months history of jaundice and upper abdominal discomfort. She is known case of SLE on hydroxycholorquine. She also gave history of pruritis and fatigue. Clinical examination revealed obesity and hepatomegaly (see lab results and reports),
Test Result Normal Values
Direct bilirubin 32 1.5-6.5 mol/L
Total bilirubin 43 3.5-16.5 mol/L
Aspartate aminotransferase 654 12-40 IU/L Alanine aminotransferase 734 5-40 IU/L Alkaline phosphatase 129 39-117 IU/L Gamma glutamyltransferase 60 6 to 37 IU/L
Random Glucose 7 3.9-5.5 mmol/L
Globulin 43 21-36 g/L
US Abdomen: Showed enlarged liver with bright coarse echogenicity. Viral Hepatitis: Negative.
ANA: 1:640
Liver Biopsy: Inter phase hepatitis with plasma cells.
Which of the following is the most appropriate next step in management?
A. Prednisolone
B. Methotrexate
C. Vitamin e with weight loss
D. Ursodeoxycholic acid

Answer: A
A 33-year-old man with 2-years history of heartburn and food regurgitation, he was taking 40 mg of pantoprazole daily with good response. He stopped it 4 months ago and started to have recurrence

of his symptoms and becoming worse. No dysphagia or weight loss or nocturnal symptoms. Clinically was normal. All routine laboratory workup was normal.
Which of the following is the best next step in his management?
A. PPI high dose (80 mg)
B. Ranitidine twice a day
C. Resume PPI at the same dose
D. PPI high dose and ranitidine

Answer: C
A 40-year-old patient presented with chronic constipation. No alarm symptoms. No other medical problems and not on any medication. Examination was normal including rectal examination. All lab workup, thyroid function, calcium, and fasting glucose were normal. Stool for occult blood was negative. Which of the following is the most appropriate treatment?
A. Senna
B. Lactulose
C. Magnesium phosphate
D. Bulking agent (psyllium)

Answer: D
A 34-year-old man patient a known case of ulcerative colitis on azathioprine 100mg/day and tapering dose of steroid presented to the clinic with 6-7 bloody bowel motions, lower abdomen pain and low grade fever. On abdominal examination, there is mild tenderness at left lower abdomen. Stool culture was negative. Stool for colostrum difficile toxins was positive (see lab results).
Blood pressure 110/70 mmH, Heart rate 100 /min, Temperature 38.4 °C, Oxygen saturation 95 % Test Result Normal Values
RBC 5 4.7-6.1 x 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 11 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 456 150-400 x 109/L
WBC 16 4.5-10.5 x 109/L
ESR 67 2-10 mm/h (Male)
3-15 mm/h (Female)
Albumin 29 34-56 g/L
Which of the following is the most appropriate treatment?
A. Oral tinidazole
B. IV cefuroxime
C. Oral vancomycin
D. IV metronidazole

Answer: C
A 34-year-old worn with progressive dysphagia diagnosed as a case of achalasia. She has no medical problem and not on any medications. She is clinically unremarkable (see lab results). Test Result Normal Values
RBC 4 4.7-6.1 x 1012/L (Male)
4.2-5.4 x 1012/L. (Female)
Hb 10 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 455 150-400 x 109/L WBC 9 4.5-10.5 x 109/L
Urea 5 2.75-7.4 mmol/L
Creatinine 98 44-115 mol/L
Which of the following is the best treatment option?
A. Fundoplication

B. Pneumatic dilatation
C. Oral isosorbide dinitrate
D. Botulinum toxin injection

Answer: B

Cardiology

A 20-year-old woman comes to the Medical Outpatient Department with a large swelling in the front of her neck that she had had for the last 3 years. She also complains of excessive sweating, occasional palpitations, and for the last month, the inability to stay in the kitchen for a long period. She has lost weight despite having a good appetite and intake. The examination of the swelling in her neck showed a soft, nodular mass measuring 8 x 15 cm with an audible bruit. Regional lymph nodes are not palpable (see lab results).

Blood pressure 130/85 mmHg
Heart rate 110 /min
Respiratory rate 20 /min
Temperature 37.0° C

Test Result Normal Values
Thyroxine (T4 free) 20.2 8.5 - 15.2 pmol/L
Thyroid-Stimulating Hormone 0.1 0.4 - 5.0 U/L
Hb 140 120-160 g/L

Which of the following is the most appropriate management for palpitations?
A. Ipodate
B. Propranolol
C. Methimazole
D. Propylthiouracil

Answer: B

A 52-year-old man was diagnosed in the Emergency Department with an acute inferior wall myocardial infarction. An intravenous morphine bolus and IV nitroglycerine infusion were administered for severe ongoing chest discomfort. 20 minutes later, and while preparing the patient for a primary percutaneous coronary intervention, his condition became worse. His JVP at this time was elevated and his lungs were clear.

Initial Emergency Department Record:
Blood pressure 120/70 mmHg
Heart rate 90 /min (regular)
Respiratory rate 22 /min
Oxygen saturation 96% (room air)

30 Minutes Later:
Blood pressure 75/40 mmHg
Heart rate 105 /min (regular)

Which of the following is the most likely cause of the hypotension?
A. Cardiac tamponade
B. Complete heart block
C. Papillary muscle rupture
D. Right ventricular infarction

Answer: D

An 81-year-old diabetic presents to the clinic with a recent onset of redness of the skin on the left lower extremity. The redness worsens when the leg is in a dependent position. On examination, pedal pulses in the left leg are diminished. The leg is cold, and there is no tenderness on palpation.

Which of the following is the most likely diagnosis?

A. Cellulitis
B. Necrotizing fasciitis
C. Arterial insufficiency
D. Superficial thrombophlebitis

Answer: C

A 55-year-old patient presents to the clinic with leg pain brought on by walking and relieved by rest. The patient reports a history of many years of smoking. Physical examination confirms loss of hair over the lower shins as well as dystrophic nails.
An ankle-brachial index 0.80

Which of the following is the most likely diagnosis?
A. Lumbar disc prolapse
B. Deep vein thrombosis
C. Chronic limb ischemia
D. Chronic venous insufficiency

Answer: C

A 15-year-old boy presents to the clinic for an annual physical assessment. The patient has normal growth and development and is able to do usual activities. His elder brother died due to a sudden cardiac death while walking to work. On physical examination, there is no cardiac murmur. Pulses are equal in all extremities.

Which condition must be excluded before any sport is allowed?
A. Bicuspid aortic valve
B. Patent ductus arteriosus
C. Ventricular septal defect
D. Hypertrophic cardiomyopathy

Answer: D

A 55-year-old diabetic and hypertensive patient presents with a history of bilateral intermittent leg pain brought on by walking for 300 meters and relieved by rest.

What is the most likely diagnosis?
A. Varicose vein
B. Diabetic foot infection
C. Peripheral neuropathy
D. Peripheral arterial disease

Answer: D

A 61-year-old woman has had type 2 diabetes for 12 years. She takes insulin and does smoke. She was recently started on enalapril 10 mg daily for hypertension but could not tolerate it because of a cough (see lab results).

Blood pressure 160/95 mmHg
Heart rate 84 /min
Test Result Normal Value
Creatinine 142 44-115 umol/L

24 Urine Analysis:
Test Result Normal Values
Creatinine clearance 66 100-140 mL/min

24-hour urine protein 1500 0-150 mg/24 hr

Which of the following is the most appropriate therapy?
A. Beta blocker
B. Thiazide diuretic
C. Calcium channel blocker
D. Angiotensin receptor blocker

Answer: D

A 70-year-old patient has a sudden onset of pain in the left lower limb. The pain is severe and associated with numbness. The patient had an acute myocardial infarction 2 weeks before and was discharged 24 hours before this presentation. The left leg is cold and pale, the right leg is normal.
Which of the following is the most likely diagnosis?
A. Acute arterial thrombosis
B. Acute arterial embolus
C. Deep vein thrombosis
D. Dissecting aneurysm

Answer: B

A patient with atrial fibrillation is on warfarin 5 mg daily, the latest INR is 7.0 on routine evaluation. There is no evidence of abnormal bleeding.
Which of the following is the most appropriate action?
A. Increase warfarin to 7.5 mg daily
B. Decrease warfarin to 2.5 mg daily
C. Continue with the same dose of warfarin
D. Hold warfarin and repeat INR next day

Answer: D

A 73-year-old man comes to the Primary Health Care Clinic with severe shortness of breath at night and swelling of the ankles. He has reduced his activity over the past year because of chest pain when he exerts himself. Physical examination confirms crackles over both lungs, enlargement of the liver, and pitting edema of the ankles. ECG, cardiac enzymes and blood electrolytes are normal.
Blood pressure 140/90 mmHg
Heart rate 95 /min
Respiratory rate 24 /min
Temperature 36.6° C

Which of the following is the most appropriate to be included in immediate management?
A. Digoxin
B. Atenolol
C. Furosemide
D. Hydrochlorothiazide

Answer: C

A 70-year-old man, with no past medical history, presents to the Emergency Department 4 hours after the onset of a severe crushing substernal chest pain radiating to the neck and left arm (see report).

Electrocardiography:
Significant ST-segment elevation in leads I, AVL, V5, and V6. Where is the most likely site of cardiac infarction?

A. Anterior
B. Inferior
C. Lateral
D. Posterior

Answer: C

A 60-year-old patient presents to the hospital, complaining of a sudden onset of chest pain radiating to the back. The blood pressure is low in the left arm, and the left femoral pulse is not palpable (see reports).

Chest X-ray:
A left pleural effusion.

ECG:
Left ventricular hypertrophy.

Which of the following is most likely to confirm the diagnosis?
A. Echocardiography
B. Cardiac isoenzymes
C. Aortic CT angiography
D. Complete blood count

Answer: C

A 46-year-old man recovering from an acute myocardial infarction 10 days ago. He develops a sharp pain behind the sternum, which is made worse by coughing. The patient gets relief when sitting forward. On examination, a scratchy superficial sound is heard at the left sternal edge, and heart sounds are muffled. There is ascites and a 2 cm raised JVP is confirmed. Which of the following ECG changes would most likely be found?
A. Atrial fibrillation
B. Sinus tachycardia
C. Diffuse ST elevation
D. Prolonged PR interval

Answer: C

A 60-year-old man is admitted to the Coronary Care Unit with an acute myocardial infarction. His hemodynamic parameters 2 hours later are:

Blood pressure 80/50 mmHg
Heart rate 40 /min
Respiratory rate 18 /min
Temperature 36.6° C
Oxygen saturation 95% on room air

Which of the following would be the appropriate management?
A. IV 0.6 mg atropine sulfate
B. Normal saline infusion
C. IV isoproterenol
D. IV dobutamine

Answer: A

A 65-year-old patient presents with 2 hours of chest pain that feels like indigestion. In the past medical history, hypertension has been diagnosed, but no evidence of myocardial

infarction or angina. This Emergency Department does not have facilities for a percutaneous transluminal angioplasty (see report).

Blood pressure 170/94 mmHg
Heart rate 104 /min
Respiratory rate 18 /min
Temperature 36.6° C
Oxygen saturation 95% on room air

ECG:
ST segment elevation in the anterolateral leads.

Which of the following is the most appropriate initial intervention?
A. ASA, streptokinase, nitroglycerine, beta-blockers
B. ASA, streptokinase, heparin, beta-blockers
C. Nitroglycerin, ASA, heparin, beta-blockers
D. ASA, nitroglycerin, beta-blockers

Answer: A

A 60-year-old man presents to the Emergency Department with heavy retrosternal chest discomfort. The discomfort started 3 days earlier as intermittent brief episodes which became more frequent and prolonged over the last 24 hours. In the Emergency Department, he experienced a similar type of discomfort (5/10 on the pain scale of 1-10) which responded well to sublingual nitroglycerin. Serial cardiac enzymes are normal (see report).

ECG:
An initial ECG showed ST segment depression of 0.2 mV in the antero-lateral chest leads, which later normalized completely.

Which of the following is the most likely diagnosis?
A. Unstable angina
B. Acute pericarditis
C. Myocardial infarction
D. Acute aortic dissection

Answer: A

A 65-year-old woman is brought to the Emergency Department of a tertiary care hospital with rapid-onset shortness of breath and left shoulder and arm ache (see image).
Blood pressure 100/50 mmHg
Heart rate 90 /min
Respiratory rate 25 /min
Temperature 36.9 °C

Which of the following is the most appropriate treatment after the initial Emergency Department management?
A. Low molecular weight Heparin
B. Reperfusion with streptokinase
C. Thrombolytic therapy with t-PA
D. Primary coronary artery stenting

Answer: D

A 75-year-old man who has been followed up routinely for hypertension and type 2 diabetes mellitus was found to have an irregular pulse, which was not found 6 months earlier. In addition to hypertension and diabetes, his past history confirms a previous transient

ischemic attack. He is currently asymptomatic and is compliant with his medications (see report).

ECG:
Atrial fibrillation at a rate of 70 beats per minute.

Which of the following is the most appropriate management plan?
A. Add digoxin to his current treatment regimen
B. Pharmacological cardioversion to sinus rhythm
C. Start anticoagulation therapy to prevent stroke
D. No further step is needed as he is asymptomatic

Answer: C

A 29-year-old woman presents to the Emergency Department complaining of being unable to sleep flat for the last 2 nights due to difficulty in breathing. She has just delivered 1 week earlier. Physical examination confirms a laterally displaced apex beat and the presence of a third heart sound. The jugular veins were distended, and bilateral lung crackles could be easily heard.

Which of the following is the most likely diagnosis?
A. Peripartum cardiomyopathy
B. Pulmonary embolism
C. Myocardial infarction
D. Pneumonia

Answer: A

A 55-year-old man had a myocardial infarction 6 days before. He suddenly develops dyspnea, cough, and frothy sputum. For the first time, a harsh systolic murmur is heard over the precordium.

Which of the following is the most likely cause of this development?
A. Ruptured aortic cusp
B. Pulmonary embolism
C. Tricuspid regurgitation
D. Ruptured papillary muscle

Answer: D

A 40-year-old man presents to the Emergency Department after fainting while exercising. He reports past episodes of chest pain, which have occurred both at rest and with exercise. On examination of the chest, there is a IV systolic ejection murmur heard at the left sternal border, which does not radiate. The intensity of the murmur increases when the patient bears down or stands up. The lung fields are clear (see report).
Blood pressure 125/75 mm Hg Heart rate 110 /min Respiratory rate 18 /min Temperature 36.6° C

ECG:
Non-specific ST and T wave changes with left atrial enlargement.

Which of the following is the most likely diagnosis?
A. Aortic stenosis
B. Pulmonic stenosis
C. Restrictive cardiomyopathy
D. Hypertrophic cardiomyopathy

Answer: D

Explanation : MVP & HCM are the only murmurs that increase with valsalva murmurs and decrease with the leg raise and squatting.

A 55-year-old man presents to the Emergency Department with continuous retrosternal chest pain.
An ECG was obtained (see image).

Which of the following is the most likely diagnosis?
A. Anterior wall ST-elevation myocardial infarction
B. Inferior wall ST-elevation myocardial infarction
C. Acute coronary syndrome with unstable angina
D. Acute Non-ST-elevation myocardial infarction

Answer: B

A 60-year-old woman with congestive heart failure complains of shortness of breath for 3 months. It is precipitated by daily activity at home. The shortness of breath awakened her several times at night. Her medication included fosinopril, carvedilol, and furosemide. On examination, there is bilateral basilar crepitation (see report).

Echocardiogram:
There is dilated left ventricular chamber. The left ventricular ejection fraction is 30%.

Blood pressure 125/80 mmHg
Heart rate 60 /min
Respiratory rate 16 /min
Temperature 37.1° C

Which of the following drugs is indicated?
A. Aspirin
B. Amlodipine
C. Spironolactone
D. Isosorbide dinitrate

Answer: C

Which of the following agents has been shown to reduce mortality in patients with congestive heart failure?
A. Digitalis
B. Enalapril
C. Furosemide
D. Procainamide

Answer: C

Which of the following is a recognized side effect of furosemide in congestive heart failure?
A. Hypokalaemia
B. Bronchospasm
C. Hypoglycaemia
D. Haemolytic anaemia

Answer: D

A 40-year-old woman presents to the Emergency Department with vomiting and generalized abdominal pain. She does not know her medications. She states that her vision has been slightly yellow tinged recently (see image).
Blood pressure 90/50 mmHg
Heart rate 60 /min
Respiratory rate 16 /min
Temperature 37° C

Which of the following is the most likely diagnosis?
A. Anterolateral MI
B. Digoxin toxicity
C. Hypokalaemia
D. Hypothermia

Answer: B

A 60-year-old patient with a previous ischemic stroke presents with a regular narrow complex tachycardia. The patient is hemodynamically stable. The decision is made to administer IV adenosine.
In the presence of which of the following should the dose of adenosine be reduced?
A. Theophylline
B. Dipyridamole
C. Chronic renal impairment
D. Significant mitral regurgitation

Answer: B

Which of the following drugs used in the treatment of chronic heart failure has been shown to increase survival?
A. Digoxin
B. Enalapril
C. Furosemide
D. Hydralazine

Answer: B

A 39-year-old man presents to the clinic with unexplained fever 19 days after an attack of acute rheumatic fever. Several of his teeth have been extracted prior to the hospital admission. Clinical examination confirms a systolic murmur and splenomegaly (see lab results).
Test Result Normal Values
WBC 14.0 4.5-10.5 x 109/L
Neutrophils 72 40-60%

Urinalysis:
Test Result Normal Values:
Erythrocytes 25 0-2 per HPF

Which of the following is the most likely diagnosis?
A. Floppy mitral valve
B. Bacterial endocarditis
C. Acute myocardial infarction
D. Recurrent rheumatic heart disease

Answer: B

A 62-year-old man presents with recurrent retrosternal chest pain, which he said has become much worse over the last 3 weeks. He describes the pain as lasting much longer than before and that it occurs even while he is sitting in a chair or in bed at night reading.

Which of the following is the most likely type of angina?
A. Unstable
B. Heberden
C. Exertional
D. Prinzmetal

Answer: A

A 52-year-old man comes to the clinic for a checkup. He has hypertension for 5 years and is currently being treated with hydrochlorothiazide 25 mg/day (see lab results).
Blood pressure 155/101 mmHg
Heart rate 84 /min
Test Result Normal Values
Sodium 141 134-146 mmol/L
Potassium 4.1 3.5-5.1 mmol/L
Creatinine 105 44-115 umol/L
Glucose, fasting 7.3 3.5-6.5 mmol/L

Which of the following is the most appropriate management modification?
A. Add a beta blocker
B. Add a Calcium channel blocker
C. Double the hydrochlorothiazide dose
D. Add an angiotensin converting enzyme inhibitor

Answer: D

A 30-year-old intravenous drug user presents with fever, night sweats, chest pain, and arthralgia. On examination, painless erythematous lesions noted on the palms of the hands; with round erythematous lesions and central clearing noted in the retina and splinter haemorrhage under the fingernails.
Which of the following is the most likely diagnosis?
A. Syphilis
B. HIV infection
C. Infectious hepatitis
D. Bacterial endocarditis

Answer: D

Which of the following agents has been shown to reduce mortality in patients with congestive heart failure?
A. Digitalis
B. Enalapril
C. Furosemide
D. Procainamide

Answer: C
Which of the following is a recognized side effect of furosemide in congestive heart failure?
A. Hypokalaemia
B. Bronchospasm
C. Hypoglycaemia
D. Haemolytic anaemia

Answer: D

A 70-year-old man with ischemic heart disease developed dizziness and confusion following accidental ingestion of a higher dose of atenolol.
Blood pressure 70/40 mmH
Heart rate 46 /min
In addition to holding atenolol and administration of intravenous fluid, which of the following drugs is the most appropriate?
A. Nifedipine
B. Glucagon
C. Salbutamol nebulizer
D. Aminophylline infusion

Answer: B
A 60-year-old man with diabetes, hypertension and intermittent claudication is diagnosed to have stable angina. Decision is made to manage him with intensive medical therapy. He is started on aspirin, rosuvastatin and nitroglycerine but still gets angina with moderate exercise. Blood pressure 130/85 mmHg
Heart rate 86 /min
Which of the following is the most appropriate additional drug?
A. Diltiazem
B. Nifedipine
C. Amlodipine
D. Metoprolol

Answer: A
A 20-year-old woman presents to the Emergency Department with palpitations for 2-hours (see image and lab results).
Blood pressure 135/80 mmHg
Heart rate 150 /min
Test Result Normal Values
Hb 138 130-170 g/L (Male)
120-160 g/L (Female)
Thyroid-Stimulating Hormone 3.2 0.4 - 5.0 U/mL
Which of the following is the most appropriate initial management?
A. Oral amiodarone
B. Carotid sinus massage
C. Intravenous adenosine
D. Electrical cardioversion

Answer: B
A 65-year-old man with diabetes is admitted to the Coronary Care Unit with diagnosis of acute ST elevation anterior myocardial infarction. He developed massive hematemesis following streptokinase infusion. Which of the following drugs would reverse the action of streptokinase?
A. Factor VIII concentrate
B. Aminocaproic acid
C. Protamine
D. Vitamine K

Answer: B
A 69-year-old man is admitted with acute chest pain and shortness of breath. Workup confirmed acute myocardial infarction. Upon discharge, he is advised to quit smoking. but he is unwilling to do so. What should be said to the patient during the counseling session?
A. "We feel that you are uncooperative"
B. "We will involve the family in this issue"

C. "If you do not stop smoking you will die"
D. "Continuing smoking means more complications*

Answer:D
A 50-year-old man who works as a bank employee is discharged following a 7-day admission for acute myocardial infarction. He was given 5 different drugs.
What further information regarding life style and daily activities should be given to him?
A. Advise him to change his job
B. Start driving 2 days after discharge
C. Refrain from sexual activities for 3 months
D. Perform mild aerobic exercise from next week

Answer:D
A 50-year-old man known to have advanced dilated cardiomyopathy is admitted to the Coronary Care Unit for the 5th time during the past 2 months. This admission!
Ate is in refractory heart failure despite being on high doses of inotropic agents and nasal CAP, Decision is made to consider him for heart transplantation.
After setting the scene, how would you break the news?
A. Start with a warning shot and deliver the news in small steps
B. Do not waste time, tell him the decision directly
C. Deal with his concerns by expressing sympathy
D. Discuss the decision with his son

Answer: A

A patient with infective endocarditis started on antibiotics. What is the simplest way in monitoring antibiotics response?
A. Blood culture
B. Daily full blood count
C. Serial echocardiography
D. Serum C-reactive protein concentration

Answer: D

A 65-year-old man presents with severe chest pain to Emergency Room. ECG showed st segment elevation of inferior leads. He received thrombolytic therapy (PA). 3 hours later, started to feel dizzy. ECG showed 2:1 AV block. IV atropine given with no effect.

Vital signs 3 hours later: Blood pressure 80/45 mmH, Heart rate 39 /min, Respiratory rate 18 /min, Oxygen saturation 90 %
Which of the following is the best management option?
A. IV fluid
B. Dopamine
C. Isoprenaline
D. Temporary pacemaker

Answer: D
A 65-year-old man presented to Emergency Room with severe chest pain. He was diagnosed as inferior MI and received thromolytic therapy (PA). Few hours later, he developed dizziness. ECG is done (see image). Blood pressure 80/50 mmg, Heart rate 41 /min
Which of the following is the most likely diagnosis?
A. 1st degree heart block

B. 2nd degree heart block
C. 3rd degree heart block
D. 4th degree heart block

Answer: B
A 34-year-old woman presented to the clinic with new onset SOB and fatigue. She is diagnosed as a case of mitral stenosis with pulmonary hypertension. She is planning to get pregnant and she is hiding the diagnosis from her husband. Which of the following is the most appropriate advice?
A. Close monitor and follow-up
B. She can go ahead with her plan
C. Call her husband and tell him the diagnosis
D. Counselling her about the risk and squally of the pregnancy

Answer: D
A 33-year-old man presented with shortness of breath, orthopnea and paroxysmal nocturnal dyspnea. Exanimation revealed collapsing pulse and wide pulse pressure. Early diastolic murmur over left sternal border and pistol-shot sound heard over femoral arteries.
Which of the following is the most likely diagnosis?
A. Coarctation of aorta
B. Aortic regurgitation
C. Cardiac tamponade
D. Ventricular septal defect

Answer: B
A 55-year-old patient comes to the clinic. He is hypertensive and on captopril. ECG showed tall, tented T waves. Which of the following is the best next step in management?
A. Reassurance
B. Order cardiac enzyme
C. Check potassium level
D. Admit him with urgent echo

Answer: C
A 70-year-old woman known to have chronic heart failure resented with sever shortness of breath, paroxysmal nocturnal dyspnea and orthopnea. On examination, she looks unwell, high
JVP with bilateral basal crackles at both lungs. S3 was heard at the apex.
Blood pressure 110/70 mmHg
Heart rate 100 /min
Oxygen saturation 88 %
Which of the following is the best next step in management?
A. 100% 02
B. Metalazone
C. IV furosemide
D. Order urgent echo

Answer: C
A 65-year-old woman known to have heart failure presented with sever shortness of breath, paroxysmal nocturnal dyspnea and orthopnea. On examination, she looks unwell, high JVP with bilateral basal crackles at both lungs. S3 was heard at the apex.
Blood pressure 100/70 mmHg
Heart rate 93 /min
Oxygen saturation 88 %
Which of the following is the most likely diagnosis?
A. Myocardial infarction

B. Pulmonary oedema
C. Right-sided heart failure
D. Acute mitral regurgitation

Answer: B
A 24-year-old athlete man attended routine clinic visit. He is asymptomatic and has no significant past medical history but found to have a resting pulse of 47 beat/min, normal blood pressure and cardiac examination. Normal CBC, renal function, LFT, and TSH. ECG normal only sinus bradycardia. Which of the following is the best advice?
A. Reassurance
B. Order urgent echo
C. Genetic counselling
D. Repeat ECG every 3 months

Answer: A; Athletes are known to have low resting heart rates
A 67-year-old woman diagnosed with heart failure being treated with loop diuretic and ACE inhibitor with good control of her symptoms. Which of the following medications helps in reducing her long-term mortality?
A. Digoxin
B. Nifedipine
C. Amiodarone
D. Spironolactone

Answer: D
A 67-year-old diabetic patient with atrial fibrillation is hesitant to start anti-coagulant due to its risk of inducing bleeding. which of the following factors is the most important in making the decision?
A. The patient has no risk for bleeding
B. The patient has mild risk of thrombosis
C. The bleeding risk is higher than thrombosis risk
D. His risk of thromboembolic events is higher than bleeding

Answer: D
A 60-year-old woman admitted to the hospital for cardiac catheterization but she is concerned about its complications. Which of the following is the best way to address her concerns?
A. Tell her it is very rare and no need to worry
B. Inform her that her husband will sign the consent
C. Listen to her concern and explain what are the expected complications
D. Listen to her concern and explain what are the expected complications and give alternative

Answer: D
A 57-year-old man with known history of asthma, congestive heart failure and peptic ulcer disease was admitted with asthma exacerbation and rapid atrial fibrillation. He was managed with nebulized salbutamol, IV digoxin, and his home medications was continued (ACE inhibitor, Ranitidine, Spironolactone). 24 hours later, his serum potassium is noted to be 2.8 mEq/L. Which of the following medications is the most likely to cause this abnormality?
A. Digoxin
B. Ranitidine
C. Salbutamol
D. Spironolactone

Answer: C
A 21-year-old woman with history of recurrent palpitations, light-headedness and 2 attacks of fainting. ECG showed short PR interval and Inferior leads Q waves. She was diagnosed as Wolf-

Parkinson-White (WPW) syndrome. She was given atenolol with no response. Which of the following is the best management option?
A. Digoxin
B. Amiodarone
C. Increase atenolo dose
D. Radiofrequency ablation

Answer: D

A 47-year-old man presents with intermittent chest pains for the past 2 days. He complains of worsening pain. On examination, he tachycardic and sweaty. ECG is done (see image). Blood pressure 133 /81 mmHg. What does the ECG show?
A. Posterior mvocardial infarction
B. Right bundle branch block
C. Ventricular tachycardia
D. Left bundle branch block

Answer: D
A 61-year-old man comes to the clinic with a 2-day history of dyspnoea on exertion. He has a long standing hypertension and a 30-pack-year smoking history. Physical examination is unremarkable except for rare crackles at the bases of the lungs. Which of the following tests would be most helpful for detecting left ventricular dysfunction?
A. B-type natriuretic peptide (BNP)
B. C-reactive protein (CRP)
C. Creatinine Kinase (CK)
D. Troponin T

Answer: A

A 77-year-old man with a history of atrial fibrillation is admitted to the hospital following an acute coronary intervention and treated with a percutaneous coronary intervention. His medication on admission were warfarin, paracetamol and allopurinol. Antithrombotic therapy immediately given after his event. Which of the following regimen should he be prescribed?
A. Continue warfarin with the addition of 2 antiplatelets
B. Stop warfarin, start 3 antiplatelets
C. Stop warfarin, start 2 antiplatelets
D. Continue warfarin monotherapy

Answer: A
A 59-year-old man presents 4 weeks after being discharged from hospital. He had been admitted with chest pain and thrombolysed for a myocardial infarction. 2 days later, he developed marked tongue and facial swelling. His vital signs are within the normal limits. Which of the following drugs is most likely to be responsible?
A. Atenolo
B. Ramipril
C. Aspirin
D. Atorvastatin

Answer: B
A 66-year-old man with hypertension comes to the clinic for a follow-up. He is currently taking a combination of losartan and amlodipine, which is failing to keep his blood pressure within target.

Examination is unremarkable. His blood test and renal function profile were within the normal limits. What is the most appropriate next step assuming he has no relevant contraindications?
A. Add ramipril 1.25mg od
B. Add atenolol 50mg od
C. Add doxazosin 1mg od
D. Add indapamide MR 1.5mg od

Answer: D

A 57-year-old man who is intolerant of aspirin is started on clopidogrel for the secondary prevention of ischemic heart disease. Which of the following drugs can make clopidogrel less effective?
A. Warfarin
B. Omeprazole
C. Beta blockers
D. Selective serotonin reuptake inhibitors

Answer: B
A 61-year-old man with suspected left ventricular heart failure. He had B-type natriuretic peptide (BNPy test, which came back as being slightly elevated.
Which of the following factors cause a falsely elevated BNP?
A. ACE inhibitor therapy
B. Furosemide
C. Obesity
D. COPD

Answer: D

A 23-year-old woman evaluated for dyspnoea and recurrent palpitation. She has a history of Fallot tetralogy repair at the age of 3. On cardiac examination, apex was normal with left parasternal impulse. S1 is single with grade 2/4 decrescendo diastolic murmur that increases with inspiration noted at left sternal border.
Blood pressure 110/70 mmg
Heart rate 94 /min
Oxygen saturation 92 %
Which of the following is the most likely diagnosis?
A. Mitral valve stenosis
B. Ventricular septal defect
C. Tricuspid valve regurgitation
D. Pulmonary valve regurgitation

Answer: D

A 22-year-old woman presents with fatigue, SOB, arthralgia and fever for 2 weeks. It is worsened with time specially the fever. She has no significant medical history apart from sore throat 3 weeks ago and dental extraction 2 months ago. Skin examination revealed petechia over lower limb. Cardiac examination demonstrated 3/6 holosystolic murmur that radiates to axilla and mild splenomegaly (see report).
Blood pressure 100/60 mmH
Heart rate 88 /min
Temperature 38.5 °C
Urine dipstick:

Showed mild proteinuna and microscopic hematuria
Which of the following is the most likely diagnosis?
A. Lymphoma
B. Infective endocarditis
C. Group A streptococcus infection
D. Systemic lupus erythermatosus

Answer: B

A 30-year-old woman is admitted with dyspnea, SOB, PND and orthopnea 3 days post vaginal delivery. She gave a history of lower limb edema. She has no significant medical history. On examination, she looks in distress with bilateral LL pitting edema, bilateral basal crackles on lung examination. On cardiac examination, apex was displaced down with pan-systolic murmur at the apex radiating to axilla and S3 gallop.
Blood pressure 90/60 mmH
Heart rate 110 /min
Temperature 37.6 °C
Which of the following is the most likely diagnosis?
A. Viral myocarditis
B. Puerperium sepsis
C. Infective endocarditis
D. Peri-partum cardiomyopathy

Answer: D

A 30-year-old woman is admitted with cough, dyspnea, chest pain, SOB, PND and orthopnea 3 days post vaginal delivery. She gave a history of lower limb edema. She has no significant medical history. On examination, she looks in distress with bilateral LL pitting oedema, bilateral basal crackles on lung examination. On cardiac examination, apex was displaced down with pan-systolic murmur at the apex radiating to axilla and S3 gallop
Blood pressure 90/70 mmHg
Heart rate 120 /min
Temperature 37.6 °C
Oxygen saturation 90 %
Which of the following is the most appropriate step in this patient management?
A. Echocardiography
B. Thallium stress test
C. ECG and Chest x-ray
D. Cardiac MRI and PFT

Answer: A

A 54-year-old man is evaluated prior elective hernia repair surgery. He is known case of bicuspid aortic valve and mechanical aortic valve replacement 4 years ago. He is asymptomatic. Cardiac examination showed grade 1/6 ejection systolic murmur at aortic area.
Blood pressure 110/70 mmHg
Heart rate 76 /min
Respiratory rate 18 /min
Oxygen saturation 95 %
Which of the following is the best management option for endocarditis prophylaxis?
A. Amoxicillin
B. Augmentin
C. Cefuroxime

D. No antibiotic

Answer: D
A 64-year-old man presented to Emergency Room with symptoms of dyspnoea, SOB, and palpitation. He is a known case of diabetic mellitus, hypertension and hyperlipidaemia. He is smoker with history of stroke. On examination, found to have irregular pulse. Cardiac and chest examination showed high JVP, S3 and bi-basal crackles. Atrial fibrillation on ECG.
Blood pressure 150/90 mmH
Heart rate 144 /min
Oxygen saturation 89 %
What is the CHADS2 score of this patient?
A. 2
B. 3
C. 4
D. 5

Answer: D

A 62-year-old man presents to Outpatient Clinic with recurrent episodes of steal chest tightness, which occurs mainly after heavy physical exertion. He has no chronic medical problems. Clinical examination and baseline ECG were normal. Which of the following is the most appropriate next step in diagnosis?
A. Exercise ECG
B. Echocardiography
C. Coronary arteriography
D. Myocardial perfusion scintigraphy

Answer:A
A 28-year-old man presents with recent onset of chest pain. The pain is retrosternal, localized, sharp and constant in intensity. He noticed increase in seventy with movement. He started active exercise program a week prior to the onset of the symptom.
Blood pressure 110/70 mmH
Heart rate 76 /min
Oxygen saturation 95 %
Which of the following is the most appropriate next step in management?
A. Atenolo
B. Ibuprofen
C. Nitroglycerine
D. Reassurance and assess after 1 week

Answer: B
A 65-year-old woman came for routine annual check-up found to have with harsh ejection systolic murmur, which propagated to the neck. Which of the following factors is the most important in deciding the time of the surgery?
A. Low pulse pressure
B. Intensity of the murmur
C. The patients symptoms
D. Left ventricular hypertrophy

Answer:C
A 65-year-old man found to have an ejection systolic murmur on clinical examination. There is no history of chest pain, shortness of breath nor syncope. ECHO confirmed aortic stenosis with good left

ventricular systolic function. Aortic valve gradient of 40 mmH.
Blood pressure 100/65 mmHg
Heart rate 67 /min
Which of the following is the most appropriate management option?
A. Follow-up
B. Anti-coagulation
C. Aortic valvoplasty
D. Surgical valve replacement

Answer:A
A 67-year-old man known to have hypertension came to Emergency Room with acute onset of palpitation. His pulse on the monitor showing pulse rate of 170/min. He have irregular pulse with normal cardiac and chest examinations.
Blood pressure 120/70 mmHg
Respiratory rate 18 /min
Oxygen saturation 91 %
Which of the following is the best treatment option?
A. Cardioversion
B. Adenosine
C. Amidorone
D. observation

Answer: C
A 38-year-old Indian man presents with progressive dyspnea for 3 months. He has past history of old tuberculosis. He has no other medical problems and not on any medications. His JVP was elevated and rises further on inspiration. Cardiac examination was normal with no murmurs. Blood pressure 105/60 mmH
Heart rate 120 /min
Oxygen saturation 93 %
Which of the following is the most likely diagnosis?
A. Myocarditis
B. Corpulmonale
C. Cardiomyopathy
D. Constrictive Pericarditis

Answer: D
A 28-year-old woman who is a known case of mitral stenosis came to the clinic asking about how critical is her condition. Echo was done and mitral valve orifice was recorded.
Which of the following mitral valve orifice measurement is consider critical?
A. <1 cm
B. <2 cm
C. <3 cm
D. <4 cm

Answer: A
A 72-year-old man presented with acute sever chest pain with an ECG revealing St segment elevation I1,I,AVF, leads was treated with thrombolysis but 2 days later acutely unwell with sever dyspnea. On examination, he has loud systolic murmur at the apex, which radiates to the axilla and bilateral crackles at the base of the lungs.
Blood pressure 90/60 mmHg
Heart rate 100 /min
Respiratory rate 25 /min

Oxygen saturation 88 %
Which of the following is the most likely diagnosis?
A. Myocarditis
B. Cardiogenic shock
C. Rupture papillary muscle
D. Acute Rt sided heart failure

Answer: C
A 65-year-old man with heart failure requires rate control to treat coexisting atrial fibrillation. Which of the following is the best treatment option?
A. Digoxin
B. Adenosine
C. Lidocaine
D. Nitroglycerine

Answer: A
Asymptomatic healthy 24-year-old woman came for a check-up visit and found to have grade II mid-diastolic murmur over the apex. Which of the following is the best next step in diagnosis?
A. Cardiac MRI
B. Echocardiography
C. Cardiac catheterization
D. Antistreptolysin O titre

Answer: B
A 70-year-old man presented with progressive dyspnea, palpitation and fatigue for 3 months. He is known to have hypertension on captopril. He underwent echo which showed poor left ventricular function so furosemide was added with good control of his symptoms. His vital signs upon the last clinic visit shown below.
Blood pressure 150/90 mmHg
Heart rate 90 /min
Respiratory rate 18 /min
Oxygen saturation 95 %
Which of the following medication should be added at this stage?
A. Digoxin
B. Carvedilol
C. Nifedipine
D. Hydralazine

Answer: B

A 60-year-old man is evaluated in the clinic for cardiac risk assessment. He is asymptomatic and physically active. He has never smoked. He has no chronic health issues and takes no medications. His physical examinations are normal. Cardiovascular risk calculation using the Pooled Cohort Equations predicts a 6.5% risk of a myocardial infarction in the next 10 years that indicate an intermediate risk of myocardial infarction. Which of the following investigations should be done next?
A. Cardiac CT angiography
B. Stress echocardiography
C. High-sensitivity C-reactive protein
D. Cardiac magnetic resonance (CMR) imaging

Answer: C

A 58-year-old woman with hypertension and type 2 diabetes mellitus presents to the clinic for routine follow-up. She has no symptoms. Medications are hydrochlorothiazide, losartan, atorvastatin, and metformin. Cardiac examination shows a grade 2/6 diastolic murmur heard best over the apex. Peripheral pulses are normal. Electrocardiogram shows sinus rhythm with nonspecific ST changes. Blood pressure 160/80 mmH
Heart rate 78 /min
Respiratory rate 18 /min
Temperature 36.6 °C
Oxygen saturation 96 %
Which of the following is the most appropriate test to perform next?
A. Transthoracic echocardiography
B. Cardiac CT angiography
C. Transesophageal echocardiography
D. No further testing

Answer: A
A 55-year-old woman presents to the clinic for her regular follow-up. 6 months ago, she presented to Emergency Department with palpitations and dyspnea and diagnosed to have atrial fibrillation. She is currently asymptomatic with no palpitations. Her medical history is also significant for a transient ischemic attack and hypertension. Her medications are warfarin, metoprolol and candesartan, Her physical examination is unremarkable. Her most recent electrocardiogram shows normal sinus rhythm. which of the following is the most appropriate management?
A. Continue warfarin
B. Discontinue warfarin
C. Continue warfarin and start aspirin
D. Discontinue warfarin and start clopidogrel

Answer: A
A 68-year-old man presented to the Emergency Department with progressive exertional dyspnoea, orthopnea and dry cough for 7 days. His medical history is significant for hypertension. Physical examination reveals raised central venous pressure, fine crackles at lung bases and hepatomegaly. His extremity examination reveals bilateral pitting edema. Echocardiogram shows a left ventricular ejection fraction of 60% and normal valves. Chest radiograph shows evidence of pulmonary edema (see lab results).
Blood pressure 160/85 mmHg
Heart rate 86 /min
Respiratory rate 18 /min
Test Result Normal Values
Sodium 140 134-146 mmol/L
Potassium 4 3.5-5.1 mmol/L
Urea 6 2.75-7.4 mmol/L
Creatinine 88 44-115 mol/L
Brain natriuretic peptide 900 0-300 ng/L
Which of the following is the most appropriate next step in the management?
A. Start calcium channel blocker agent
B. Start B-Blocker agent
C. Start spironolactone
D. Start furosemide

Answer: D
A 26-year-old pregnant woman visits the clinic for regular follow-up. She is asymptomatic. She is in her 30th week of pregnancy. Her physical examination reveals a mid-systolic ejection murmur at

the right upper sternal border. The murmur radiates to carotids. No diastolic murmur is heard. Which of the following is the most likely diagnosis?
A. Aortic stenosis
B. Mitral stenosis
C. Tricuspid stenosis
D. Physiologic murmur of pregnancy

Answer:A
A 28-year-old man is evaluated in the Emergency Department for severe persistent chest pain for 1-day. The pain is constant but exacerbated when leaning forward and not associated with other symptoms. Physical examination is unremarkable apart of a friction rub at the left lower sternal border. Electrocardiogram shows diffuse, concave upward ST-segment elevations and PR-segment depression most prominent in leads V1 through V6.
Which of the following is the most appropriate treatment?
A. Prednisone
B. Nitroglycerin
C. Warfarin
D. Ibuprofen

Answer:D
A 58-year-old woman was admitted with a diagnosis of acute myocardial infarction received appropriate treatments and now she is ready for discharge. She is so concern about the secondary prevention of infarction. Which of the following statements is the most appropriate?
A. Anti-platelet agents should be used for short period to reduce the risk of recurrent infarction
B. Calcium channel blockers should be used indefinitely to reduce the risk of recurrent infarction
C. Hormonal replacement therapy should be given to postmenopausal women to prevent coronary events
D. Angiotensin-converting enzyme (ACE) Inhibitors should be used indefinitely by patient with clinically evident heart failure

Answer:D
A 55-year-old woman with non-decompensated systolic heart failure and ejection fraction of 25% presents to the clinic for routine follow-up. Which of the following medications is the most appropriate management?
A. Lisinopril
B. Amlodipine
C. Furosemide
D. Spironolactone

Answer: A
A 45-year-old woman is evaluated for a 3-week history of progressive shortness of breath that limit her activities. Medical history is significant for asthma and moderate mitral regurgitation. Her only medication is an albuterol inhaler as needed. On physical examination, her vital signs are normal. Cardiac examination reveals a grade 3/6 holosystolic murmur radiating to the axilla Lungs are clear to auscultation. Which of the following is the most appropriate diagnostic test?
A. Spirometry
B. Transthoracic echocardiogram
C. Transesophageal echocardiogram
D. Helical (or spiral) computed tomography (CT) of the chest

Answer:C

A 65-year-old man presents to the Emergency Department with exertional syncope and progressive shortness of breath. Clinical examination reveals fine crackles at both lung bases and mid-systolic murmur at right upper sternal boarder that radiated to carotids. Chest radiography demonstrated clear evidence of pulmonary edema.
Which of the following statements about the assessment and diagnosis is the most accurate?
A. The murmur will be augmented by standing position
B. The main treatment for symptomatic patient is diuretics
C. Electrocardiogram (ECG) will show right bundle branch block (RBBB)
D. Presence of symptoms of congestive heart failure are poor prognosis

Answer: D
A 48-year-old woman is seen in the clinic because of recurrent severe nocturnal chest pain for several months that rapidly resolved with sublingual nitrate. She had a normal cardiac catheterization and positive ergonovine echocardiography testing. She does not want to use multiple medications and she accept to use only one drug as a long-term maintenance therapy. Which of the following drugs is the best strategy for long-term management of this patient?
A. Lisinopril
B. Carvedilol
C. Nifedipine
D. Aspirin

Answer: C; Vasospastic Angina

A 65-year-old man presents to the clinic with progressive exertional shortness of breath that associated with syncope. His examinations show a mid-systolic murmur at right upper sternal boarder that radiates to carotids. Chest x-ray shows cardiomegaly. ECG reveals left ventricular hypertrophy.
Which of the following is the most appropriate management?
A. Start antihypertensive with tight control of his blood pressure
B. Surgical valvular replacement
C. Start vasodilators
D. Start diuretics

Answer: B
A 28-year-old man presents to the Emergency Department with sudden onset of chest pain that reduced with sitting forward associated with shortness of breath class II. No history of fever. Examinations reveal a normal heart sound with friction rub. His cardiac enzymes are within normal range. ECG shows diffuse ST segment elevation.
Blood pressure 130/70 mmH Heart rate 90 /min Respiratory rate 24 /min Temperature 37.9 °C Oxygen saturation 95 %
What is the most appropriate treatment for this patient?
A. Start antibiotics
B. Start corticosteroids
C. Start anticoagulation
D. Start non steroid anti-inflammatory drug

Answer: D
A 55-year-old man with congestive heart failure on multiple treatments came to the clinic for follow up. The physician discussed with him the importance of compliance and the effect of the treatments on his life. Which of the following medications is associated with reduced mortality rate in congestive heart failure?
A. Digoxin therapy
B. Diuretic therapies

C. anticoagulation therapies
D. Angiotensin-converting enzyme (ACE) inhibitors

Answer: D

A 29-year-old man is seen in the pre-employment clinic for routine exam, He is asymptomatic. Physical examination revealed a systolic murmur that heard at the apex. The murmur is augmented with sustained handgrip and reduced with Valsalva manoeuvre. Which of the following is the most likely cause for this murmur?
A. Mitral regurgitation
B. Aortic regurgitation
C. Mitral stenosis
D. Aortic stenosis

Answer: A

A 52-year-old man develops sudden severe retrosternal chest pain that radiates to his back. Within minutes, he became unconscious. He is a heavy smoker and has history of hypertension. Blood pressure 85/56 mmHg Heart rate 92 /min. Which of the following is the most likely diagnosis?
A. Vasovagal attacks
B. Pulmonary embolism
C. Tear in the aortic intima
D. Acute myocardial infarction

Answer: C
A 23-year-old man is brought to the hospital after an episode of collapse. He states that on a few occasions while playing football he has had some chest pain, breathlessness and felt as he was going to faint. Last night during training, he collapsed and blacked out for a few seconds. Clinical examination revealed a harsh ejection systolic murmur with a palpable systolic thrill at the left steal edge; and a prominent jerky carotid pulse. Which of the following is the most likely diagnosis?
A. Hypertrophic obstructive cardiomyopathy
B. Wolff-parkinson-white syndrome
C. Brugada syndrome
D. Long QT syndrome

Answer:A
A 68-year-old man is seen for a follow-up. He has longstanding breathlessness and symptoms of cardiac failure. The ECG confirmed a diagnosis of heart failure due to left ventricular dysfunction. His currently on enalopril 10 mg OD, simvastatin 40 mg OD, furosemide 40 mg OD and omeprazole 20 mg OD. Examination revealed that the chest is clear and heart sounds are normal. There is no peripheral edema. Blood pressure 125/83 mmHg, Heart rate 76 /min. Which of the following is the most appropriate next step in the management at this stage?
A. Add in digoxin
B. Add in losartan
C. Add in bisoprolol
D. No additional medication indicated

Answer: C
A 78-year-old man diagnosed with heart failure due to left ventricular dysfunction. His currently on aspirin 75 mg OD, simvastatin 40 mg OD and furosemide 40 mg OD. On examination, heart sounds are normal, his chest is clear and there is no peripheral edema. Which of the following is the most appropriate drug therapy to initiate at this stage?
A. Digoxin

B. Enalopril
C. Hydralazine
D. Spironolactone

Answer: B
A 55-year-old man presents with a 3-month history of exertional chest pain. He has previous history of myocardial infarction. Which of the following is modifiable risk factors in the treatment of ischemic heart disease?
A. Family history
B. Smoking
C. Gender
D. BMI

Answer: B
A 78-year-old woman presents with exertional syncope. On clinical examination, she has a slow rising carotid pulse and a loud ejection systolic murmur at the upper right sternal edge, radiating to the carotids. Which of the following is the most likely cause of these findings?
A. Mitral stenosis
B. Aortic stenosis
C. Mitral valve prolapse
D. Tricuspid regurgitation

Answer:B
A 43-year-old healthy man concerned about heart disease. Which of the following is the most important risk factor of ischemic heart disease?
A. BMI of 31.2 kg/m2
B. Waist circumference of 103 cm
C. Blood pressure of 132/82 mmHg
D. 2 separate fasting plasma glucose of 8.2 and 8.3 mmol/L.

Answer: D
A 72-year-old man presented with an episode of right-sided weakness that lasted 10 minutes and fully resolved and he is clinical stable. He has no other medical illness. On clinical examination, he is in atrial fibrillation. Blood pressure 110/70 mmHg, Heart rate 95 /min, Temperature 36.6 °C. Which of the following is the most appropriate next step in the management?
A. Aspirin
B. Warfarin, INR range 3-4
C. Warfarin, INR range 2-3
D. No additional drug treatment

Answer: C
A 58-year-old man comes to the clinic for a follow-up. He has been discharged after an acute inferior myocardial infarction. He is diabetic type 2, on metformin 1g BD, aspirin 75 mg. atorvastatin 10 mg OD and lisinopril 10 mg OD. Examination revealed bilateral basal crackles on auscultation of the chest. Blood pressure 148/88 mmH, Heart rate 76 /min, BMI 24 kg/m2
Which of the following is the most appropriate medication to add to his regime?
A. Bisoprolol
B. Amlodipine
C. Losartan
D. Warfarin

Answer: A

A 61-year-old woman comes to the clinic complaining of palpitations. She has also noticed some weight loss over the last 2 months. On clinical examination, she has an irregularly irregular pulse and a fine tremor. Her ECG shows atrial fibrillation. Which of the following is the next most appropriate investigation to do?
A. Carotid Ultrasound
B. Thyroid function tests
C. Exercise tolerance test
D. 24 hours ECG monitoring

Answer: B
A 66-year-old man comes for follow-up. He has shortness of breath on exercise, which have increased over the past few months. He became unable to walk more than 200-300 meters on the corridor and cannot climb the stairs easily. He has history of hypertension and currently on amlodipine 5 mg OD and indapamide 2.5 mg OD. On examination, he has bilateral basal crackles and mild pitting ankle edema (see lab results).
Blood pressure 188/96 mmH
Heart rate 79 /min
Test Result Normal Values
Hb 133 130-170 g/L (Male)
120-160 g/L (Female)
Sodium 138 134-146 mmol/L
Potassium 4.2 3.5-5.1 mmol/L
Creatinine 133 44-115 mol/L
Which of the following is the most appropriate next therapy?
A. Spironolactone
B. Furosemide
C. Bisoprolol
D. Lisinopril

Answer:D
A 66-year-old man presents with a diastolic murmur best heard at the left steal edge. On examination, the apex beat is displaced outwards. Which condition typically presents with these findings?
A. Coarctation of the aorta
B. Aortic regurgitation
C. Mitral regurgitation
D. Aortic stenosis

Answer: B
A 64-year-old man is diagnosed with persistent atrial fibrillation. His past medical history includes hypertension and type 2 diabetes. He has no contraindications to any antithrombotic treatments. What antithrombotic treatment should be considered first line for this patient?
A. Aspirin and Clopidogrel
B. Clopidogrel
C. Warfarin
D. Aspirin

Answer: C
A 77-year-old woman comes to the clinic for a check-up. She has an irregular pulse when the nurse checked her vitals. An ECG has confirmed persistent atrial fibrillation .She is completely asymptomatic. She has a past medical history of migraine and asthma and on salbutamol PRN for asthma and PRN ibuprofen for her migraines. Other clinical examination is normal. Blood pressure

138/85 mmg, Heart rate 88 /min. What antithrombotic treatment should be considered first line in this patient?
A. Aspirin
B. Clopidogrel
C. Aspirin and dipyridamole
D. Oral anticoagulation

Answer: D
A 63-year-old smoker man presents with pain in his left leg, which is exacerbated by exercise and relieved by rest. Examination revealed weak pulses in the left leg compared to the right. The ankle brachial pressure Index is 0.84. Which of the following is the most likely diagnosis?
A. Peripheral arterial disease with critical ischaemia
B. Normal peripheral arterial disease
C. Peripheral arterial disease
D. Incompressible vessels

Answer: C

A 65-year-old man with diabetes and hypertension is admitted with extensive anterior ST elevation myocardial infarction (STEMI). had successful angioplasty and stenting to left anterior descending artery. Aspirin, clopidogrel, bisoprolol and enalapril were started (see lab results).
Test Result Normal Values
Cholesterol (HDL) 1.0 > 1.03 mmol/L
Cholesterol (LDL) 3.36 <3.36 mmol/L Total Cholesterol 5.0 <5.1 mmol/L
Triglycerides 2.2 <2.16 mmol/L
Which of the following is recommended?
A. No additional medication is indicated
B. Atorvastatin
C. Fenofibrate
D. Niacin

Answer: B
A 55-year-old man with diabetes presented to the Cardiology Clinic with vague chest pain with exertion for 2 months (see report).
Electrocardiogram: Normal.
Which of the following is the most appropriate next step in management?
A. Assurance
B. Exercise ECG testing
C. Coronary angiogram
D. Adenosine myocardial perfusion imaging

Answer: B
A 60-year-old man with poorly controlled hypertension presents with exertional dyspnea and occasional paroxysmal nocturnal dyspnea. No chest pain or palpitation. On examination, he is comfortable at rest; the JVP is 3 cm above sternal border, loud A2 and left ventricular S4, no murmur (see reports).
Blood pressure 160/100 mmH
Heart rate 76 /min
Respiratory rate 18 /min
Oxygen saturation 95 %
Electrocardiogram: Shows sinus rhythm and left ventricular hypertrophy.

Echocardiogram: Normal cardiac chambers size, concentric left ventricular hypertrophy and normal left ventricular ejection fraction.
Which of the following is the most likely diagnosis?
A. Left vntricular diastolic dysfunction
B. Restrictive cardiomyopathy
C. Silent myocardial ischemia
D. Constrictive pericarditis

Answer: A
A 30-year-old man is found to have cardiomegaly on chest X-ray during pre-employment check- up. He denies chest pain, dyspnea, orthopnea or PND. Physical examination unremarkable (see reports).
Blood pressure 120/80 mg
Heart rate 76 /min
Respiratory rate 18 /min
Electrocardiogram: Normal.
Echocardiogram: Dilated left ventricle with ejection fraction of 40%.
Which of the following is the best management?
A. Digoxin
B. Lisinopril
C. Furosemide
D. Echocardiogram in 6 months

Answer: B
A 45-year-old man with previous history of rheumatic heart disease presents with worsening dyspnea, orthopnea and PND. Physical examination shows holosystolic murmur in the apex and bibasilar crackles on lung auscultation. His symptoms improved with furosemide, spironolactone, enalapril and carvedilol (see report).
Echocardiogram: Showed mildly dilated left ventricle, LVEF 45% and severe rheumatic mitral regurgitation.
Which of the following is the best next step in management?
A. Digoxin
B. Losartan
C. Follow-up in 6 months
D. Mitral valve replacement

Answer: D; Patient is symptomatic so regardless of EF, repair the valve
A 70-year-old man is found to have ejection systolic murmur in the aortic area during routine medical check-up. He denied symptoms of dyspnea, chest pain or syncope. Physical examination revealed slow raising carotid pulsation; normal JVP and normal lung auscultation (see reports).
Blood pressure 110/70 mmHg
Heart rate 76 /min
Respiratory rate 18 /min
Electrocardiogram: Normal sinus rhythm with LVH.
Echocardiogram: Shows concentric LVH, normal ejection fraction and severely stenosed aortic valve.
Which of the following is the next step in management?
A. Aortic valve replacement
B. Follow-up in 6 months
C. Furosemide
D. Enalapril

Answer: B; Patient is asymptomatic with normal EF
A 30-year-old woman with history of rheumatic heart disease since childhood presented with exertional dyspnea and occasional paroxysmal nocturnal dyspnea. Physical examination

showed JVP of 5 cm above sternal angle, loud S1, loud P2 and mid diastolic rumbling murmur in the apex. Fine basal crackles are present on lung auscultation. Which of the following is the most likely diagnosis?
A. Mitral stenosis
B. Aortic stenosis
C. Tricuspid stenosis
D. Pulmonary stenosis

Answer: A
A 60-year-old man presents with abdominal distension dyspnea and orthopnea. The JVP is markedly elevated and fails to descend during inspiration. Lower limbs showed massive pitting edema up to the thighs. Heart sounds are normal. There are bibasilar crackles on lung auscultation. Abdominal examination showed hepatomegaly and positive shifting dullness (see report).
Echocardiogram: Both atria are dilated. Left and right ventricles are normal in size and LVEF is 60%. No valve lesions.
Which of the following has the highest diagnostic value?
A. Holter monitor
B. Cardiac CT scan
C. Coronary angiogram
D. Exercise ECG testing

Answer: B
An 18-year-old boy presents with recurrent syncope with exercise. Physical examination showed normal JVP, jerky carotid pulsations and thrusting apex. There is an ejection systolic murmur, which is best heard in the left parasternal area that increases in intensity with standing and reduced during handgrip (see report).
Blood pressure 110/70 mmHg
Heart rate 76 /min
Electrocardiogram: Shows left ventricular hypertrophy. Which of the following is the most likely diagnosis?
A. Aortic stenosis
B. Pulmonary stenosis
C. Ventricular septal Defect
D. Hypertrophic obstructive cardiomyopathy

Answer: D
A young man with diagnosis of hypertrophic cardiomyopathy presented with recurrent exertional chest pain and syncope. Which of the following is the best management?
A. Nifedipine
B. Metoprolol
C. Hydralazine
D. Nitroglycerine

Answer: B; First line medical therapy for Sx HOCM, is BB, Next is CCB (Verapamil)
A 65-year-old woman with diabetes and hypertension, presented with retrosternal chest pain at rest for 3 hours. Physical examination is normal except for S4 gallop (see lab result and report). Blood pressure 110/70 mmHg, Heart rate 76 /min, Oxygen saturation 95 %
Test Result Normal Values
Troponin 10 < 0.2 pg/L
Electrocardiogram: T-wave inversion in leads V2-V5.
Which of the following is the most likely diagnosis?
A. Stable angina
B. Unstable angina

C. Elevation myocardial infarction
D. Non ST elevation myocardial infarction

Answer: D
A 60-year-old man with diabetes and hypertension is admitted to the Coronary Care Unit with diagnosis of unstable angina. Physical examination is normal except for S4 gallop. Aspirin, bisoprolol, enoxaparn, atorvastatin and nitrate are started (see lab result and reports).
Blood pressure 110/70 mmHg, Heart rate 76 /min
Respiratory rate 18 /min
Test Result Normal Values
Troponin .001 < 0.2 pg/L
Electrocardiogram: Showed ST depression in leads II, III, AVF. Echocardiogram: Normal.
What additional medication is indicated?
A. Clopidogrel
B. Candesartan
C. Streptokinase
D. Spironolactone

Answer: A
A 30-year-old man presents with burning retrosternal chest pain associated with dyspnea. His history is significant only for common cold 2 weeks ago (see image).
Which of the following clinical signs is likely to be present?
A. Murmur
B. Pericardial rub
C. Cannon a-wave
D. Fourth heart sound

Answer: B
A 70-year-old man presents with dyspnea, fatigue and presyncope. On examination, the JVP showed Cannon a-wave and variations in the intensity of the first heart sound (see image).
Which of the following is the definitive treatment?
A. Implantable cardioverter defibrillator
B. Permanent pacemaker
C. Oral theophylline
D. Oral metoprolol

Answer: B; Cannon a-wave indicates severe AV dissociation (3rd degree Heart block)
A 64-year-old woman with advanced breast cancer complains of dyspnea on exertion. On examination, her systolic blood pressure is noted to markedly decrease during inspiration. Heart sounds are distant to auscultation. Chest is clear (see report). Blood pressure 95/65 mmH, Heart rate 100 /min, Respiratory rate 24 /min, Temperature 36.6 °C, Oxygen saturation 95 %
Chest x-ray: Marked cardiomegaly with clear lung fields.
Which of the following is the most likely diagnosis?
A. Hypertrophic obstructive cardiomyopathy
B. Restrictive cardiomyopathy
C. Dilated cardiomyopathy
D. Cardiac tamponade

Answer: D
A 65-year-old woman with diabetes and hypercholesterolemia presents to the Emergency Department with chest tightness associated with sweating and nausea of 4-hour duration. Physical examination showed elevated JVP, which increases further during inspiration, S4 gallop and clear

lung fields on chest auscultation. Aspirin and clopidogrel were given. Results of cardiac enzymes are pending (see image).
Blood pressure 90/55 mmH
Heart rate 76 /min
Respiratory rate 18 /min
Oxygen saturation 95 %
Which of the following immediate action should be taken?
A. Chest X-ray
B. Echocardiogram
C. Coronary CT angiogram
D. ECG with right chest leads

Answer: D
A 25-year-old woman presented with dyspnea and cyanosis. Physical examination revealed elevated JVP with predominant a-wave and left parasternal heave. There is an ejection systolic murmur best heard in the left second intercostal space that increases in intensity during inspiration. An ejection click is also heard in the same area
Blood pressure 110/70 mmHg
Heart rate 76 /min
Respiratory rate 20 /min
Which of the following is the most likely diagnosis?
A. Aortic stenosis
B. Coarctation of aorta
C. Ventricular septal defect
D. Congenital pulmonary stenosis

Answer: D
A 70-year-old man is admitted for stomach resection for gastric cancer. On the 4TH postoperative day. the patient developed retrosternal chest pain accompanied with dyspnea and perfuse sweating. Physical examination is remarkable only for S4. Aspirin, clopidogrel and nitrate were administered (see report)
Blood pressure 110/70 mmHg, Heart rate 76 /min, Respiratory rate 18 /min, Temperature 36.6 °C
Oxygen saturation 95 %
Electrocardiogram: Shows ST elevation in leads V2-V5.
Which of the following is the most appropriate step in management?
A. Thrombolytic therapy
B. Primary angioplasty
C. Abciximab
D. Warfarin

Answer: B
A 70-year-old man with hypertension, diabetes and aortic sclerosis discovered to have atrial fibrillation on routine health check-up. He denies dyspnea, chest pain or palpitations. Which of the following drugs is recommended to prevent systemic embolization?
A. Aspirin
B. Warfarin
C. Clopidogrel
D. Rivaroxaban

Answer: B; In SMLE consider any Valvular Heart Disease an indication for Warfarin.
A 22-year-old woman presents with retrosternal burning chest pain, which is worse during breathing and coughing. Precordial auscultation showed an added sound of 3 components; systolic, diastolic and presystolic (see image). Which of the following is the best initial treatment?

A. Tissue plasminogen activator
B. Enoxaparin
C. Ibuprofen
D. Prednisone

Answer: C
In a patient with heart failure due to dilated cardiomyopathy presenting with fast atrial fibrillation, which of the following drugs is the most appropriate to control the heart rate?
A. Digoxin
B. Diltiazem
C. Ivabradine
D. Propranolol

Answer: A
A 40-year-old man is admitted with palpitations of 4-hour duration. He had similar attack a month ago but did not seek medical advice. His initial electrocardiogram showed atrial fibrillation with controlled ventricular rate. He subsequently converted to sinus rhythm. Which of the following drugs is the most appropriate to maintain sinus rhythm?
A. Digoxin
B. Bisoprolol
C. Verapamil
D. Amiodarone

Answer: D
An 18-year-old girl is recently diagnosed to have rheumatic fever with carditis. Echocardiogram showed mild mitral stenosis with mild regurgitation. To what age the rheumatic fever prophylaxis is recommended?
A. 40
B. 28
C. 23
D. 21

Answer: A; For 10 years or till the age of 40 whichever is longer
Which of the following is an absolute contraindication for thrombolvtic therapy?
A. Pregnancy
B. Intracranial hemorrhage 2 years ago
C. Cardiopulmonary resuscitation for 10 min
D. Blood pressure at presentation of 180/110 mmH

Answer: B
A 28-year-old woman presents to the Cardiology Outpatient Department with recurrent palpitations for the last 6 months. She describes the palpitations as fast heartbeat, which starts and ends suddenly. Each attack lasts 5-10 min. No associated chest pain, dyspnea, syncope or presyncope (see lab result and report).
Test Result Normal Values
Hb 132 130-170 g/L (Male)
120-160 g/L (Female)
Thyroid-Stimulating Hormone 3.8 0.4 - 5.0 jU/mL.
Electrocardiogram: Normal sinus rhythm with rate of 70/min.
Which of the following is the best next investigation?
A. Exercise ECG testing
B. Electrophysiology
C. Echocardiogram

D. Holter monitor

Answer: D
An elderly patient is admitted with diagnosis of unstable angina. He is known to have atrial fibrillation and taking warfarin. The attending resident administered heparin but did not check the INR, which came back as 6.8. The patient developed bleeding from the gums and hematuria. His son is upset, and he requests an explanation.
Which of the following is the most appropriate response?
A. Say that you are sorry for what happened and request him to calm down
B. Apologize and reassure him that you will start an investigation
C. Tell him that this is an expected side effect of the drug
D. Inform him that the situation is under control

Answer:B
A 60-year-old man with hypertension and diabetes undergone coronary angiogram, which showed 3-vessel disease requiring coronary bypass surgery. The patient refused surgery. Being the resident with the team, what is the appropriate communication tool to be used?
A. Use the SPIKES protocol
B. Tell the son to speak to the patient
C. Respect autonomy and continue medical treatment
D. Apply the counselling in the presence of the consultant

Answer: D
A patient who presented with heart failure had an echocardiogram, which showed severe rheumatic mitral regurgitation requiring valve replacement. He is waiting to know the result. Which of the following is the most appropriate way to break the news?
A. Avoid asking him about what he already knows
B. Tell him all the details from the beginning
C. Express more sympathy than empathy
D. Deliver the information in small steps

Answer: D
A 53-year-old widow with dilated cardiomyopathy presented to the Cardiology Clinic with decompensated heart failure. She lives alone, has no insurance and cannot afford the cost of medications. During counselling, which of the following is the most appropriate plan?
A. Arrange for a charity organization to help her financially
B. Involve the social worker and the hospital support system
C. Refer her to the Emergency Department
D. Advise her to avoid salt and stop smoking

Answer: B
A 79-year-old man with multiple co-morbidities is hospitalized for dissecting aneurysm and needs urgent surgery. He lives alone and he has no written will or advanced directive. His condition is very critical in the ICU. What is the best way to arrange the consent for surgery?
A. The attending medical team should sign the consent form
B. The social worker is asked to complete the consent form
C. Proceed for emergency and delay the consent
D. Request a court decision

Answer: A

A 48-year-old man presents with central and severe chest pain, which started around 1 hour ago. There is no radiation of the pain or associated shortness-of-breath. He has had some similar fleeting

pains over the past 2 weeks but these settled spontaneously after a few seconds. Which of the following is the most likely diagnosis?
A. Gastro-oesophageal reflux disease
B. Myocardial Infarction
C. Pneumothorax
D. Anxiety

Answer: B
A 64-year-old man, who has hypertension, presents with increasing shortness of breath. His symptoms are worse at night. On examination, there is bilateral basal crackles and a third heart sound (S3). Which of the following is the most likely diagnosis?
A. Heart failure
B. Bronchial asthma
C. Pulmonary fibrosis
D. Obstructive sleep apnea

Answer: A
A 39-year-old man presents with shortness of breath for 2 weeks. The symptoms associated with activity and relieved by rest. He is otherwise healthy, takes no medications and denies chest pain or pressure. Which of the following is the most likely diagnosis?
A. Classical angina
B. Anginal equivalent
C. Musclo-skeletal pain
D. Myocardial infarction

Answer: B
A 77-year-old man with poorly controlled hypertension presents with a 6-months history of fatigue and dyspnea on exertion. He denies edema, orthopnea, chest pain and palpitations. His only medication is hydrochlorothiazide (25 mg) once daily. On examination, he has a grade 2/6 midsystolic ejection murmur, which is loudest at the right upper steal border. An S4 is heard at the apex, there is no S3. His lungs are clear and there is no peripheral edema. Which of the following is the most likely diagnosis?
A. Aortic regurgitation
B. Mitral regurgitation
C. Aortic stenosis
D. Mitral stenosis

Answer: C
A 49-year-old smoker man has type 2 diabetes mellitus, morbid obesity and a recent diagnosis of symptomatic peripheral arterial disease. He is started on atorvastatin (Lipitor), offered a supervised exercise program and discussed smoking cessation and interventions.
Which of the following should be recommended to prevent cardiovascular events?
A. Aspirin
B. Warfarin
C. Atenolol
D. Enoxaparin SC injection

Answer: A
A 34-year-old woman with a history of prolonged QT syndrome presents with severe acute bacterial sinusitis. Which of the following antibiotics should be avoided?
A. Amoxicillin
B. Moxifloxacin
C. Clarithromycin

D. Amoxicillin/clavulanate

Answer: C
A 68-year-old woman is admitted to the hospital with first episode of acute heart failure. She has a history of hypertension treated with a thiazide diuretic. An echocardiogram reveals no evidence of valvular disease and no segmental wall motion abnormalities. Left ventricular hypertrophy is noted, and her ejection fraction is 55%.
Which of the following is the most likely cause of her heart failure?
A. Hypertrophic cardiomyopathy
B. Diastolic dysfunction
C. Systolic dysfunction
D. High-output failure

Answer: B
A 76-year-old woman with a history of hypertension and osteoporosis presents to the clinic for preoperative clearance for an upcoming eye surgery. She complains of progressively worsening fatigue over the past 8 to 10 months. She states she often feels dizzy but denies syncope. Her current medications include alendronate and hydrochlorothiazide. Which of the following is the most appropriate action?
A. An exercise treadmill test
B. A 48-hour holter monitor
C. Referral to a cardiologist for evaluation
D. Clearance for eye surgery with no further evaluation

Answer: C
A 61-year-old man complains of shortness of breath. He has a history of hypertension, type 2 diabetes and hyperlipidemia. He was a smoker for more than 30 years but he quit 4 years ago. On physical examination, he is not in respiratory distress, but he has diffuse wheezing bilaterally over lower lobes of his lungs. His cardiac examination demonstrates an S4 and distended jugular veins. Which of the following is the best treatment to relieve his symptoms?
A. Anticoagulant therapy
B. Steroid therapy
C. Diuretics
D. Aspirin

Answer: C
A 38-year-old man has episodic chest pain. He states that the pain feels like tightness, located right behind his sternum, lasts less than 3 minutes and is relieved with rest. He takes no medications. He has no family history of coronary disease and has never smoked. His ECG in the office is normal. What is the most appropriate test to exclude cardiac ischemia?
A. Cardiac CT
B. Exercise ECG
C. Resting echocardiogram
D. Creatine kinase enzyme

Answer: B
A 55-year-old man has congestive heart failure due to left ventricular systolic dysfunction. He has no history of medications allergy or interaction, Which of the following is the best first-line treatment option?
A. ACE inhibitors and diuretics
B. Hydralazine and B-Blockers
C. B-Blockers and diuretics
D. Nitrates and -Blockers

Answer: A
A 58-year-old man has chronic atrial fibrillation. Which of the following is this patient at increased risk for?
A. Acute myocardial infarction
B. Cerebrovascular accident
C. Sudden cardiac death
D. Ventricular fibrillation

Answer: B
A 55-year-old man presents to the clinic with history of atrial fibrillation. What is the drug of choice to prevent complications of his condition?
A. Prophylactic warfarin
B. Prophylactic heparin
C. Prophylactic lidocaine
D. No drug is recommended

Answer: A
A 59-year-old man is admitted to the hospital with infero-lateral myocardial infarction. Which of the following medications would improve his survival?
A. Hydrochlorothiazide
B. Metoprolol
C. Morphine
D. Warfarin

Answer: B
A 56-year-old man has a low-pitched diastolic rumble heard at the apex, a widened pulse pressure and a grade 3/6 diastolic decrescendo murmur at the left steal border.
Which of the following is the most likely diagnosis?
A. Hypertrophic cardiomyopathy
B. Pulmonic insufficiency
C. Aortic regurgitation
D. Mitral stenosis

Answer: C
A 60-year-old man, with type 2 diabetes, presents with persistent diffuse neck pain for the past hour, diaphoresis, shortness of breath, a normal troponin I level, and an electrocardiogram that reveals only nonspecific ST-segment and T-wave changes.
Which of the following is the most appropriate next step in management?
A. Administer oral ibuprofen
B. Recheck the troponin I level in 6 hours
C. Perform a right-sided electrocardiogram
D. Administer aspirin and sublingual nitroglycerin

Answer: D
A 42-year-old man presented with worsening pleuritic, positional chest pain. Examination revealed pericardial rub, a clear lung examination, and no evidence of lower-extremity edema ECG shows non-diagnostic findings. Which of the following is the most appropriate next step for diagnosis?
A. D-dimer testing
B. Exercise stress testing
C. Referral to echo-cardiograph specialist
D. Reassure the patient about insignificant symptoms

Answer: D
A 26-year-old woman had 2 episodes of syncope in the past 2 months during micturition and coughing. Physical examination and electrocardiogram are unremarkable. Which of the following is the most appropriate management options?
A. Transthoracic echocardiogram
B. Reassurance and education
C. Holter monitor
D. Tilt-table test

Answer: B
A 22-year-old woman with vasovagal syncope comes to the clinic. She has a normal physical examination and a normal electrocardiogram. Which of the following management approaches is most appropriate?
A. Reassurance and explanation
B. Echocardiogram
C. Tilt-table study
D. Holter monitor

Answer: A

Rheumatology

A 53-year-old woman presents to the Outpatient Department. She has been unwell for 6 months with arthralgia, fever, weight loss, and more recently has developed pleuritic chest pains. She also has synovitis of the small joints of the hands. There were no rashes or nail fold infarcts. Heart sounds are normal (see lab results and reports).

Blood pressure 110/70 mmHg
Heart rate 70 /min
Respiratory rate 18/min

Test Result Normal Values
Hb 89 120-160 g/L
WBC 3.2 4-10.5 x 109/L
Platelet count 99 150-400 × 109/L
Neutrophils 80 40-60%
Lymphocytes 17 20-40 %
MCV 90 80-95 f1
C-reactive peptide 99 <8.2 mg/L
ESR 56 3-15 mm/h
Clotting time 7 5-10 min
Ferritin 920 22-340 pg/L
ANA 1:320 =1:40

Pulmonary Angiography CT:
Confirms a small left pleural effusion. There is no consolidation, pulmonary embolus, and no lymphadenopathy.

ECG:
Sinus tachycardia.

Which of the following the most likely diagnosis?
A. Infective endocarditis
B. Adult-onset Still's disease
C. Systemic lupus erythematosus

D. Rheumatoid arthritis with vasculitis

Answer: C

2. A 47-year-old patient presents with 12 hours of severe pain and swelling of the right first metatarsophalangeal joint. On examination, the affected joint is erythematous and tender. Blood pressure 110/65 mmHg, Heart rate 105 /min, Respiratory rate 12 /min, Temperature 38°C.

Which of the following is the most likely cause?
A. Sodium urate crystals
B. Staphylococcus aureus
C. Disseminated gonorrhea
D. Calcium pyrophosphate crystals

Answer: A

3. A 52-year-old patient presents with isolated pain and swelling of the left knee. Arthrocentesis demonstrates innumerable needle-shaped negatively birefringent crystals.
Which of the following is the most appropriate treatment?
A. Probenecid
B. Methotrexate
C. Indomethacin
D. Sulfinpyrazone

Answer: C

4. A 32-year-old woman complains of severe joint pain in the fingers of both hands, especially in the morning during the winter months. Examination confirms deformity of her fingers.

Which of the following is the most likely diagnosis?
A. Gout
B. Haemophilia
C. Rheumatoid arthritis
D. Primary degenerative osteoarthritis

Answer: C

5. A 74-year-old woman complains of a deep, aching pain of the hip, which is aggravated by walking. She states that after a busy day, the pain keeps her awake at night, and each morning her hip is stiff for several hours.
Which of the following is the most likely cause of the pain?
A. Depression
B. Osteomyelitis
C. Osteoporosis
D. Osteoarthritis

Answer: D

6. A 60-year-old woman complains of pain in her right knee for the last 3 years. The pain is worse with activity and relieved by rest and analgesics. Physical examination reveals that she has a varus knee deformity and tenderness over the medial joint line.
Which of the following is the most likely diagnosis?
A. Inflammatory arthritis
B. Medial meniscus tear
C. Septic arthritis
D. Osteoarthritis

Answer: D

7. A 58-year-old man presents to the Outpatient Clinic with very severe pain, swelling, and redness of the first metatarsophalangeal joint. Some joint fluid is withdrawn (see lab results and reports).

Test Result Normal Values
Colour yellow clear
Clarity opaque transparent
WBC per mm3 100,000 < 200
PMNs 75 < 25%

Fluid Exam:
Negatively birefringent rate crystals are seen on polarizing examination of the synovial fluid.

Hands X-ray:
Juxta-articular punched-out sclerotic erosions involving the first metatarsophalangeal and interphalangeal joints with associated periarticular soft tissue swelling.

Which of the following treatments should be avoided in the current situation?
A. NSAID
B. Colchicine
C. Allopurinol
D. Corticosteroids

Answer: C

8. Which of the following crystals is responsible for the arthritic symptoms of pseudogout?
A. Calcium urate
B. Calcium oxalate
C. Calcium chloride
D. Calcium pyrophosphate

Answer: D

16. A 45-year-old woman comes to the clinic complaining of episodes of pain with swelling and warmth of her hands and feet for the past 3 weeks. During these episodes, lasting hours to days, it is difficult for her to walk or to prepare meals. On physical examination, she has painful 1 cm subcutaneous nodules on the extensor surfaces of her elbows.

Which of the following laboratory findings would confirm the diagnosis?
A. HLA-B27
B. ANA positive at 1:256
C. Anti-CCP antibody positive
D. Markedly decreased serum complement

Answer: C

A 45-year-old woman with systemic lupus erythematosus comes for routine follow-up visit. She was asymptomatic and currently on no medications. Her condition is stable for several years (see lab results).
Blood pressure 140/90 mmH
Test Result Normal Values
24 hr urine protein 35 0.8-14 mg/di (Male)
2.5-11 mg/di (Female)

Protein 500 0-150 mg/24 hr
RBC 5000 up to 4,000 in 1 ml
Urea 20 2.75-7.4 mmol/L
Creatinine 160 44-115 mol/L
Which of the following is the most appropriate next step in management?
A. Anti Ds-DNA
B. Renal biopsy
C. Doppler renal ultrasound
D. Repeat urine analysis in next visit

Answer: B
A 35-year-old woman with rheumatoid arthritis who maintained on methotrexate 15 mg weekly. She is currently on remission. 4 months ago, her liver function tests were normal but the most recent liver function tests are shown (see lab results).
Test Result Normal Values
Aspartate aminotransferase 300 12-40 IU/L
Alanine aminotransferase 280 5-40 IU/L
Albumin 30 34-56 g/L
What is the most appropriate action to be done for her abnormal liver function tests?
A. Do ultrasound for the liver
B. Start glucocorticoids
C. Stop methotrexate
D. Order liver biopsy

Answer: C

A 22-year-old woman presented to the clinic with 2-months history of polyarthralgia, skin rash, mucosal ulcers and fatigue. Her clinical examinations revealed malar rash over her face and active arthnitis at several joints (see lab results).
Test Result Normal Values
Hb 100 130-170 g/L. (Male) 120-160 g/L (Female)
Platelets count 100 150-400 x 109/L WBC 3 4.5-10.5 x 109/L
ESR 35 2-10 mm/h (Male)
3-15 mm/h (Female)
Lupus anticoagulant positive negative
Urea 5 2.75-7.4 mmol/L
Creatinine 78 44-115 mol/L
C-reactive peptide 5 <8.2 mg/L
Complement C3 0.3 0.7-1.5 g/L
Complement C4 0.13 0.15-0.45 g/L
Which of the following test obtained at this visit is most likely to confirm the diagnosis?
A. Anti-Ro (SSA) antibodies
B. Antinuclear antibodies (ANA)
C. Anti-double-stranded DNA (anti-DNA)
D. Anti- ribonucleoprotein (RNP) antibodies

Answer: B
A 30-year-old woman presented to the clinic with polyarthralgia, malar rash and mucosal ulcerations for 2-months. Examination revealed active malar rash and arthritis affecting several joints. She has positive anti-nuclear antibody with high titer (see lab results).
Test Result Normal Values

Hb 110 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 280 150-400 x 109/L WBC 10 4.5-10.5 x 109/L
ESR 65 2-10 mm/h (Male)
3-15 mm/h (Female)
Urea 4 2.75-7.4 mmol/L
Creatinine 60 44-115 mol/L
Appearance clear clear
Protein absent absent
Leukocytes 2 0-3 per high power field
Erythrocytes 0 0-2 per high power field
Complement C3 0.6 0.7-1.5 g/L
Complement C4 0.13 0.15-0.45 g/L
Alkaline phosphatase 50 39-117 TU/L
Alanine aminotransferase 18 5-40 IU/L
Aspartate aminotransferase 20 12-40 IU/L
Which of the following is the best treatment option at this time?
A. Hydroxychloroquine and methotrexate
B. Hydroxychloroquine and azathioprine
C. Hydroxychloroquine and cyclophosphamide
D. Hydroxychloroquine and mycophenolate mofetil

Answer: A
A 30-year-old woman presented to the clinic with polyarthralgia, malar rash and mucosal ulcerations for 2-months. Examination revealed active malar rash and arthritis affecting several joints. She has positive anti-nuclear antibody with high titer (see lab results).
Test Result Normal Values
Hb 100 130-170 g/L (Male) 120-160 g/L (Female)
Platelets count 90 150-400 × 109/L WBC 3.5 4.5-10.5 x 109/L
ESR 60 2-10 mm/h (Male)
3-15 mm/h (Female)
Urea 5 2.75-7.4 mmol/L
Creatinine 150 44-115 mol/L
Protein 1200 0-150 mg/24 hr
Complement C3 0.4 0.7-1.5 g/L
Complement C4 0.1 0.15-0.45 g/L
Which of the following is the best high dose treatment option at this time?
A. Steroid and mycophenolate mofetil
B. Steroid and hydroxychloroquine
C. Steroid and methotrexate
D. Steroid and cyclosporine

Answer: A
A 50-year-old man with active rheumatoid arthritis presented to the clinic complaining of polyarthralgia and significant early morning stiffness. His current treatments are methotrexate, prednisone and hydroxychioroquine. Examination revealed active arthritis at several joints (see lab results).
Test Result Normal Values
Hb 160 130-170 g/L (Male)
120-160 q/L (Female)
Platelets count 500 150-400 x 109/L WBC 10 4.5-10.5 x 109/L
ESR 80 2-10 mm/h (Male)
3-15 mm/h (Female)

Urea 5 2.75-7.4 mmol/L
Creatinine 75 44-115 mol/L
C-reactive peptide 18 <8.2 mg/L Rheumatoid factor 600 <58 kIU/L Aspartate aminotransferase 37 12-40 IU/L Alanine aminotransferase 35 5-40 IU/L Albumin 50 34-56 g/L
Which of the following is the most appropriate additional treatment option?
A. Azathloprine
B. Adalimumab
C. Cyclosporine
D. Mycophenolate mofetil

Answer: B
A 38-year-old man presented to the clinic complaining of recurrent painful swelling affecting right knee and left ankle for 6-months despite using non-steroidal anti-inflammatory drugs. His past history revealed urinary tract infection 2-weeks prior to onset of his joint pain. On examination, he has active arthritis at right knee and left ankle (see lab results).
Test Result Normal Values
Hb 160 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 450 150-400 x 109/L WBC 12 4.5-10.5 x 109/L
ESR 65 2-10 mm/h (Male)
3-15 mm/h (Female)
Urea 4 2.75-7.4 mmol/L
Creatinine 100 44-115 mol/1.
C-reactive peptide 20 <8.2 mg/L
Rheumatold factor 13 <58 kJU/L
Color clear clear or light yellow
Protein absent absent
Nitrite absent absent
Leukocytes 2 0-3 per high power field
Which of the following is the most appropriate treatment?
A. Sulfasalazine
B. Methotrexate
C. Azithromycin
D. Cyclosporine

Answer: A; For patient with reactive arthritis for more than 6 months use Sulfasalazine.
A 30-year-old woman with systemic lupus erythematosus is complaining of arthralgia and skin rash for 3 months. Examination revealed active arthritis at two small joints of right hand and malar rash over her face (see lab results).
Test Result Normal Values
Hb 110 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 190 150-400 x 109/L WBC 6 4.5-10.5 x 109/L
ESR 35 2-10 mm/h (Male)
3-15 mm/h (Female)
Urea 5 2.75-7.4 mmol/L
Creatinine 65 44-115 mol/L
Which of the following advice should be presented to the patient about hydroxychloroquine?
A. Interferes with mycophenolate mofetil
B. Should be stopped during pregnancy
C. Should be stopped during lactation
D. Efficacy is reduced by smoking

Answer: D
A 50-year-old woman presented to the clinic with polyarthralgia affecting distal and proximal interphalangeal joints bilaterally for 3-years. She does not have significant early morning stiffness. Clinical examination revealed no active arthritis but there was non-tender hard nodules over some distal interphalangeal (see lab results).
Test Result Normal Values
Hb 140 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 190 150-400 × 109/L
WBC 11 4.5-10.5 x 109/L
ESR 15 2-10 mm/h (Male)
3-15 mm/h (Female)
C-reactive peptide 5 <8.2 mg/L
Rheumatoid factor 30 <58 klU/L
Urea 4 2.75-7.4 mmol/L
Creatinine 75 44-115 mol/L
Which of the following is the most likely rheumatological disease?
A. Seronegative rheumatoid arthritis
B. Hand osteoarthritis
C. Polyarticular gout
D. Reactive arthritis

Answer: B
A 53-year-old woman presented to the clinic with polyarthralgia affecting distal and proximal interphalangeal joints bilaterally for 3-years. She does not have significant early morning stiffness. Clinical examination revealed no active arthritis (see lab results).
Test Result Normal Values
Hb 140 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 200 150-400 × 109/L
WBC 8 4.5-10.5 x 109/L
ESR 12 2-10 mm/h (Male)
3-15 mm/h (Female)
Urea 4 2.75-7.4 mmol/L
Creatinine 75 44-115 umol/L
C-reactive peptide 5 <8.2 mg/L
Which of the following is the best management option at this stage?
A. Methotrexate therapy
B. Surgical intervention
C. Weight loss
D. Finger splints

Answer: D
A 65-year-old man presented to the clinic with 3-year history of progressive worsening of bilateral knee pain and stiffness that occurs at rest and at night. He has no history of trauma. Paracetamol and ibuprofen resulted in mild improvements of his joint pain. Upon examination, there is no warmth, erythema or effusion and there is no compromise in joints stability (see lab results).
Blood pressure 140/90 mmHg Temperature 36.6 °C
Weight 90 kg
Height 160 cm
Test Result Normal Values
Hb 125 130-170 g/L. (Male) 120-160 g/L (Female)
Platelets count 300 150-400 × 109/L WBC 9 4.5-10.5 x 109/L

ESR 15 2-10 mm/h (Male)
3-15 mm/h (Female)
Blood urea nitrogen 9 2.8 to 8.9 mmol/L
Creatinine 115 44-115 mol/L
C-reactive peptide 6 <8.2 mg/L
Which of the following is the most appropriate next investigation?
A. MRI of knee joints
B. Bone scintigraphy
C. Ultrasonography for knee joints
D. Standing plain radiography of knee joints

Answer: D
A 30-year-old man presented to the clinic with left knee pain for one month. 3 months ago, he had an episode of right knee active arthritis that was aspirated and then improved. Synovial fluid aspiration showed leukocyte count 40,000/cu mm, and negative cultures. Clinical examination revealed a swollen and tender left knee only. He has not had rashes or eye symptoms. Knee radiographs are normal (see lab results).
Test Result Normal Values
Hb 140 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 450 150-400 x 109/L WBC 10 4.5-10.5 x 109/L
ESR 55 2-10 mm/h (Male)
3-15 mm/h (Female)
Urea 4 2.75-7.4 mmol/L
Creatinine 85 44-115 mol/L
C-reactive peptide 16 <8.2 mg/L
Rheumatoid factor 35 <58 kIU/L
Which of the following is the most appropriate initial treatment?
A. Ibuprofen
B. Methotrexate
C. Infliximab
D. Amoxicillin

Answer: A
A 23-year-old woman with systemic lupus erythematosus (SLE) presented to the clinic with a 3-month history of lower-extremity swelling and shortness of breath. She has not had chest pain, palpitations, or fevers. On physical examination, she has jugular venous distention and engorgement with inspiration. Heart sounds are distant; a prominent early diastolic sound without gallops or murmurs is heard. Breath sounds are normal. Abdominal examination reveals shifting dullness. Pitting edema to the level of the knees is noted (see lab results).
Blood pressure 110/70 mmHg Heart rate 95 /min Respiratory rate 25 /min Temperature 36.6 °C Oxygen saturation 90 %
Test Result Normal Values
Hb 90 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 450 150-400 x 109/L WBC 3 4.5-10.5 x 109/L
Creatinine 150 44-115 mol/L
Albumin 28 34-56 g/L
Complement C3 0.5 0.7-1.5 g/.
Complement C4 0.1 0.15-0,45 g/L.
Aspartate aminotransferase 100 12-40 IU/L
Alanine aminotransferase 94 5-40 IU/L
Which of the following SLE association most likely explains the current condition?

A. Constrictive pericarditis
B. Myocardial infarction
C. Liver failure
D. Renal failure

Answer: A
A 30-year-old woman with systemic lupus erythematosus presented to Emergency Department with seizures and fatigue for 2-days. No history of fever or skin rash. Musculoskeletal and neurologic examinations are normal. She has negative septic workup.
Brain MRI revealed multiple hyperintense lesions (see lab results).
Blood pressure 120/70 mmg Temperature 37.2 °C Oxygen saturation 95 %
Test Result Normal Values
Hb 110 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 200 150-400 x 109/L WBC 6 4.5-10.5 x 109/L
ESR 35 2-10 mm/h (Male) 3-15 mm/h (Female)
APTT 30 30-40 sec
INR 1 0.8-1.2
Lupus anticoagulant negative negative
Urea 6 2.75-7.4 mmol/L
Creatinine 100 44-115 mol/L
Pressure 300 200-250 mmH20 (Male) 300-400 mmH2O (Female)
Cells 20 0-3 /pL
Total protein (Women) 0.65 0. 15-0.45 g/L
Anti SS antibodies 35 negative
Complement C3 0.5 0.7-1.5 g/L.
Complement C4 0.1 0.15-0.45 g/L.
Which of the following is the most appropriate treatment in addition to corticosteroids?
A. Cyclophosphamide
B. Azathioprine
C. Phenytoin
D. Heparin

Answer: A
A 50-year-old man presented to the clinic with a 15-year history of multiple joints pain with daily significant moming stiffness that lasts for 3 hours. He has been admitted 3 times during the last year to the local hospital for treatment of bacterial pneumonia. Physical examination reveals multiple rheumatoid nodules and an enlarged spleen. He has active arthritis affecting multiple metacarpophalangeal (MCP) and proximal interphalangeal (PIP) joints symmetrically. Hands x- ray show severe erosive disease of the MCP and PIP joints.
Bone marrow aspiration and biopsy reveals an excess of immature forms and maturation arrest with normal differential (see lab results).
Test Result Normal Values
Hb 90 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 100 150-400 x 109/L
WBC 2 4.5-10.5 x 109/L
ESR 80 2-10 mm/h (Male)
3-15 mm/h (Female)
Rheumatoid factor 400 <58 klU/L
Which of the following is the most likely diagnosis?
A. Systemic lupus erythematosus
B. Feltys syndrome

C. Lymphoma
D. Tuberculosis

Answer: B; A triad of Neutropenia, Splenomegaly, and Arthritis is Felty's Syndrome.
A 40-year-old man presented to Emergency Department with fever and painful swelling in the left knee for 2 days. 2 weeks ago, he had watery, non-bloody diarrhea and abdominal pain for 5- days after eating from a restaurant. On physical examination, left knee is warm and tender, with a moderate effusion. The right wrist is mildly swollen and tender. The left achilles tendon is mildly tender and swollen at its insertion into the calcaneus. Blood, stool and synovial cultures are negative. However, testing for Clostridium difficile toxin is positive (see lab results).
Blood pressure 140/90 mmHg
Heart rate 76 /min
Temperature 38.5 °C
Test Result Normal Values
ESR 75 2-10 mm/h (Male)
3-15 mm/h (Female)
WBC 18 4.5-10.5 x 109/L.
Creatinine 110 44-115 mol/L
Uric acid 260 200-410 mol/L (Male)
140-360 mol/L (Female)
C-reactive peptide 20 <8.2 mg/L
Rheumatoid factor 35 <58 klU/L
Which of the following is the most likely arthritis?
A. Septic
B. Gonococcal
C. Reactive
D. Peripheral associated with inflammatory bowel disease

Answer: C
A 32-year-old woman smoker presented to the clinic with 6-months history of polyarthraigia, malar rash and fatigue. Examination revealed active malar rash over her face and lower limb pitting edema up to the knees. She has positive antinuclear antibody with high titer (1:1280). She is started on hydroxychloroquine and mycophenolate mofetil (see lab results).
Test Result Normal Values
Hb 110 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 200 150-400 x 109/L WBC 3.5 4.5-10.5 x 109/L
ESR 45 2-10 mm/h (Male)
3-15 mm/h (Female)
Urea 5 2.75-7.4 mmol/L
Creatinine 75 44-115 mol/L
24 hr urine protein 750 0.8-14 mg/di (Male) 2.5-11 mg/dl (Female)
Complement C3 0.4 0.7-1.5 g/L
Complement C4 0.1 0.15-0.45 g/L
Which of the following non-pharmacological intervention would be most important at this time?
A. Weight loss
B. Smoking cessation
C. Low protein diet
D. Reduce physical activities

Answer: B
A 27-year-old man with systemic lupus erythematosus (SLE) is seen in the clinic because of worsening fatigue and exertional shortness of breath. He has had no fevers, arthritis, ulcers, rash,

chest pain or palpitations. His current treatments are hydroxychloroquine and mycophenolate mofetil. Systemic examination is unremarkable (see lab results).
Blood pressure 130/70 mmHg Respiratory rate 18 /min Temperature 37.3 °C
Test Result Normal Values
Hb 78 130-170 g/L (Male)
120-160 g/L (Female)
Reticulocyte 7 0.2-1.2 %
Platelets count 300 150-400 x 109/L WBC 6 4.5-10.5 x 109/L
Creatinine 85 44-115 mol/L
Haptoglobin 0.2 0.9-1.4 g/L
Iron 11 11.7-31.8 molL (Male) 9-30.4 umol/L (Female)
Ferritin 500 20-300 g/L. (Male) 20-120 pg/L (Female)
Lactate dehydrogenase 1200 60-160 IU/L
Which of the following anemias is most likely associated with SLE at this stage?
A. Anaemia of chronic disease
B. Iron deficiency
C. Haemolytic
D. Pernicious

Answer: C
A 55-year-old man with rheumatoid arthritis for 6-years on methotrexate 15 mg weekly and prednisone 5 mg daily presented to the Emergency Department with a 2-day history of left knee pain and swelling. On examination, left knee is swollen, red, warm and tender. Left knee aspiration revealed high white blood cells 35000 cells/mm and the culture is pending (see lab results).
Blood pressure 105/70 mmHg Heart rate 96 /min Temperature 39 °C
Oxygen saturation 97 %
Test Result Normal Values
Hb 125 130-170 g/I (Male)
120-160 g/L (Female)
Platelets count 450 150-400 x 109/L WBC 18 4.5-10.5 x 109/L
ESR 80 2-10 mm/h (Male)
3-15 mm/h (Female)
C-reactive peptide 28 <8.2 mg/L
Rheumatoid factor 600 <58 klU/L
Which of the following is the most appropriate immediate management?
A. Intravenous antibiotics
B. Intravenous high dose steroids
C. Intra-articular steroid injection
D. Oral non-steroidal anti-inflammatory drugs

Answer: A
A 65-year-old-woman presented to the clinic with a 5-day history of pain and swelling in the left knee. Her pain aggravated by walking and relieved by rest. No history of small joints involvement or significant morning stiffness. On examination, left knee is swollen but not warm or erythematous. Left knee X-ray shows osteophytes and joint space narrowing.
Joint aspiration reveals 280 leukocytes/mm (see lab results).
Blood pressure 140/80 mmHg
Temperature 36.6 °C
BMI 35 kg/m2
Test Result Normal Values
Hb 125 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 300 150-400 x 109/L

WBC 11.5 4.5-10.5 x 109/L
ESR 8 2-10 mm/h (Male)
3-15 mm/h (Female)
Which of the following is the most likely arthritis?
A. Osteoarthritis
B. Rheumatoid
C. Gouty
D. Septic

Answer:A
A 35-year-old man with rheumatoid arthritis for 6 months on methotrexate 20 mg weekly and ibuprofen daily presented to the clinic with increasing joint pain in both hands and feet associated with morning stiffness for 2-hours. On examination, he shows signs of active arthritis at both wrists. Hand X-rays show erosions at multiple joints (see lab results).
Test Result Normal Values
Hb 120 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 450 150-400 x 109/L WBC 8 4.5-10.5 x 109/L
ESR 60 2-10 mm/h (Male)
3-15 mm/h (Female)
C-reactive peptide 18 <8.2 mg/L
Rheumatoid factor 500 <58 kIU/L
Which of the following is the most appropriate additional treatment?
A. Adalimumab
B. Azathioprine
C. Cyclophosphamide
D. Hydroxychloroquine

Answer: A
A 28-year-old woman presented to the clinic with 2-month history of pain and swelling in the hand joints with morning stiffness for 2-hours. She is 3-months postpartum, and taking care of her infant. On examination, she has active arthritis at second and fourth proximal and metacarpophalangeal joints and both wrists (see lab results).
Test Result Normal Values
Hb 120 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 450 150-400 x 109/L WBC 6 4.5-10.5 x 109/L
ESR 70 2-10 mm/h (Male)
3-15 mm/h (Female)
C-reactive peptide 18 <8.2 mg/L
Which of the following is the most likely diagnosis?
A. Viral arthritis
B. Reactive arthritis
C. Polyarticular gout
D. Rheumatoid arthritis

Answer: D
A 75-year-old man presented to the clinic with a 5-year history of right knee pain that worsens with walking and is relieved with rest. Paracetamol 1 gm 3 times a day resulted in mild improvements. He also has hypertension, diabetes mellitus and chronic kidney diseases. On examination, his right knee is mildly swollen but not tender or warm with significant limitation of his flexion movements. Right knee radiography shows osteophytes and significant joint space narrowing (see lab results).
Test Result Normal Values

ESR 8 2-10 mm/h (Male)
3-15 mm/h (Female)
Urea 8.5 2.75-7.4 mmol/L
Creatinine 190 44-115 mol/L
Which of the following is the most appropriate treatment?
A. Celecoxib
B. Ibuprofen
C. Tramadol
D. Prednisone

Answer: C; Because of renal impairment.
A 30-year-old woman presented to the clinic with 4-month history of muitiple joint pain and morning stiffness for 2-hours. On examination, she has active arthritis at multiple small joints of both hands and feet. Hands radiography show erosions at several small joints of both hands (see lab results).
Test Result Normal Values
Hb 110 130-170 g/L. (Male) 120-160 g/L (Female)
Platelets count 450 150-400 x 109/L WBC 10 4.5-10.5 x 109/L
ESR 65 2-10 mm/h (Male)
3-15 mm/h (Female)
Urea 4 2.75-7.4 mmol/L
Creatinine 67 44-115 mol/L
C-reactive peptide 25 <8.2 mg/L
Rheumatoid factor 450 <58 klU/L
Which of the following is the most appropriate management?
A. Prednisone 60 mg daily
B. Hydroxychloroquine 400 mg daily
C. Indomethacin, 25 mg three times daily
D. Hydroxychloroquine 400 mg daily and methotrexate 15 mg weekly

Answer: D
A 28-year-old man with systemic lupus erythematosus presented to the clinic with a 1-month history of joint pain and moring stiffness. He has no current history of skin rash or other systemic features. His current treatment consists of hydroxychloroquine 400 mg daily. On examination, he has active arthritis at both wrist joints and second proximal interphalengeal joint of both hands (see lab results).
Test Result Normal Values
Hb 125 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 220 150-400 x 109/L WBC 5 4.5-10.5 x 109/L
ESR 55 2-10 mm/h (Male)
3-15 mm/h (Female)
Urea 4.5 2.75-7.4 mmol/L
Creatinine 78 44-115 mol/L
C-reactive peptide 12 <8.2 mg/L
Complement C3 0.9 0.7-1.5 g/L
Complement C4 0.1 0.15-0.45 g/L
Which of the following agents should be added to control the arthritis?
A. Methotrexate
B. Sulfasalazine
C. Azathioprine
D. Cyclosporine

Answer: A

Hematology/ Oncology

6. A 26-year-old woman has become increasingly tired for the past 2 months. During the past week, she has noted purple blotches on her skin. On physical examination, there are petechial hemorrhages on her trunk and extremities (see lab results).

Test Result Normal Values
Hb 72 120-160 g/L
HCT 0.22 0.37-0.48
MCV 92 80-95 fl
WBC 1.5 4.5-10.5 x 109/L
Platelet count 10 150-400 x 109/L

Which of the following is the most likely diagnosis?
A. Aplastic anemia
B. Sickle cell anemia
C. Hemolytic anemia
D. Iron deficiency anemia

Answer: A

7. A 26-year-old woman presents for pre-marital counseling (see lab results).

Test Result Normal Values
Hb 110 120-160 g/L
HCT 0.33 0.37-0.48
MCV 67 80-95 fl
Hb A2 4.5 <3.5 %

Sickling Negative

Which of the following is the most likely diagnosis?
A. Beta Thalassemia minor
B. Beta Thalassemia major
C. Sickle Beta thalassemia
D. Beta Thalassemia intermedia

Answer: A

8. A 55-year-old man presents to the clinic with headaches and increased tiredness for the last 4 months. On physical examination, he is febrile and his spleen tip is palpable (see lab results and report).

Test Result Normal Values
Hb 200 130-170 g/L
HCT 0.67 0.42-0.52
MCV 93 80-95 f1
WBC 12 4.5-10.5 x 109/L
Platelets 420 150-400 x 109/L

The serum erythropoietin: Very low levels.

Which of the following is the most likely diagnosis?
A. Myelofibrosis
B. Polycythemia rubra vera

C. Congenital heart disease
D. Essential thrombocythemia

Answer: B

9. A 26-year-old man has been diagnosed with sarcoidosis 2 years before; he now presents with mild pallor (see lab results and report)

Test Result Normal Values
Hb 100 130-170 g/L
HCT 0.29 0.42-0.52
MCV 79 80-95 fl
WBC 7.5 4.5-10.5 x 109/L
Platelet count 200 150-400 x 109/L
Ferritin 915 22-561 ng/L

Hemoglobin electrophoresis: Normal.

Which of the following is the most likely diagnosis?
A. Sideroblastic anemia
B. Beta Thalassemia minor
C. Iron deficiency anemia
D. Anemia of chronic disease

Answer: D

Explanation: IDA present with low ferritin not high.

A 35-year-old woman presents with bilateral leg deep vein thrombosis. She has a history of recurrent stillbirths (see lab result).
Test Result Normal Value
APTT 55 30-40 sec
Which of the following is the most likely diagnosis?
A. Antiphospholipid syndrome
B. Antithrombin deficiency
C. Protein S deficiency
D. Protein C deficiency

Answer: A

A 40-year-old Healthy Primigravida with a history previous deep venous thrombosis.
Which of the following is the most appropriate management?
A. Aspirin
B. Heparin
C. Enoxaparin
D. No anticoagulation

Answer: D

A 20-year-old man with sickle cell anemia presents with right-sided chest pain and shortness of breath for 2 days. It was associated with pain in both legs and back. On examination, fine crackles were heard in the right middle and lower zones (see image).
Blood pressure 125/75 mmHg
Heart rate 92 /min
Respiratory rate 25/min

Temperature 37.9° C
Oxygen saturation 91% on room air

Which of the following is the most likely diagnosis?
A. Viral pericarditis
B. Pulmonary embolism
C. Acute chest syndrome
D. Pneumocystis carinii pneumonia

Answer: C

A 20-year-old man with sickle cell anemia presents with right-sided chest pain and shortness of breath for 2 days. It is associated with pain in both legs and back. On examination, fine crackles are heard in the right middle and lower zones (see image).
Blood pressure 125/75 mmHg
Heart rate 92 /min
Respiratory rate 25 /min
Temperature 37.9° C
Oxygen saturation 91% on room air

What is the most effective drug to reduce this Incidence?
A. Folic acid
B. Hydroxyurea
C. 5-azacytidine
D. Erythropoietin

Answer: B

A 42-year-old woman presents with tiredness and lethargy. She has tingling in her hands and feet. Investigations confirm a megaloblastic anemia. She had a total gastrectomy 3 years ago.

Which of the following vitamin deficiencies is most likely to cause these symptoms?
A. B1
B. B2
C. B6
D. B12

Answer: D

A 73-year-old woman is brought to Emergency Room after a fall at home. She is diagnosed with left hip fracture (see lab results). Weight 82 kg
Test Result Normal Values
Sodium 136 134-146 mmol/L
Potassium 4.2 3.5-5.1 mmol/L
Creatinine 68 44-115 mol/L
What is the best order by the admitting orthopedic surgeon to prevent deep vein thrombosis?
A. Aspirin 81 mg po daily
B. Enoxparin 40 mg SC daily
C. Fondaparinux 10 mg SC daily
D. Heparin Sulfate 10,000 units IV BID

Answer: B

A 20-year-old man is noted to bleed excessively during an elective dental extraction. Following the procedure, examination reveals petechial skin hemorrhages. Blood results showed factor VIII, activity 87% (see lab results).
Test Result Normal Values
Hb 125 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 288 150-400 × 109/L
WBC 8 4.5-10.5 x 109/L
Prothrombin time 13.5 10-13 sec
APTT 38 30-40 sec
Which of the following is the most likely diagnosis?
A. Idiopathic thrombocytopenic purpura
B. Von Willebrands disease
C. Haemophilia B
D. Haemophilia A

Answer: B
A 26-year-old man with sickle cell anemia presents with pallor, lethargy and headache (see lab results).
Test Result Normal Values
Hb 44 130-170 g/L (Male)
120-160 g/L (Female)
Reticulocyte 2.8 0.2-1.2 %
Which of the following is the most likely diagnosis?
A. Sequestration crisis
B. Thrombotic crisis
C. Haemolytic crisis
D. Aplastic crisis

Answer: D
A 66-year-old woman presents with lethargy for the past 2 weeks. Her past medical history includes polymyalgia rheumatic and ischemic heart disease. Clinical examination is unremarkable (see lab results).
Test Result Normal Values
Hb 131 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 144 150-400 x 109/L WBC 21 4.5-10.5 x 109/L Neutrophils 44 40-60% Lymphocytes 60 20-40 %
Which of the following is the most likely diagnosis?
A. Chronic lymphocytic leukemia
B. Infectious mononucleosis
C. Transient viral illness
D. Lymphoma

Answer: A

A 48-year-old woman had terminal ileum resection due to Crohn's disease. She presents this time with difficulty in walking. She is anemic and has lost position sense bilaterally. Which of the following is the most likely diagnosis?
A. Aplastic anaemia
B. Folate deficiency
C. Haemolytic anaemia

D. Vitamin B12 deficiency

Answer: D

A 25-year-old woman presented with few months history of increase forgetfulness, fatigue and numbness of lower limbs. She is a known case of crohn's disease, post terminal ilium resection 2 years ago. She is on azathioprine. She denies abdominal pain or diarrhoea (see lab results).
Test Result Normal Values
RBC 4.5 4.7-6.1 x 1012/L. (Male) 4.2-5.4 x 1012/L (Female)
Hb 90 130-170 g/L (Male) 120-160 g/L (Female)
MCH 38 28-33 pg/cell
MCV 99 80-95
Platelets count 199 150-400 x 109/L
A deficiency of which of the following nutrients would be responsible for her current clinical presentation?
A. Zing
B. Thiamine
C. Selenium
D. Vitamin B12

Answer: D

A 39-year-old man with asymptomatic hereditary hemochromatosis presented to the clinic for a check-up (see lab results).
Test Result Normal Values
Alkaline phosphatase 42 39-117 IU/L
Alanine aminotransferase 20 5-40 IU/L Aspartate aminotransferase 22 12-40 IU/L Ferritin 860 20-300 g/L (Male)
20-120 pg/L (Female)
Which of the following is the most appropriate management?
A. Give penicillamine
B. Referral to upper GI endoscopy
C. Repeat the serum ferritin after 6 month
D. Initiate regular therapeutic phlebotomy

Answer:D

A 59-year-old woman comes to the clinic with headache, dyspnea on exertion and occasional palpitations (see lab results).
Test Result Normal Values
Hb 95 130-170 g/L (Male)
120-160 g/L (Female)
MCV 68 80-95 f1
Platelets count 530 150-400 x 109/L
Total iron binding capacity 30 11.6-31.3 mol. (Male) 9-30.4 pmol/L (Female)
Ferritin 3 20-300 g/L (Male)
20-120 pg/L (Female)
Which of the following is the most likely diagnosis?
A. B12 deficiency
B. Folate deficiency
C. Iron deficiency anemia
D. Anemia of chronic disease

Answer: C
A 65-year-old-woman has decreased concentration (see lab results).
Test Result Normal Values
Hb 89 130-170 g/L (Male)
120-160 g/L (Female)
MCV 106 80-95 f1
Platelets count 129 150-400 x 109/L
Vitamin B12 (Women) 0.46 0.12-0.66 mol/L
Folic Acid (Women) 0.03 <0.1 umol/L
Which of the following would be found on further testing?
A. High methyl malonic acid and normal homocysteine
B. High methyl malonic acid and high homocysteine
C. Normal methyl malonic acid and high homocysteine
D. Low methyl malonic acid and low homocysteine

Answer: C
A 46-year-old man has dyspnea on exertion and palpitations. On examination, he has patches of vitiligo and has absent reflexes with positive Babinski in the lower limbs (see lab results).
Test Result Normal Values
Hb 91 130-170 g/L (Male)
120-160 g/L (Female)
MCV 103 80-95 f1
Platelets count 129 150-400 × 109/L
Ferritin 120 20-300 g/. (Male)
20-120 g/L (Female)
Vitamin B12 (Men) 124 160-900 ng/L
Folic Acid (Men) 6 <4 ug/L
Which of the following is the most appropriate management?
A. Oral iron supplement
B. Oral b12 supplement
C. Folate supplement
D. Parenteral b12 supplement

Answer: D
A 35-year-old woman has fever and headache for 2 days. On examination, she has a scattered petichiae over her body. Her platelets are 32. Peripheral smear done (see image and lab results).
Temperature 38.6 °C
Test Result Normal Values
Platelets count 32 150-400 x 109/L
Hb 11.2 130-170 g/L (Male)
120-160 g/L (Female)
MCV 102 80-95 f1
Reticulocyte 4.9 0.2-1.2 %
Lactate dehydrogenase 690 60-160 IU/L
Indirect bilirubin 43 3.2-12.1 umal/L
Total bilirubin 65 3.5-16.5 mol/L
Creatinine 66 44-115 mol/L
APTT 36 30-40 sec
INR 1.1 0.8-1.2
Prothrombin time 11 10-13 sec
Which of the following is the most likely diagnosis?
A. Immune thrombocytopenia (ITP)

B. Hemolytic uremic syndrome (HUS)
C. Thrombotic thrombocytopenic purpura (TTP)
D. Disseminated intravascular coagulopathy (DIC)

Answer: C
A 38-year-old woman has fever and headache for 2 days. On examination, she has a scattered petichiae over her body. Peripheral smear done (see image and lab results).
Temperature 38.6 °C
Test Result Normal Values
Hb 112 130-170 g/L (Male)
120-160 g/L (Female)
Reticulocyte 5.6 0.2-1.2 %
Platelets count 32 150-400 x 109/L Indirect bilirubin 48 3.2-12.1 mol/L Total bilirubin 76 3.5-16.5 mol/L
Lactate dehydrogenase 690 60-160 IU/L APTT 33 30-40 sec
INR 1.0 0.8-1.2
Prothrombin time 10 10-13 sec
Which of the following is the most appropriate treatment?
A. Intravenous immunoglobulin and prednisone
B. Transfuse platelets
C. Plasma exchange
D. Argatroban

Answer: C
A 48-year-old man develops right popliteal vein thrombosis post-sleeve surgery. He is started on enoxaparin and warfarin. What is the target therapeutic international normalization ratio range for this patient?
A. 1-1.9
B. 2-3
C. 3.1-4
D. No need for monitoring

Answer: B
A 53-year-old woman admitted to the hospital with pneumonia. She has diabetes, hypertension and is on hemodialysis for end-stage renal disease (see lab results).
Test Result Normal Values
Hb 98 130-170 g/L (Male)
120-160 g/L (Female)
Platelets count 151 150-400 x 109/L
APTT 33 30-40 sec
INR 0.9 0.8-1.2
Prothrombin time 9.8 10-13 sec
Creatinine 460 44-115 mol/L
Which of the following is the most appropriate venous thromboembolism prophylaxis for this patient?
A. Graded compression stockings
B. Unfractionated heparin
C. Fondaparinux
D. Enoxaparin

Answer: B
A 38-year-old woman admitted with pyelonephritis. 2 days after admission, she develops sudden severe shortness of breath (see report).

Blood pressure 84/50 mmHg
Heart rate 124 /min
Oxygen saturation 84 %
CT-pulmonary angiogram: A saddle embolus at the bifurcation of the pulmonary artery and multiple segmental pulmonary emboli bilaterally.
which of the following is the most appropriate treatment?
A. Unfractionated heparin and warfarin
B. Enoxaparin
C. Alteplase
D. Apixaban

Answer: C
A 59-year-old man has diabetes, hypertension, dyslipidemia and chronic renal impairment. He was admitted 5 days ago to the Coronary Care Unit for a non-ST-elevation myocardial infarction (NSTEMI). He is receiving unfractionated heparin. His labs upon admission were normal. Day 5 of his admission, labs as shown (see lab results).
Test Result Normal Values
Hb 13.1 130-170 g/L (Male) 120-160 g/L (Female)
Platelets count 65 150-400 x 109/L WBC 10.3 4.5-10.5 x 109/L Creatinine 88 44-115 umol/L
APTT 36 30-40 sec
INR 1.1 0.8-1.2
Prothrombin time 11 10-13 sec
Which of the following is the most likely cause of his thrombocytopenia?
A. Immune thrombocytopenia (ITP)
B. Heparin induced thrombocytopenia (HIT)
C. Thrombotic thrombocytopenic purpura (TTP)
D. Disseminated intravascular coagulopathy (DIC)

Answer: B
A 38-year-old woman has fever and headache for 2 days. Peripheral smear done (see image and lab results).
Heart rate 110 /min
Temperature 38.9 °C
Test Result Normal Values
Hb 112 130-170 g/L (Male)
120-160 g/L (Female)
Reticulocyte 3.2 0.2-1.2 %
Platelets count 123 150-400 x 109/L Fibrinogen 1.1 2-4 g/L
APTT 68 30-40 sec
INR 1.8 0.8-1.2
Prothrombin time 24 10-13 sec
Direct bilirubin 24.5 1.5-6.5 mol/L
Total bilirubin 48.3 3.5-16.5 mol/L
Lactate dehydrogenase 390 60-160 IU/L
Which of the following is the most likely diagnosis?
A. Immune thrombocytopenia (ITP)
B. Heparin induced thrombocytopenia (HIT)
C. Thrombotic thrombocytopenic purpura (TTP)
D. Disseminated intravascular coagulopathy (DIC)

Answer: D

A 62-year-old woman has atrial fibrillation for which she has started on warfarin. Few weeks after, she fell down and developed a subdural hematoma that requires evacuation (see lab result).
Test Result Normal Values
INR 3.9 0.8-1.2
Which of the following is the most appropriate management?
A. Vitamin K
B. Fresh frozen plasma
C. Activated factor VII (fviia)
D. Fresh frozen plasma and vitamin K

Answer: D
An 18-year-old-boy is receiving chemotherapy for acute lymphoblastic leukemia. On day 17 of chemotherapy, he has fever. He has no focus of infection (see lab results). Heart rate 110 /min, Temperature 38.9 °C,
Test Result Normal Values
Platelets count 25 150-400 x 109/L WBC 0.6 4.5-10.5 x 109/L Neutrophils 60 40-60% Lymphocytes 20 20-40 %
Which of the following is the most appropriate management?
A. Send blood, urine, sputum cultures
B. Send blood, urine, sputum cultures and start paracetamol
C. Send blood, urine, sputum cultures and start oral broad spectrum antibiotics
D. Send blood, urine, sputum cultures and start intravenous broad spectrum antibiotics

Answer: D
A 43-year-old-woman has fever, night sweats and weight loss. She has lost 12 Kg in the last 3 month (unintentional). She had no contact with a tuberculosis patient. On examination, she has bilateral cervical lymph nodes, largest 3 cm X2 cm. There is also inguinal lymphadenopathy
and the spleen is palpable 3 fingers below the costal margin.
Which of the following is the most appropriate next step in management?
A. Fine needle aspiration of the lymph node
B. Excisional biopsy of the lymph node
C. Bone marrow biopsy
D. Splenectomy

Answer: B
A 63-year-old man has chronic lymphocytic leukemia started on chemotherapy. His initial lymphocyte count was 73,000. 2 days after chemotherapy, he presents with fatigue and dry mouth (see lab results).
Test Result Normal Values
WBC 5.7 4.5-10.5 x 109/L
Lymphocytes 60 20-40 %
Potassium 5.5 3.5-5.1 mmol/L
Phosphate, inorganic 1.48 0.82-1.51 mmol/L Creatinine 139 44-115 mol/L
Uric acid 620 200-410 umol/L (Male)
140-360 mol/L (Female)
Which of the following electrolyte abnormality is associated with the above condition?
A. Hypocalcimia
B. Hypercalcimia
C. Hyponatremia
D. Hypernatremia

Answer: A

A 59-year-old man has unsteady gait. On examination, he has decreased proprioception in the lower limbs with positive planter reflexes (positive Babinski) and absent ankle and knee's reflexes. He has anemia. Peripheral smear done (see image).
Which of the following is the most likely diagnosis?
A. Alcoholism
B. Iron deficiency
C. B12 deficiency
D. Hypothyroidism

Answer: C
A 91-year-old man with poor performance status, dementia, and multiple medical problems who has been diagnosed with an adenocarcinoma of an unknown primary site and has upper abdominal lymphadenopathy, liver lesions, and pulmonary nodules. Which of the following is the most appropriate management?
A. Exploratory laparotomy with biliary bypass procedure
B. Excisional biopsy to identify the primary site
C. Palliative systematic chemotherapy
D. Comfort care

Answer: D
A 75-year-old man has lung cancer comes to the clinic to manage his back pain. A month ago, his oncologist told him that the prognosis was poor and that he is unlikely to survive for more than a year (see lab result).
Test Result Normal Values
Calcium ionised 3.2 1.1-1.3 mol/L
Which of the following is the most appropriate next step in management?
A. Intrvenous isotonic saline
B. Subcutaneous morphine
C. Oral alendronate
D. Oral furosemide

Answer: A

Endocrinology

A 65-year-old patient presents with a 3-day history of seizures (see lab results).

Test Result Normal Values
Sodium 112 134-146 mmol/L
Potassium 3.9 3.5-5.1 mmol/L
Urea 3.4 2.75-7.4 mmol/L
Serum osmolality 240 280-300 mmol/kg

24 Urine Analysis:
Test Result Normal Values
Osmolality 860 280-910 mOsm/Kg H20
Which of the following is the most likely diagnosis?
A. Conn's syndrome
B. Addison's disease
C. Cushing's syndrome
D. Inappropriate antidiuretic hormone secretion

Answer: D

Explanation: conn’s syndrome & cushing’s present with hypo K.

A 36-year-old patient presents with a 6-month history of decreased libido and weak erections (see lab results).

Test Result Normal Values
Prolactin 2100 < 870 pmol/L
Follicle-stimulating hormone 12 5-20 IU/L
Luteinizing hormone 18 5-22 U/L

Which of the following is the next most useful investigation?
A. Abdominal/pelvic CT scan
B. Fasting blood sugar
C. Testosterone level
D. Brain MRI

Answer: D

A 40-year-old man presents to the doctor with a history of central weight gain, change in facial appearance, poor libido, easy bruising, and back pain. Examination reveals pigmentation of the skin, a buffalo hump, edema, proximal muscle wasting, and an elevated blood pressure.

Which of the following is the most likely diagnosis?
A. Cushing's disease
B. Adrenal adenomas
C. Adrenal carcinomas

Answer: A

Explanation: Pigmentation of the skin indicates a high level of POMC, a precursor for ACTH and melatonin. This phenomenon is seen in conditions like Addison's disease, where elevated ACTH levels contribute to skin pigmentation. Therefore, if the fourth option is Cushing syndrome, don’t choose it.

Cushing disease & addison Sx = high ACTH
Cushing syndrome = Low ACTH

A 30-year-old man presents with a history of increased urine output, especially during the night and increased thirst for the past 2 months (see report).

Water deprivation test:
Serum osmolality above normal without adequate concentration of urine osmolality; no response to desmopressin administration.

Which of the following is the most likely diagnosis?
A. Nephrogenic diabetes insipidus
B. Surreptitious diuretic abuse
C. Central diabetes insipidus
D. Hysterical polydipsia

Answer: C

Explanation: Wrong answer, should be nephrogenic since there is no response to the ADH analog!!

A 60-year-old patient complains of irritability, heat intolerance and recent onset of heart palpitations. The neck examination is normal. (see lab results).

Test Result Normal Values
Thyroid-Stimulating Hormone 0.1 0.4 - 5.0 U/L
Thyroxine (T4 free) 30.4 8.5 - 15.2 pmol/L
Triiodothyronine (T3 free) 9.5.3.5 - 6.5 pmol/L

Which of the following is the most likely diagnosis?
A. Sick euthyroid syndrome
B. Primary hypothyroidism
C. Thyrotoxicosis
D. T3 Toxicosis

Answer: C

A patient with hypothyroidism, on 0.025 mg of L-thyroxin, complains of fatigue, cold intolerance, and weight gain (see lab result).

Blood Pressure 120/60 mmH
Heart Rate 57 /min
Respiratory Rate 16 /min
Temperature 36.6° C

Test Result Normal Value
TSH 19.9 mU/L 0.4-6.5 mU/L

Which of the following is the most appropriate treatment?

A. Increase L-thyroxin and check TSH in 6 weeks
B. Increase L-thyroxin and check TSH in 3 weeks
C. Decrease L-thyroxin and check TSH in 6 weeks
D. Decrease L-thyroxin and check TSH in 3 weeks

Answer: A

A 35-year-old man complains of discomfort in both breasts with milk production when expressed manually for a month. It has been associated with headache and low sexual interest. He has been married for 3 years and his wife has not been able to become pregnant. On examination, he has mild acne on his face (see lab results).
Weight 97 kg
Height 173 cm

Test Result Normal Values
Thyroxine (T4 free) 8.4 8.5 - 15.2 pmol/L
Thyroid-Stimulating Hormone 6.8 0.4 - 5.0 U/L
Follicle-stimulating hormone 5.8 5-20 IU/L
Luteinizing hormone 7.1 5-22 U/L
Prolactin 2350 < 870 pmol/L

which of the following is the most likely visual field defect?
A. Junctional scotoma
B. Bitemporal hemianopia
C. Superior quadrantanopia
D. Homonymous hemianopia

Answer: B

A 32-year-old woman complains that her menstrual cycle has stopped for the last 9 months. She also has discomfort in both breasts with milk production when expressed manually. Her sexual drive has been decreased for the last year. She has been married for 3 years and failed to become pregnant. The physical examination is normal and pregnancy test is negative (see lab results).

Weight 81 kg
Height 165 cm

Test Result Normal Values
Thyroxine (T4 free) 9.1 8.5 - 15.2 pmol/L
Thyroid-Stimulating Hormone 5.2 0.4 - 5.0 yU/L
Follicle-stimulating hormone 5 5-20 IU/L
Luteinizing hormone 10 5-22 U/L
Prolactin 2450 < 870 pmol/L

Which of the following is the most appropriate investigation?
A. Brain MRI
B. Neck ultrasound
C. Pelvis ultrasound
D. Breast Mammogram

Answer: A

A 44-year-old woman presents with a neck swelling for the last 2 months. It is associated with frequent bowel movement, increased appetite and weight loss. Over the last 6 months, her period has stopped and she has noticed a milky discharge from her nipples. On physical examination, there is a small diffuse goiter. Pregnancy test is negative (see lab results).

Blood pressure 120/75 mmH
Heart rate 80 /min
Respiratory rate 17 /min
Temperature 37.1° C
Test Result Normal Values
Thyroxine (T4 free) 19.5 8.5 - 15.2 pmol/L
Thyroid-Stimulating Hormone 8.7 0.4-5.0 mU/L

Which of the following is the most appropriate radiological imaging?

A. Pituitary MRI
B. Thyroid ultrasound
C. Ovarian ultrasound
D. Breast mammogram

Answer: A

1. A 50-year-old woman with major depression complains of excessive urination and thirst for 2 months. She had a right mastectomy 1 year ago. 3 months before, lung metastases were diagnosed. The physical examination is normal except for the surgical scar (see lab results).

Blood pressure 115/75 mmHg
Heart rate 80/min
Respiratory rate 16 /min
Temperature 37.1° C

Test Result Normal Values
Sodium 150 134-146 mmol/L
Potassium 3.8 3.5-5.1 mmol/L
Urea 6.7 2.75-7.4 mmol/L
Creatinine 111 44-115 mol/L
Fasting blood sugar 6.8 3.5-6.5 mmol/L

24 Urine Analysis
Test Result Normal Values
Osmolality 110 280-910 mOsm/Kg H20

Which of the following is the most likely diagnosis?
A. Diabetes insipidus
B. Adipsic hypernatremia
C. Psychogenic polydipsia

Answer: A

Explanation:
Diabetes insipidus: Hypernatremia >> (water wasting)
SIADH & Psychogenic polydipsia = hyponatremia >> (No water wasting or too much water consuming)

2. A 28-year-old woman presents with a painful neck swelling for the last week. The swelling has been increasing over time. She has a sore throat and difficulty in swallowing. She has frequent bowel motions. On a physical examination, there is a diffuse tender anterior neck swelling, and both hands show a fine tremor (see lab results).

Blood pressure 140/90 mmHg
Heart rate 105 /min
Respiratory rate 18 /min
Temperature 38.1° C

Test Result Normal Values
Thyroxine (T4 free) 26.4 8.5 - 15.2 pmol/L
Thyroid-Stimulating Hormone 0.03 0.4 - 5.0 mU/L
ESR 58 3-15 mm/h

Which of the following is the most appropriate treatment?
A. Prednisone
B. Methimazole
C. Propylthiouracil
D. Radioactive iodine

Answer: A

3. A 45-year-old woman comes to the clinic for a follow-up. She was started on levothyroxine for treatment of hypothyroidism 2 weeks before (see lab result).

Test Result Normal Value
TSH 7.5 0.4-6.5 mU/L

Which of the following would be the best treatment plan?
A. Same dose, check TSH level in 1 to 2 months
B. Reduce dose, check TSH level in 1 to 2 months
C. Reduce dose, check TSH level in 48 to 72 hours
D. Stop treatment until TSH normal, and then restart

Answer: A

4. A 23-year-old pregnant patient complains of irritability and sweating. On examination, she is thin and appears anxious. Her thyroid gland is nodular, diffusely enlarged, but not tender. She has brisk deep tendon reflexes. Her uterus is appropriate for gestational age (see lab results and report).

Blood pressure 115/70 mmH
Heart rate 110 /min
Respiratory rate 12 /min
Temperature 37.1 °C

Test Result Normal Values
Thyroid-Stimulating Hormone 0.1 0.4 - 5.0 mU/L
Antithyroid Antibody negative negative

Thyroid ultrasound:
Diffuse nodular enlargement. No cyst identified.

Which of the following is the best next step in management?
A. Thyroid biopsy
B. Radioactive iodine
C. Subtotal thyroidectomy
D. Antithyroid medication

Answer: D

5. A patient has presented with a 2 cm solid thyroid nodule on the Ultrasound. Which of the following is the best next step in the management?
A. Repeat the ultrasound after 6 months
B. Thyroid scan with Iodine
C. Fine needle aspiration
D. CT neck

Answer: C

A 60-year-old diabetic patient presents for follow-up. Currently, the patient takes metformin that maintains fasting blood glucose levels lower than 5.5 mol/L with HbA1c level of 5.8%. Physical examination is normal, including a lower extremity neurologic examination. The patient has regular eye exams that show no signs of retinopathy.

Blood pressure 149/90 mmHg (three readings)
Heart rate 76 /min
Respiratory rate 18 /min
Temperature 36.6° C

Which of the following is the most appropriate management?
A. No change in therapy
B. Add ACE inhibitor
C. Add beta-blocker
D. Add sulfonylurea

Answer: B

A 25-year-old patient is diagnosed with diabetic ketoacidosis (see lab results).
Blood pressure 80/50 mmHg
Heart rate 140 /min
Respiratory rate 26 /min
Temperature 36.6° C
Test Result Normal Values
Random Glucose 52.7 3.9 - 5.5 mol/L
Potassium 6 3.5-5.1 mmol/L
Bicarbonate 5 21-28 mmol/L

Which of the following is the best initial management?

A. Dopamine drip to raise blood pressure above 90 mm/Hg
B. 2L of normal saline with Kcl 20 mEq/L intravenously
C. 20-unit regular insulin intravenously, followed by 5% dextrose in water at 200 mL/h
D. Normal saline 2 L as an intravenous bolus and insulin intravenous infusion at 0.1 U/kg/hr

Answer: D

A 55-year-old man with a family history of type 2 diabetes undergoes screening (see lab results).

Blood pressure 130/85 mmH
Heart rate 80 /min
Height 185 cm
Weight 91 kg

Test Result Normal Values
Glucose, fasting 7.4 3.5-6.5 mmol/L
Hemoglobin A1C 6.3 4.7-5.6%

Which of the following should be recommended?

A. Random plasma glucose
B. Haemoglobin A1C after 6 weeks
C. Fasting plasma glucose after 3 months
D. 2-hour 75 g oral glucose tolerance test

Answer: D

Explanation: for asymptomatic patients two abnormal tests must be obtained to diagnose DM 2, in this case the Hg A1c is in the prediabetic range. So we should perform 2h OGTT.

A 55-year-old man with a family history of type 2 diabetes undergoes screening. He gives the history of polyuria (see lab results).
Blood pressure 130/85 mmHg
Heart rate 80 /min
Height 185 cm
Weight 91 kg

Test Result Normal Values
Glucose, fasting 7.0 3.5-5.5 mmol/L
Hemoglobin A1C 7.1 5.7-6.4%

Which of the following is the most likely diagnosis?
A. Prediabetes
B. Type 2 diabetes
C. Impaired glucose tolerance
D. Mature onset diabetes of youth

Answer: B

Explanation: explained above, but be careful and memorise the normal, prediabetes, diabetes value range because they listed different values range compared to the previous Qs!!

A 45-year-old man is diagnosed with type 2 diabetes 3 months ago, and started on metformin 500 mg TDS, comes for his first follow-up (see lab results).

Blood pressure 125/75 mmHg
Heart rate 80 /min

Test Result Normal Values
Glucose, fasting 7.4 3.5-6.5 mmol/L
Creatinine 125 44-115 mol/L
HbA1C 6.8 4.7-5.6 %

Which of the following investigations is recommended to assess his renal function annually?
A. Serum creatinine
B. Urine microalbumin
C. Urine albumin/creatinine ratio
D. Estimated glomerular filtration rate

Answer: C

A 55-year-old woman has had type 2 diabetes for the past 20 years. She is on an 1800 kcal/ day diet and takes metformin 850 mg 3 times a day (see lab results).

Blood pressure 130/85 mmHg
Heart rate 80 /min

Test Result Normal Values
Glucose, fasting 7.1 3.5-6.5 mmol/L
HbA1C 7.8 4.7-5.6 %

What should be the target Hemoglobin A1C percentage?
A. <5.0
B. <5.5

C. <6.0
D. <7.0

Answer: D

Which of the following findings in the urine would indicate diabetic nephropathy in type 2 diabetes?

A. Hematuria
B. Proteinuria
C. Hyaline casts
D. Red blood cell casts

Answer: B

A 28-year-old woman developed gestational diabetes during her second pregnancy. She gave birth to a healthy baby 6 months ago. There is a strong family history of type 2 diabetes and now wants to be evaluated for diabetes (see lab results).
Height 166 cm
Weight 91 kg

Test Result Normal Values
Glucose, fasting 6.5 3.5-6.5 mmol/L
HbA1C 5.5 4.7-5.6 %

Which of the following is the most appropriate therapy?
A. Insulin
B. Acarbose
C. Glyburide
D. Metformin

Answer: D

A 38-year-old man is unable to lose weight despite being on an intensive lifestyle management plan of a low-calorie diet and exercise (see lab results).

Blood pressure 145/95 mmHg
Heart rate 80 /min
Height 173 cm
Weight 125 kg

Test Result Normal Values
Glucose, fasting 7.1 3.5-6.5 mol/L
HbA1C 8.1 4.7-5.6 %

Which of the following is the most appropriate advice to reduce weight?
A. Bariatric surgery
B. Intensify exercise
C. Further reduce calories in diet
D. Weight-reducing medications

Answer: A
A 66-year-old woman has recently been found to have high blood pressure. She had a cholecystectomy 15 years ago and has osteoarthritis in the knees for which she uses ibuprofen (see lab results).

Blood pressure 160/95 mmHg
Heart rate 84 /min

Test Result Normal Values
Creatinine 138 44-115 mol/L
Sodium 142 134-146 mmol/L
Potassium 2.9 3.5-5.1 mmol/L
Chloride 100 97-108 mmol/L
Bicarbonate 31 21-28 mmol/L

Which of the following is the most likely Diagnosis?
A. Essential hypertension
B. Pheochromocytoma
C. Primary hyperaldosteronism
D. NSAIDs induced hypertension

Answer: C

Explanation: NSAIDs induced hypertension cause Hyperkalemia due to reduction in renal plasma flow. + Normal Na value due to the aldosterone escape phenomenon of conn syndrome “primary hyperaldosteronism”
A 28-year-old patient presents with a history of 10 days of palpitations, sweating and neck discomfort (see lab results).
Blood pressure 110/70 mmHg
Heart rate 116 /min
Respiratory rate 18 /min
Temperature 37.7° C
Oxygen saturation 95% on room air

Test Result Normal Values
Thyroxine (T4 free) 76.5 8.5 - 15.2 pmol/L
Thyroid-Stimulating Hormone 0.15 0.4 - 5.0 U/L
WBC 5.2 4.5-10.5 x 109/L
ESR 73 2-10 mm/h

Which of the following is the most likely diagnosis?
A. Graves' disease
B. Subacute thyroiditis
C. Hashimoto's thyroiditis
D. Multinodular toxic goiter

Answer: B

A 72-year-old patient is evaluated for mid-thoracic back pain. The patient does not smoke cigarettes and rarely drinks alcohol. The only medications taken are aspirin, calcium supplements, and a multivitamin. Very few dairy products are consumed. On physical examination, the patient has a mild dorsal kyphosis with mild tenderness over the mid and lower thoracic regions (see lab results and report).

Test Result Normal Values
Thyroid-Stimulating Hormone 8.2 0.4 - 5.0 U/L
Calcium 2.45 2.15-2.62 mmol/L
Phosphate 1.26 0.82-1.51 mmol/L
Albumin 35 34-56 g/L
Alkaline phosphatase 290 39-117 U/L
T-Score (Bone mineral Spine L1-4) -1.9 > -1
T-Score (Bone mineral Total hip) -2.1 > -1

24 Urine Analysis:
Test Result Normal Values

Urinary calcium 5.1 < 7.5 mmols /day
X-ray Spine:
Compression fractures at T8, L2 and L3.

Which of the following is the most likely diagnosis?
A. Paget's disease
B. Osteomalacia
C. Osteoporosis
D. Osteopenia

Answer: C

A 30-year-old woman with controlled hypothyroidism comes for follow-up. She had been sir thyroxin 175 mcg, but the dose was increased 3 months ago to 200 mcg. Physical examination is normal (see lab results).

Blood pressure 130/80 mmHg
Heart rate 80 /min
Respiratory rate 18 /min
Temperature 37.2° C

Test Result Normal Values
Thyroxine (T4 free) 12.4 8.5-15.2 pmol/L
Thyroid-Stimulating Hormone 17.2 0.4-5.0 U/L

What is the most likely explanation for these lab findings?
A. Small dose
B. Ectopic thyroid
C. Poor medication adherence
D. Secondary hypothyroidism

Answer: C

A 45-year-old man was being investigated for hypertension. He was prescribed a drug for his lipid disorder but returned 1 week later with intense face flushing after taking the tablet prescribed. He was then advised to take a full dose of aspirin 3 times daily instead of a low dose once daily. This gave complete relief (see lab results).
Tests at first visit

Test Result Normal Values
Cholesterol (HDL) 0.72 <1.03 mol/L
Cholesterol (LDL) 5.30 <3.36 mol/L
Total Cholesterol 6.12 <5.1 mol/L
Triglycerides 2.20 <2.16 mmol/L
Alanine aminotransferase 35 5-40 IU/L
Creatinine kinase 205 24-195 IU/L

Which of the following drugs was he prescribed?
A. Niacin
B. Clofibrate
C. Atorvastatin
D. Cholestyramine

Answer: A

Explanation: Niacin causes vasodilation (flushing) Via prostaglandin, that is why we give aspirin because it blocks PG production.

35-year-old women comes to the clinic complaining of infertility for 3 years. She notices excessive breast milk production for 4 months. The physical examination is unremarkable (see lab results and report).

Blood pressure 130/80 mmHg
Heart rate 80 /min
Respiratory rate 18 /min
Temperature 37.2° C

Test Result Normal Values
Thyroxine (T4 free) 12.4 8.5 - 15.2 pmol/L
Thyroid-Stimulating Hormone 1.2 0.4 - 5.0 mU/L
Prolactin 1452 < 870 pmol/L

Brain MRI:
Well-defined pituitary mass about 0.7 cm, which represents an adenoma.

Which of the following is the most appropriate treatment?
A. Surgery
B. Cabergoline
C. Observation
D. Radiotherapy

Answer: B

Explanation: Dopamine agonist is the first line treatment for prolactinoma + this is microadenoma (less than 1 cm)

A 69-year-old man is evaluated during a routine examination. He reports his blood pressure to be around 150/84 mmHg on several occasions. He is known to have mild benign prostatic hyperplasia, and hyperlipidemia for which he is taking simvastatin. Cardiac examination is normal, and the remainder of the physical examination is unremarkable. Laboratory studies show a normal chemistry panel; a urine dipstick demonstrates no blood or protein. An electrocardiogram shows no evidence of left ventricular hypertrophy
Blood pressure 147/86 mmHg Heart rate 80 /min Respiratory rate 14 /min Temperature 36.6 °C
Oxygen saturation 95 %
BMI 31 kg/m2
Which of the following is the most appropriate next step in management?
A. Repeat blood pressure determination in 4 months
B. Lifestyle modifications and weight loss
C. Start Amlodipine
D. Start B-blockers

Answer: B
A 60-year-old man is evaluated during a follow-up visit for management of newly diagnosed hypertension and diabetes mellitus started on metformin and gliclazide but not started on antihypertensive therapy. Physical examination is unremarkable (see lab results).
Blood pressure 155/95 mmH
Heart rate 80 /min
BMI 32.5 kg/m2
Test Result Normal Values
Blood urea nitrogen 5 2.8 to 8.9 mmol/L
Creatinine 100 44-115 mol/L
Random Glucose 6 3.9-5.5 mmol/L
HA1C 6.9 4.7-5.6 %

Protein Present absent
Glucose Absent absent
Which of the following is the most appropriate antihypertensive treatment for this patient?
A. Lisinopril
B. Atenolol
C. Amlodipine
D. Hydrochlorothiazide

Answer: A
A 20-year-old woman is diagnosed of type 1 diabetes mellitus after presenting with diabetic ketoacidosis. Her blood sugars are now stable and she is ready for discharge. Which of the following is the best insulin regime she should be offered as outpatient?
A. Twice daily NPH insulin
B. Once daily mixtard insulin
C. Basal-bolus insulin regimen with once-daily insulin glargine
D. Rapid-acting insulin analogue before each meal with no longer acting insulin

Answer: C
A 70-year-old woman known to have type 2 diabetes mellitus. She is currently taking metformin 1g bid physical examination is unremarkable (see lab results).
Test Result Normal Values
Creatinine 80 44-115 mol/L
Glucose, fasting 6 3.5-6.5 mmol/L
Random Glucose 7.5 3.9-5.5 mmol/L
HbA1C 6.9 4.7-5.6 %
Which of the following is the most appropriate next step in management plan?
A. Increase dose of metformin
B. Add glimepiride
C. Add Sitagliptin
D. No changes

Answer: D
A 30-year-old woman presents with secondary amenorrhea after stopping the oral contraceptive pill 6 months ago. She gets regular headaches and visual disturbance. On examination, she has visual fields showed bilateral defects in the upper outer quadrants (see lab results and report).
Test Result Normal Values
Thyroid-Stimulating Hormone 0.1 0.4 - 5.0 wU/mL. Thyroxine (T4 free serum) 7.5 8.5 - 15.2 pmol/L
Follicle-stimulating hormone 0.2 5-20 IU/L (Follicular phase) 15-35 IU/L (Mid cycle peak)
5-15 IU/L (Luteal phase)
50-100 IU/L (Postmenopausal women)
5-20 IU/L (men)
Luteinizing hormone 0.3 3-15 U/L (Male)
5-22 U/L (Follicular phase)
5-22 U/L (Luteal phase)
30-250 U/L (ovulatory phase)
> 30 U/L (Postmenopausal)
Prolactin 950 < 652 pmol/L (Male)
< 870 pmol/L (Female)
Cortisol 8 a.m. 30 138-635 mol/L
Pituitary MRI: 4 cm pituitary mass compressing optic chiasm. Which of the following is the definitive treatment option?
A. Octreotide
B. Bromocriptine

C. Stereotactic radiotherapy
D. Trans-sphenoidal surgery

Answer: D
A 32-year-old woman was admitted 2 days ago complaining of feeling unwell with vomiting, postural dizziness and weight loss. Synacthen test confirmed the diagnosis with Addison's disease. She was started on treatment with hydrocortisone 3 times a day, as well as fludrocortisone on a daily basis. She has been taking hydrocortisone 10mg at 9:00 and then 5mg at 12:00 and 15:00. On discharge, she informs the medical team that she sometimes does shift-work.
What is the most appropriate advice to give regarding steroid dosing for night shifts?
A. On shift days omit taking steroids
B. Take doses at 09:00, 12:00 and 15:00 regardless of shift patterns
C. On shift days take doses at 09:00, but then take other doses at night
D. Adjust to take first dose at waking, then doses at 3 and 6 hours from starting

Answer: D
A 34-year-old woman presents with increasing symptoms of heat intolerance, diarrhea and anxiety over the past couple of weeks. She is 28 weeks pregnant with her first baby and has a past medical history of hyperthyroidism, currently treated with 15 mg carbimazole. She has no other past medical history. She does not smoke, drink alcohol nor take any other drugs (see lab results).
Blood pressure 110/70 mmHg Heart rate 120 /min Respiratory rate 18 /min Temperature 37.6 °C
Oxygen saturation 95 %
Test Result Normal Values
Thyroid-Stimulating Hormone 0.1 0.4 - 5.0 U/mL
Triiodothyronine (T3 free serum) 12 3.5 - 6.5 pmol/L
Thyroxine (T4 free serum) 25 8.5 - 15.2 pmol/L
Which of the following is the most appropriate next step in management?
A. Refer patient for immediate caesarean section
B. Increase carbimazole dose to 20mg once daily
C. Switch carbimazole to propyithiouracil
D. Refer for a thyroidectomy

Answer: B
A 40-year-old woman is 6 weeks pregnant, presented to Endocrinology Clinic with diarrhea, heat intolerance and palpitation for the last month. She has no past medical history but she has strong family history of thyroid disease (see lab results).
Heart rate 90 /min
Respiratory rate 18 /min
Temperature 37.2 °C
Test Result Normal Values
TSH 0.001 0.4 - 5.0 yU/mL
Free T4 45 8.5 - 15.2 pmol/L
TSH receptor antibodies Positive negative
Which of the following is the most appropriate initial management?
A. Propranolol
B. Carbimazole
C. Propyithiouracil
D. Total thyroidectomy

Answer: C; First line treatment of graves in pregnant women.
A 45-year-old woman is referred to the Endocrinology Clinic due to abnormal thyroid function tests. Otherwise, she is asymptomatic. On examination, there is no evidence of goiter (see lab results).
Blood pressure 139/85 mmHg

Heart rate 66 /min
Temperature 36.6 °C
BMI 29 kg/m2
Test Result Normal Values
Thyroid-Stimulating Hormone 15 0.4 - 5.0 pU/mL.
Thyroxine (T4 free serum) 6.5 8.5 - 15.2 pmol/L
Which of the following is the most appropriate management plan?
A. Reassure and discharge
B. Arrange for US of the neck
C. Repeat blood tests in 6 months
D. Start levothyroxine replacement

Answer: D; TSH is more than 10
A 27-year-old woman presents with 2-day history of generally unwell, nausea, vomiting and lethargy. She is a known type 1 diabetic using a long-standing subcutaneous insulin regime. She was started on treatment for diabetic ketoacidosis with intravenous fluids and fixed rate insulin. Her blood sugar is reviewed 3 hours after treatment was initiated. On admission, blood tests are done (see lab results).
Test Result Normal Values
ABG HCO3- 8 22-28 mmol/L
pH 7.2 7.36-7.45
Random Glucose 32 3.9-5.5 mmol/L
What should be the aim in managing hyperglycemia in a diabetic ketoacidosis patient?
A. Reduce blood glucose to under 14 mol/I as quickly as possible
B. Reduce blood glucose by 3mmol/I per hour
C. Reduce blood glucose by 6 mmol/ per hour
D. Aim blood glucose above 18 mmol/

Answer: B
A 20-year-old man with type 1 diabetes presents to the Emergency Department with vomiting and abdominal pain. He has been suffering from diarrhea for 2 days and then became severely dehydrated and started to vomit. He has not had his insulin for the last 24 hours. Normally, he has a basal bolus regime with Glargine as a long-acting insulin, and Humalog as short-acting. On admission, he is found to have acidosis, elevated serum ketones and elevated blood glucose, and was diagnosed with diabetic ketoacidosis. He is started on IV fluid
What insulin should he be prescribed now?
A. Fixed rate IV insulin as well as long-acting insulin
B. Sliding scale SC insulin
C. Fixed rate IV insulin
D. Insulin mixtard bid

Answer: A
A 55-year-old man presents with excessive sweating and headache. An oral glucose tolerance test and MRI brain were ordered to confirm the diagnosis of acromegaly
What other investigation will be required in the future?
A. Transesophageal echocardiography
B. No further investigations required
C. Colonoscopy
D. CT abdomin

Answer: C
A 55-year-old man is referred to the Diabetes Clinic after presenting with polyuria and polydipsia with strong family history of type 2 diabetes mellitus. Physical examination is unremarkable (see lab

results).
Blood pressure 136/85 mmHg
Heart rate 76 /min
Temperature 36.6 °C
BMI 34 kg/m2
Test Result Normal Values
Glucose, fasting 9 3.5-6.5 mmol/L
HA1C 7.8 4.7-5.6 %
Which of the following is the most appropriate first-line treatment?
A. Dietary advice, exercise and metformin
B. Dietary advice and exercise
C. Sulphonylurea
D. Exenatide

Answer: A
A 43-year-old man comes to the clinic for a follow-up. He has been recently diagnosed with type 2 diabetes and has been following a plan of lifestyle measures to improve his diet and increase his level of exercise (see lab result).
Blood pressure 129/80 mmHg
BMI 32 kq/m2
Test Result Normal Values
HA1C 6.9 4.7-5.6 %
Which of the following is the most appropriate treatment option now?
A. Insulin
B. Metformin
C. Sulphonylurea
D. Further diet and exercise

Answer: B
A 40-year-old man was diagnosed with primary hyperparathyroidism based on the biochemistry results (elevated calcium, low phosphate and raised PTH levels) and parathyroid ultrasound images, which showed left lower parathyroid adenoma
Which of the following is considered a valid indication for consideration for elective parathyroidectomy?
A. Age >50 years
B. Evidence of osteoporosis
C. Normal renal function test
D. High parathyroid hormone level

Answer: B
An 18-year-old woman presented to the Emergency Room with spasms of hands along with twitching around the mouth. Her general physical and systemic examination was unremarkable except for pallor, dental caries and small body built. She had prolonged QT interval on ECG
(see lab results).
Test Result Normal Values
Calcium 1.6 2.15-2.62 mmol/L
Phosphate, inorganic 0.7 0.82-1.51 mmol/L
Parathyroid hormone (intact PTH levels) 4.5 1.1-5.3 pmol/L
25-Hydroxy Vitamin D3 (Women) 10 25-90 mol/L
Which of the following is the most appropriate initial treatment approach?
A. Oral calcium
B. Cholecalciferol
C. Intravenous calcium

D. Oral phosphate supplementation

Answer: C

A 70-year-old man was hospitalized 6 days ago for a 4-day history of acute leg ischemia treated with angioplasty and stenting. He is now asymptomatic. He has a history of stage 3 chronic kidney disease and hypertension that is well-controlled with diltiazem and captopril. Other medications are aspirin and clopidogrel. On physical examination, the patient is febrile and normal blood pressure. The remainder of the examination is unremarkable (see lab results).
Test Result Normal Values
Creatinine 140 44-115 mol/L
Total Cholesterol 5.9 <5.1 mmol/L
LDL (cholesterol) 4.9 <4.0 mmol/L
Which of the following is the most appropriate therapy for secondary prevention of cardiovascular disease in this patient?
A. Niacin
B. Ezitimibe
C. Rosuvastatin
D. No additional treatment

Answer: C

A 43-year-old woman is evaluated during a follow-up visit. She is overweight and has hypertension and type 2 diabetes mellitus, both of which are well controlled. For several years, she has attempted to lose weight through various commercial diets; dietician-monitored, calorie-restricted diets; and physical activity but her weight has remained stable. Head, neck, lung, and heart examinations are normal. Blood pressure 110/70 mmHg
Heart rate 76 /min Respiratory rate 13 /min Temperature 36.6 °C BMI 28,5 kg/m2
In addition to continuing calorie restriction and exercise, which of the following is the most appropriate management to help this patient achieve weight loss with tolerable side effect?
A. Orlistat
B. Lorcaserin
C. Phentermine-topiramate
D. Roux-en-Y gastric bypass

Answer: A; Bariatric Surgery is not indicated for this patient

An 18-year-old woman complains of dizziness feeling light-headed frequently with standing up and weight loss around 5 kg over the last 5 months. She has noticed a recent scar on the back of her hand, which has started to turn very dark.
Blood pressure 90/60 mmHg, Heart rate 99 /min
What is the most appropriate investigation to confirm the diagnosis?
A. Low-dose dexamethasone test
B. Abdominal ultrasound scan
C. Cortisol measurement
D. Synacthen test

Answer: D

A 67-year-old man presents with insomnia, irritability and palpitation for 3 months. He is known to have hypertension, depression and atrial fibrillation. He is on amiodarone, fluoxetine and enalopril. Clinically, he is unremarkable. Blood pressure 130/70 mmHg, Heart rate 76 /min, Oxygen saturation

95 %
Which of the following is the most appropriate next step?
A. Add propranolol
B. Measure thyroxin level and TSH
C. Substitution of antidepressant drug
D. Refer him for psychiatric assessment

Answer: B

A 35-year-old woman 9 weeks pregnant presented with a history of palpitations, heat intolerance and anxiety for several weeks, otherwise she feels well. Her only medication is a prenatal vitamin. On physical examination, she has warm and moist skin with fine tremor. There is no ophthalmology. Examination of the neck shows a diffusely enlarged thyroid with an audible bruit. Cardiopulmonary and abdominal examinations are unremarkable (see lab results).
Blood pressure 120/70 mmH
Heart rate 115 /min
Respiratory rate 18 /min
Temperature 36.6 °C
Oxygen saturation 95 %
BMI 22 kg/m2
Test Result Normal Values
Thyroid-Stimulating Hormone < 0.01 0.4 - 5.0 yU/mL
Triiodothyronine (T3 free serum) 14 3.5 - 6.5 pmol/L
Thyroxine (T4 free serum) 35 8.5 - 15.2 pmol/L
Which of the following is the most appropriate treatment?
A. Observation
B. Propylthiouracil
C. Radioactive iodine
D. Thyroidectomy

Answer: B
A 60-year-old man came for follow-up evaluation of Cushing syndrome. He presented with new. onset diabetes mellitus and a 15.1-kg weight gain over the last 6 months. Medical history is otherwise unremarkable. He is currently taking no medications and has had no exposure to exogenous glucocorticoids in the past year. On physical examination, he has facial plethora, central obesity, and bilateral supraclavicular fat pads. There are purple abdominal striae measuring 1 cm wide and multiple ecchymoses on the extremities. Initial laboratory studies as following and a 24-hour urine free cortisol level was greater than 3 times the upper limit of normal (see lab results).
Blood pressure 165/92 mmHg
Heart rate 88 /min Respiratory rate 15 /min Temperature 36.9 °C BMI 29 kg/m2
Test Result Normal Values
ACTH < 1 2-11 pmol/L
Cortisol 8 a.m. 248 mol/I post 138-635 mol/L
1 mg overnight dexamethasone test
Which of the following is the most appropriate next diagnostic test?
A. Inferior petrosal sinus sampling
B. Late night salivary cortisol level
C. CT scan of the adrenal glands
D. MRI pituitary

Answer:C

A 33-year-old woman is evaluated in the Emergency Department for the sudden severe headache, vomiting and right eye vision loss she has had mild headaches that come and go over the past 2 years and for amenorrhea for the past 2 years. On physical examination, she has loss of vision in her right eye, other cranial nerves are intact. Strength and sensation in all extremities are normal, as are her speech and gait (see reports).
CT of the head: Shows acute pituitary hemorrhage.
Pituitary MRI: Shows a 4.1 x 2.5 x 3.2-cm pituitary mass compressing the optic chiasm and the right cavernous sinus with central hemorrhage.
Blood pressure 120/65 mmH
Heart rate 99 /min
Respiratory rate 15 /min
Temperature 37 °C
BMI 25 kg/m2
Which of the following is the most appropriate immediate management in the Emergency Room?
A. IV hydrocortisone
B. Assess pituitary function
C. Repeat imaging in 2 weeks
D. Urgent transsphenoidal pituitary decompression

Answer: A
A 39-year-old man found to have thyroid nodule during an annual physical examination. He has no personal or family history of any chronic medical disease, and he takes no medications. On physical examination, vital signs are normal. Palpation of the thyroid reveals left non-tender thyroid nodule, the remainder of the gland is unremarkable, and there is no palpable cervical lymphadenopathy. Other physical examination findings are normal (see lab results and report). Test Result Normal Values
Thyroid-Stimulating Hormone 1.7 0.4 - 5.0 wU/mL
Thyroxine (T4 free serum) 13.5 8.5 - 15.2 pmol/L
Ultrasound of the neck: Shows a left 3-cm hypochoic nodule with internal microcalcifications.
Which of the following is the most appropriate next step in management?
A. Fine-needle aspiration of the nodule
B. CT with contrast of the neck
C. Levothyroxine therapy
D. Thyroid scan

Answer: A
A 29-year-old woman is evaluated for a 10-day history of severe neck pain with heat intolerance, palpitations, and insomnia. Medical history is significant only for a viral upper respiratory tract infection 4 weeks ago. She takes no medications. On physical examination, she appears anxious and she has sinus tachycardia. There is no proposis or lid lag. Examination of the thyroid show a normal-sized tender gland. 24-Hour radioactive iodine 5% (low) uptake (see lab results).
Test Result Normal Values
Thyroid-Stimulating Hormone < 0.01 0.4 - 5.0 wU/mL
Triiodothyronine Reverse (RT3) 4.8 0.9 - 2.8 mol/L
Thyroxine (T4 free serum) 34 8.5 - 15.2 pmol/L
Which of the following is the most appropriate treatment?
A. Propranolol
B. Methimazole
C. Propylthiouracil
D. Radioactive iodine

Answer: A

A 70-year-old woman was admitted 1 week ago to the Intensive Care Unite with pneumonia requiring intubation, intravenous fluids, and dopamine support for her blood pressure. On physical examination, her skin is warm and dry, no proposis, examination or the neck shows a normal-sized thyroid without nodules. Cardiovascular examination reveals regular tachycardia otherwise unremarkable systemic examination (see lab results).
Blood pressure 90/57 mmH Heart rate 120 /min Respiratory rate 18 /min Temperature 39 °C
Test Result Normal Values
Thyroid-Stimulating Hormone 0.1 0.4 - 5.0 U/mL
Triiodothyronine (T3 free serum) 2 3.5 - 6.5 pmol/L
Thyroxine (T4 free serum) 11 8.5 - 15.2 pmol/L
Which of the following is the most likely cause of this patients abnormal thyroid function?
A. Grave's disease
B. Subacute thyroiditis
C. Hashimoto thyroiditis
D. Euthyroid sick syndrome

Answer: D
A 49-year-old woman is evaluated because of fatigue and weight gain although she follows a daily diet control and exercises 3 times per week for 45 minutes. She has hypercholesterolemia requiring statin therapy. Her mother had hypothyroidism. On physical examination, she has dry skin, thyroid gland mildly enlarged with a diffusely nodular texture. Laboratory studies are with similar results for TSH and T4 were obtained 4 months ago (see lab results).
Blood pressure 143/93 mmHg Heart rate 66 /min
BMI 29 kg/m2
Test Result Normal Values
Thyroid-Stimulating Hormone 7.0 0.4 - 5.0 pU/mL.
Thyroxine (T4 free serum) 9.5 8.5 - 15.2 pmol/L
Thyroid peroxidase antibody (TPOAb) 150 <35 lU/mL
Which of the following is the most appropriate next step in management?
A. Thyroid ultrasound
B. Start levothyroxine therapy
C. Repeat serum TSH measurement in 12 months
D. Schedule thyroid radioactive iodine uptake and scan

Answer: B
A 29-year-old man is evaluated for a diagnosis of hypercalcemia. 5 days ago, he was treated of a kidney stone in the Emergency Department. Physical examination is unremarkable (see lab results), Blood pressure 140/79 mmHg, Heart rate 85 /min, Respiratory rate 13 /min, BMI 30 kq/m2 Test Result Normal Values
Sodium 139 134-146 mmol/L
Potassium 4.0 3.5-5.1 mmol/L
Chloride 99 97-108 mmol/L
Bicarbonate 23 21-28 mmol/L
Calcium 2.95 2.15-2.62 mol/L
Phosphate, inorganic 0.76 0.82-1.51 mmol/L
Parathyroid hormone (intact PTH levels) 200 1.1-5.3 pmol/L
Blood urea nitrogen 9.5 2.8 to 8.9 mmol/L
Creatinine 125 44-115 mol/L
Which of the following is most likely cause for his hypercalcemia?
A. Milk alkali syndrome
B. Primary hyperparathyroidism
C. Secondary hyperparathyroidism
D. Elevated 1,25-dihydroxyvitamin D levels

Answer: B
A 35-year-old woman is evaluated for amenorrhea, weight gain, facial hair, alopecia, difficulty walking and easily bruising for the last 6 months. Her medical history is significant for psoriasis for which she is **using clobetasol cream (topical steroid).** On physical examination, vital signs all normal, she has round face, hirsutism, mild facial acne, central obesity, and a few wide purple striae on the back of her arms as well as her psoriatic skin lesions (see lab results).
Test Result Normal Values
Sodium 139 134-146 mmol/L
Potassium 3.5 3.5-5.1 mmol/L
Blood urea nitrogen 6.7 2.8 to 8.9 mmol/L Creatinine 96 44-115 mol/L
ACTH 1.4 2-11 pmol/
Cortisol 8 a.m. 55 138-635 mol/L
Which of the following is the most likely diagnosis?
A. Ectopic adrenacorticotropic hormone production
B. Iatrogenic cushing syndrome
C. Adrenocortical carcinoma
D. Cushing syndrome

Answer: B
A 17-year-old man is evaluated in the Emergency Department for polyuria, polydipsia, polyphagia and an unintentional weight loss over the past month. He has abdominal pain and drowsiness for the last 12 hours. His medical and family history are unremarkable. He is not of any medications. On physical examination, he looks sick with dry mucous membrane, tachypnea, and tachycardia with clear chest auscultation. Abdominal examination shows diffuse mild tenderness with normal bowel sounds. Intravenous 0.9% saline is initiated. An electrocardiogram shows sinus tachycardia 120/min. Chest radiograph is normal (see lab results).
Blood pressure 95/53 mmHg
Heart rate 119 /min
Respiratory rate 20 /min
Temperature 37.2 °C
Oxygen saturation 95 %
BMI 19 kg/m2
Test Result Normal Values
Sodium 130 134-146 mmol/L
Potassium 4.4 3.5-5.1 mmol/L
Chloride 96 97-108 mmol/L
Bicarbonate 10 21-28 mmol/L
Blood urea nitrogen 15 2.8 to 8.9 mmol/L
Creatinine 115 44-115 mol/L
Random Glucose 25 3.9-5.5 mmol/L
ABG HCO3- 10 22-28 mmol/L
pH 7.10 7.36-7.45
RBC 5 4.7-6.1 x 1012/L (Male)
4.2-5.4 x 1012/L (Female)
Hb 130 130-170 g/L (Male)
120-160 g/L (Female)
WBC 14 4.5-10.5 x 109/L
Which of the following is the most appropriate next step in the management?
A. Administer intravenous ceftriaxone
B. Initiate intravenous insulin therapy
C. Administer intravenous potassium chloride
D. Administer intravenous sodium bicarbonate

Answer: B
A 45-year-old man is evaluated for recent onset type 2 diabetes mellitus and hypertension for the last 1 year. On physical examination, he has moon like face, dorsocervical fat pad and striae measuring 8 to 12 mm wide are noted on his upper arms and abdomen. Laboratory studies reveals the following as well as 24-Hour urine cortisol excretion which was 5 times upper the normal cut value and CT scan of the abdomen with and without contrast reveals a 5-cm right adrenal mass with contrast washout at 10 minutes is 60% (see lab results).
Blood pressure 162/93 mmHg, Heart rate 88 /min, Respiratory rate 14 /min, Temperature 36.8 °C, Oxygen saturation 96 %
BMI 30 kg/m2
Test Result Normal Values
ACTH 1.3 2-11 pmol/L
Cortisol 8 a.m. 1400 138-635 mol/L
Which of the following is the most appropriate next step in the management?
A. Chemotherapy
B. Surgical excision
C. Mitotane therapy
D. Radiation therapy

Answer: B
A 54-year-old man presents for follow-up evaluation after diagnosed with type 2 diabetes mellitus 12 weeks ago. He was managed with lifestyle modifications and he come back with blood glucose log for the past 2 weeks. On physical examination, there is no evidence of diabetic retinopathy or peripheral neuropathy. Except for his blood glucose parameters, basic laboratory studies obtained at the time of her initial diagnosis were normal (see lab results).
Blood pressure 124/79 mmHg Heart rate 84 /min Respiratory rate 13 /min Temperature 36.9 °C Oxygen saturation 95 %
BMI 27 kg/m2
Initial test before lifestyle modifications: Test Result Normal Values
HA 1C 8.5% 4.7-5.6 %
Blood glucose log for the past 2 weeks: Test Result
preprandial blood glucose 150 to 160 mg/dL. (8.3-8.9 mol/L)
2-hour postprandial blood glucose 190 to 200 mg/dL. (10.5-11.1 mmol/L.)
In addition to lifestyle modifications, which of the following is the most appropriate initial management?
A. Dapagliflozin
B. Metformin
C. Sitagliptin
D. Glipizide

Answer: B
A 37-year-old man with type 1 diabetes mellitus presents to the office with history of prolonged hypoglycemia during intense exercise, despite eating a meal prior to the activity. He is using insulin glargine and insulin aspart with meals (see lab result).
Test Result Normal Value
H6A1C 7.0 4.7-5.6 %
Which of the following is the most appropriate management of this patient's hypoglycemia on the days that he exercises?
A. Decrease meal-time insulin aspart dose prior to exercise, continue insulin glargine dose
B. Increase meal-time protein prior to exercise, continue current insulin doses
C. Switch insulin aspart to a sliding-scale regimen, continue insulin glargine dose
D. Discontinue insulin glargine, continue insulin aspart dose

Answer: A
A 60-year-old man is evaluated because of fatigue, constipation and cold intolerance for the last 8 months. Medical history is significant for Hodgkin lymphoma treated with radiation therapy 3 years ago and there is no family history of thyroid disorders. On physical examination, he has dry skin with mild periorbital edema and normal thyroid exam (see lab results).
Test Result Normal Values
Hb 120 130-170 g/L (Male) 120-160 g/L (Female)
Sodium 129 134-146 mmol/L
Thyroid-Stimulating Hormone 5.0 0.4 - 5.0 U/mL.
Which of the following will establish diagnosis of hypothyroidism?
A. Repeat TSH measurement in 4 weeks
B. Free thyroxine (T4) measurement
C. Thyroid ultrasound
D. Thyroid scan

Answer: B
A 30-year-old woman is admitted to the hospital for an elective adrenalectomy for non-ACTH dependent Cushing syndrome confirmed by a contrast-enhanced adrenal CT scan finding of benign right adrenal adenoma.
Which of the following is the most appropriate perioperative management?
A. Postoperative mitotane
B. Perioperative hydrocortisone
C. Postoperative fludrocortisone
D. Preoperative phenoxybenzamine

Answer: B
A 36-year-old woman is evaluated during her first prenatal visit for Medical history of hypothyroidism, for which she has taken levothyroxine 125 g/d, for the last 2 years. On physical examination, no goiter or cervical lymphadenopathy. Cardiovascular, pulmonary and abdominal examinations are unremarkable (see lab results).
Blood pressure 100/73 mmHg
Heart rate 87 /min
Respiratory rate 13 /min
Temperature 36.5 °C
Oxygen saturation 97 %
BMI 23 kg/m2
Test Result Normal Values
Thyroid-Stimulating Hormone 4.4 0.4 - 5.0 jU/mL
Thyroxine (T4 free serum) 9.5 8.5 - 15.2 pmol/L
Which of the following is the most appropriate treatment of this patient's hypothyroidism now?
A. Increase levothyroxine dose
B. Continue levothyroxine dose
C. Decrease levothyroxine dose
D. Discontinue levothyroxine dose

Answer: A
A 59-year-old man is admitted to the hospital for evaluation of chest pain. His medical history is significant for type 2 diabetes mellitus, coronary artery disease, hypertension, and hyperlipidemia. He manages his diabetes as an outpatient with diet, exercise, and metformin. His other medications are aspirin, metoprolol, atorvastatin, and sublingual nitroglycerin as needed (see lab results).
Test Result Normal Values
HbA1C 8.5 4.7-5.6 %

Plasma Glucose 9.4-14.9 mmol/L
Which of the following is the most appropriate diabetes treatment for this patient while hospitalized?
A. Glipizide
B. Metformin
C. Sliding-scale insulin
D. Basal and prandial insulin

Answer: D
A 23-year-old woman is evaluated after a recent diagnosis of polycystic ovary syndrome. She is concerned about hirsutism and irregular menses. She is not planning for pregnancy at this time and takes no medications. On physical examination, she has mild hirsutism and the remaining of her physical examination, including pelvic examination, are normal.
Blood pressure 120/70 mmH Heart rate 76 /min Respiratory rate 15 /min Temperature 36.7 °C
BMI 36 kg/m2
Which of the following is the most appropriate treatment?
A. Metformin
B. Spironolactone therapy
C. Combined oral contraceptive pills
D. Levonorgestrel intrauterine system

Answer: C
A 61-year-old man with a 20-year history of type 2 diabetes mellitus is evaluated for bilateral burning sensation in his feet for the last 8 months, which worsens at night. His haemoglobin Ac levels have remained less than 7.0% for the last 1 years but were between 9.0% and 10.0% before adding insulin therapy to metformin 1 years ago. His medical history includes coronary artery disease, non-proliferative diabetic retinopathy, hypertension, and hyperlipidemia. Medications are aspart insulin, glargine, aspirin, metoprolol, atorvastatin, and lisinopril. On physical examination, findings are compatible with distal symmetrical polyneuropathy.
Which of the following is the most appropriate management of this patient's neuropathy?
A. Amitriptyline
B. Physiotherapy
C. Tight blood sugar control
D. Vitamin B12 supplements

Answer: A
A 34-year-old woman is evaluated for history of galactorrhea, worsening fatigue, malaise and irregular excessive menses for the last 5 months. She complains of headaches that occur about 2 to 3 times per week. Her physical examination is unremarkable. Pregnancy test was negative (see lab results).
Blood pressure 120/86 mmHg
Heart rate 62 /min
Respiratory rate 13 /min
BMI 28 kg/m2
Test Result Normal Values
Thyroid-Stimulating Hormone 16 0.4 - 5.0 pU/mL
Thyroxine (T4 free serum) 9.2 8.5 - 15.2 pmol/L
Prolactin 1000 < 652 pmol/L (Male)
< 870 pmol/L (Female)
Human chorionic gonadotropin 0.0 < 5 miU/mL
Which of the following is the most appropriate management?
A. Begin cabergoline
B. Begin levothyroxine

C. Order pituitary MRI
D. Observation and retest thyroid function test in 6 months

Answer: B
A 19-year-old woman is evaluated for management of type 1 diabetes. She was diagnosed 4 months ago after presenting to the Emergency Department with diabetic ketoacidosis. Her hemoglobin A1c level at the time of diagnosis was 14.5%. She was discharged from the hospital on a basal and prandial insulin regimen with minor adjustments required as an outpatient. Recently she started to have episodes of both fasting and postprandial hypoglycemia. Medications are insulin glargine and insulin apart. Her systemic examinations are unremarkable (see lab results).
Blood pressure 115/77 mmH Heart rate 80 /min
BMI 22 kg/m2
Test Result Normal Values
HA1C 6.0 4.7-5.6 %
Random Glucose 4.5 3.9-5.5 mmol/L
Which of the following is the most appropriate management for her hypoglycemia?
A. Decrease insulin glargine, decrease insulin aspart
B. Continue same doses of insulin glargine and aspart
C. Discontinue insulin glargine, discontinue insulin aspart
D. Discontinue insulin glargine, change insulin aspart to a sliding-scale regimen

Answer: A
A 44-year-old man is evaluated for an 8-month history of decreased libido and severe erectile dysfunction, including absence of morning erections. He and his wife would like to conceive. His medical history is otherwise unremarkable, and he takes no medications. On physical examination, vital signs are normal with unremarkable clinical examination (see lab results and report).
Test Result Normal Values
Thyroid-Stimulating Hormone 2.5 0.4 - 5.0 wU/mL
Follicle-stimulating hormone 1.2 5-20 IU/L (Follicular phase)
15-35 IU/L (Mid cycle peak)
5-15 IU/L (Luteal phase)
50-100 IU/L (Postmenopausal women)
5-20 IU/L (men)
Testosterone 4.2 10.4-41.6 mol/L (Male)
0.7-2.8 mol/L (Female)
Luteinizing hormone 1.0 3-15 U/L (Male)
5-22 U/L (Follicular phase)
5-22 U/L. (Luteal phase)
30-250 U/L (ovulatory phase)
> 30 U/L (Postmenopausal)
Prolactin 1500 < 652 pmol/L (Male)
< 870 pmol/L (Female)
MRI of the pituitary:
Reveals a 0.8-cm anterior pituitary mass consistent with an adenoma.
Which of the following is the most appropriate treatment?
A. Testosterone replacement therapy
B. Clomiphene citrate
C. Cabergoline
D. Sildenafil

Answer: C
A 79-year-old man was admitted to the Intensive Care Unit with features of septicaemia due to severe urinary tract infection. His blood tests showed thyroid dysfunction that was attributed to a

non-thyroidal iliness or sick euthyroid state.
Which one of the following is the common typical pattern of thyroid dysfunction seen in a sick euthyroid illness?
A. Elevated levels of T3 and T4; low levels of reverse T3
B. Elevated levels of T3, T4, and reverse T3
C. Low levels of T3 and T4; elevated levels of reverse T3
D. Low levels of T3 and T4; normal level of reverse T3

Answer: C
A 24-year-old man presented to thyroid clinic with a 5-month history of feeling unwell, weight loss, anxiety, and panic attacks. He also complained of redness and grittiness of eyes, with painful eye movements. On examination, he had bilateral tremors, proposis, and lid retraction, Which one of the following factors is associated with increased risk for the development of severe Grave's orbitopathy?
A. No response to thionamides
B. Free T4 and T3 levels
C. Male sex
D. Smoking

Answer: D
A 39-year-old man incidentally found to have a 2-cm left adrenal gland mass while she underwent CT of the abdomen to investigate the cause of her pelvic pain.
Which one of the following is the most common cause of an incidentally detected adrenal mass?
A. Sub-clinical cushing's disease
B. Adrenocortical carcinoma
C. Non-functional adenoma
D. Conn's syndrome

Answer: C
A 40-year-old woman was incidentally detected to have a 13-mm pituitary macro-adenoma on a CT of the head arranged to evaluate cause for her recurrent headaches. Apart from the headaches that she had suffered for the last 8 months, she had no history of any significant medical illness and was not on any regular medications. Her general physical and systemic examination was unremarkable. Which of the following is the most appropriate next step in her management?
A. Anterior pituitary hormone profile
B. Discharge from follow-up
C. Neurosurgical referral
D. Insulin tolerance test

Answer: A
A 25-year-old woman complains of dizziness and feeling light-headed. She is a teacher and most frequently experiences this when she stand up in the class. She has never fainted. She lost 5 kg, but attributes this to eating healthy food. She has noticed a recent scar on the back of her hand, which has started to turn very dark as well as generalized skin hyperpigmentation. What is the most appropriate investigation to confirm the diagnosis?
A. Synacthen test (ACTH stimulation test)
B. Abdominal ultrasound (US) scan
C. Low-dose dexamethasone test
D. Am cortisol measurement

Answer: A
Dipeptidyl peptidase inhibitors 4 (DPP4) are novel oral hypoglycaemic agents that inhibit the enzyme DPP4 and prevent breakdown of endogenous incretins, which will improve blood sugar control in

patients with diabetes mellitus type 2.
Which of the following DPP4 inhibitor does not require any dose modification in renal disease?
A. Vildagliptin
B. Linagliptin
C. Saxagliptin
D. Sitagliptin

Answer: B
A 50-year-old woman recently diagnosed with papillary thyroid carcinoma.
Which of the following is an example of positive communication skill while explaining the management option of her disease?
A. Forgetting to say thank you
B. Ask a patient to repeat your instructions
C. Interrupt a patient when she ask questions
D. Inform patients when you cannot spend any more time with them

Answer: B
A 30-year-old woman recently diagnosed with Cushing's syndrome. which of the following will be considered as negative communication skill while explaining for her the course of her disease?
A. Speaking slowly with clear language
B. Looking directly at a patient when speaking
C. Interrupt a patient when he/she does not understand
D. Encouraging the patient to ask questions

Answer: C
At which estimated GFR level, metformin should be stopped in diabetic patients?
A. GFR between 45-60
B. GFR between 30-45
C. GFR less than 30
D. GFR less than 15

Answer: C
Which of the following osteoporosis treatment has FDA approval to treat malignancy induced hypercalcemia?
A. Denosumab
B. Alendronate
C. Ibandronate
D. Recombinant human 1-34 PTH (teriparatide)

Answer: A
A 35-year-old pregnant woman with gestational age of 30-weeks comes for a follow-up. She was diagnosed to have gestational diabetes mellitus (GDM) at week 24, she was maintained on diet control but unfortunately her blood sugar reading between (150-200 mg/di). The physician decided to start insulin but she is totally refusing injections and she asks about any safe oral medication to be used in pregnancy.
Which of the following oral hypoglycemic agents has safety data and recommended to be used in pregnancy?
A. Glipizide
B. Sitagliptin
C. Glyburide
D. Rosiglitazone

Answer: C

A 44-year-old man with a strong family history of type 2 diabetes and cardiovascular disease was recently diagnosed with pre-diabetes. He came for advice regarding healthy eating patterns and physical activity that will help him to prevent type 2 diabetes (see lab result).
BMI 31 kg/m2
Test Result Normal Values
HbA1C 6.1 4.7-5.6 %
In helping this individual to achieve and maintain his target weight loss, which of the following is the best advice?
A. Reducing caloric intake
B. Exercising 30 minutes per day on 5 days per week
C. Diet low in protein and relatively higher in carbohydrates
D. Exercise for at least 30 minutes most days of the week and low carbohydrate diet

Answer: D
A 29-year-old woman comes to the clinic for a follow-up. She is diagnosed with gestational diabetes mellitus (GDM) at week 24. She is asking for an advice regarding the best dietary habits she can follow to control her blood sugar and avoid insulin injection.
Which of the following is the best diet option?
A. Low fat diet
B. Mediterranean diet
C. Low glycemic index diet
D. Energy restriction is not recommended during pregnancy

Answer: C
A 62-year-old woman receives a diagnosis of prediabetes based on routine lab work showing mildly elevated glycated hemoglobin (HbA1c. She has had obesity for many years. She eats a diet rich in processed foods. She rarely exercises. Physical exam reveals mildly elevated blood pressure along with trace acanthosis nigricans at the nape of the neck.
Blood Pressure 145/87 mmH
BMI 30 kg/m2
What advice should be given to reduce her risk of developing diabetes, in addition to increasing her physical activity to >150 minutes per week?
A. Low glycemic index diet to produce 5-10 % weight loss
B. Low fat diet to produce 5-10 % weight loss
C. Diet is less effective than metformin to reduce the risk of diabetes
D. Low carbohydrate diets are more likely than low fat diets to prevent diabetes in the long-term

Answer: B
A 40-year-old man comes to the clinic for evaluation and management of obesity. His obesity is complicated by type 2 diabetes mellitus (T2DM), hypertension, sleep apnea, dyslipidemia, chronic low back pain, osteoarthritis of the knees, and depression. He was prescribed medications for these problems but that he is not compliant to treatments. His wife accompanied him to the visit was anxious about his health and she want the doctor to council him for the best solutions he can do.
BMI 55 kg/m2
Which of the following is the most appropriate advice will help in treating this patient?
A. Emphasizing the importance of adherence to his medications
B. Referral to an orthopedic surgeon for knee arthroplasties
C. Addition of pharmacotherapy for obesity
D. Referral for bariatric surgery

Answer: D
A 39-year-old woman with type 1 diabetes for 20 years was married recently; she and her husband want to get pregnant. The woman has hypertension treated with an angiotensin converting enzyme

(ACE) inhibitor and a diuretic with maintenance of blood pressure at 118/80 mmHg. Her insulin doses are all administered through a continuous insulin infusion pump using insulin aspart. She reported that she would try to become pregnant next month (see lab results). Test Result Normal Values
24 hr urine protein
2000 0.8-14 mg/di (Male)
2.5-11 mg/dI (Female)
Creatinine 88 44-115 mol/L
HbA1C 8.0 4.7-5.6 %
What is the best advice at this moment to control her blood sugar and prevent congenital malformation?
A. Eliminate snacks and begin to take 3 heavy meals daily
B. Substitute an angiotensin receptor blocker for her ACE inhibitor
C. Her age and risk factors indicate the need to begin taking a statin at this time
D. Delay conception until her HbA1c is as close to normal as possible, if this can be safely achieved

Answer: D
A 25-year-old woman pregnant at 15 weeks presents to the clinic complaining of palpitation, tremor and diarrhea for the last 3 weeks. On examination, she looks anxious, she has graves ophthalmopathy and goiter. Other systemic examination are unremarkable. Diagnosis of grave's disease was confirmed but she is refusing to start any treatment while she is pregnant (see lab results).
Blood pressure 110/65 mmHg Heart rate 125 /min Respiratory rate 18 /min Temperature 37.6 °C BMI 20 kg/m2
Test Result Normal Values
Thyroid-Stimulating Hormone < 0.01 0.4 - 5.0 U/mL
Triiodothyronine (T3 free serum) 10.0 3.5 - 6.5 pmol/L
Thyroxine (T4 free serum) 35.0 8.5 - 15.2 pmol/L
Which of the following is the most appropriate action?
A. Agree with her opinion and discharge her from the clinic
B. Call her husband and force her to start the treatment
C. Give her follow up after 2 months and repeat her thyroid function test
D. Explain clearly the risk for her and the fetus and try to convince her to start the treatment Immediately

Answer: D
A 45-year-old man presented to the clinic with large thyroid nodule otherwise asymptomatic, that was proven to be papillary thyroid carcinoma. The physician is planning to send him for total thyroidectomy but the patient is afraid from the surgical complication. The treating physician needs to create a lasting impression at the end of the consultation.
Which of the following is the most appropriate action?
A. Avoid answering all patient's questions
B. Patient's agreement for referral is not required
C. Avoid discussing the serious surgical complications
D. Briefly explain the action plan with possible expected and unexpected outcome

Answer: A
A 22-year-old man developed traumatic paraplegia secondary to a road traffic accident about 8 months ago, requiring prolonged immobilization. His general physical and systemic examination were unremarkable and intact PTH levels (see lab results).
Test Result Normal Values
Calcium 2.75 2.15-2.62 mmol/L
Phosphate, inorganic 0.9 0.82-1.51 mmol/L

Parathyroid hormone 1.0 1.1-5.3 pmol/L
Which of the following is the most probable underlying aetiology for his hypercalcaemia?
A. Malignancy
B. Immobilization
C. Multiple myeloma
D. Primary hyperparathyroidism

Answer: B
A 40-year-old man was referred to the endocrine clinic with symptoms of reduced libido and lack of energy for the last 10 months. On examination, he had normal general physical and systemic examination (see lab results and report).
BMI 40 kg/m2
Test Result Normal Values
Thyroid-Stimulating Hormone 0.3 0.4 - 5.0 wU/mL. Thyroxine (T4 free serum) 7.5 8.5 - 15.2 pmol/L
Follicle-stimulating hormone 2.5 5-20 IU/L (Follicular phase) 15-35 IU/L (Mid cycle peak)
5-15 IU/L (Luteal phase)
50-100 IU/L (Postmenopausal women)
5-20 IU/L (men)
Testosterone 8.0 10.4-41.6 mol/L (Male)
0.7-2.8 nmol/L (Female)
Luteinizing hormone 1.3 3-15 U/L (Male)
5-22 U/L (Follicular phase)
5-22 U/L (Luteal phase)
30-250 U/L (ovulatory phase)
> 30 U/L (Postmenopausal)
Prolactin 450 < 652 pmol/L (Male)
< 870 pmol/L (Female)
MRI brain:
Showed 2.5 cm pituitary adenoma.
Which of the following is the most likely diagnosis?
A. Morbid obesity
B. Macroprolactinoma
C. Primary hypothyroidism
D. Non-functioning pituitary adenoma

Answer: D
An 18-year-old boy was referred to the Endocrine Clinic with features of short height and weight gain. On examination, he had increased BMI and central adiposity together with purple striae on lower abdomen (see lab results).
Test Result Normal Values
ACTH 20 2-11 pmol/L
Cortisol 8 a.m. 600 138-635 mol/L
Cortisol 4 p.m. 550 83-441 mol/L
Cortisol 1 hr after cosyntropin > 500 > 498 mol/L
Which of the following is the most appropriate next step in his management?
A. Bilateral inferior petrosal sinus sampling
B. CT abdomen and thorax
C. MRI of the pituitary
D. Octreotide scan

Answer: C
A 25-year-old man was referred to the Endocrine Clinic with a recent history of fragility fracture. He had no significant past medical history and was on no medication. On examination, he had sparse

facial and axillary hair growth. His systemic examination was unremarkable.
BMI 23 kg/m2
Which of the following investigations useful in establishing the aetiology of his fragility fractures?
A. Calcitonin level
B. DEXA bone densitometry scan
C. Insulin like growth factor (IGF1)
D. Testosterone and gonadotrophin levels

Answer: D
A 40-year-old man was incidentally detected to have a benign 2.0-cm right adrenal adenoma on CT of the abdomen, which was done to evaluate his symptom of abdominal pain. On examination, he had unremarkable general physical and systemic examination. Blood pressure 130/70 mmHg, BMI 26 kq/m2 Which of the following is the most appropriate next step in management?
A. MIBG scan
B. MRI adrenals
C. Surgical referral
D. Dexamethasone suppression test (overnight) and urinary metanephrines

Answer: D
A 35-year-old man was found to have a 5.5-cm left adrenal mass with irregular borders and a density of > 10 HU on non-contrast CT of the abdomen. On contrast-enhanced CT of the abdomen, there is <30% contrast wash out at 15 minutes. In view of the suspicious radiological features of the mass, the patient was referred to the endocrine surgical team for an elective adrenalectomy. Which of the following investigations should be done prior to an elective adrenalectomy?
A. 24-hour urinary fractionated metanephrines
B. Aldosterone to plasma renin activity ratio
C. Fine needle aspiration cytology
D. PET scan

Answer: A
Explanation: If Non-CECT shows attenuation of 10 or more you must obtain 24-Urinary metanephrines

Nephrology

10. A 30-year-old man comes to the Outpatient Clinic because of the recent development of marked peripheral edema. He has no significant past medical history. On examination, he has 4+ edema of the lower extremities (see lab results and reports).

Blood pressure 105/80 mmHg

Test Result Normal Values
Albumin 18 34-56 g/L
Urea 5 2.8-8.9 mmol/L
Creatinine 85 44-115 mol/L

Dipstick analysis:
4+ proteins.

Urinalysis:
Oval fat bodies, and some hyaline casts and occasional RBCs.

Which of the following is the most likely diagnosis?
A. Interstitial nephritis
B. Nephritic syndrome
C. Nephrotic syndrome
D. Ischemic nephropathy

Answer: A

Explanation: Only because there are RBCs.

9. A 29-year-old woman is diagnosed with rapidly progressive glomerulonephritis. Which of the following pathological changes is characteristic of this diagnosis?
A. Dilated proximal tubules with inflammation
B. Crescent formation within Bowman's space
C. IgA deposition in glomerular capillaries
D. Fat globules in urine sediment

Answer: B

A 75-year-old type 2 diabetic undergoes a bowel resection for colon cancer. He is well prior to the operation with well-controlled diabetes and no other underlying condition. The operation is successful and the patient is given postoperative insulin and IV dextrose. 2 days after the operation, he becomes very agitated (see lab results).

Test Result Normal Values
Sodium 124 134-146 mmol/L
Potassium 3.2 3.5-5.1 mmol/L
Creatinine 80 44-115 umol/L
Blood urea nitrogen 4.6 2.8-8.9 mmol/L
Serum osmolality 270 280-300 mmol/kg

24 Urine Analysis
Test Result Normal Value
Osmolality 310 280-910 mOsm/Kg H20

Which of the following is the most likely cause?

A. Water overload
B. Addison's disease
C. Diabetic nephropathy
D. Inappropriate antidiuretic hormone

Answer: D

This is a case of euvolemic hypotonic hyponatremia. The way to differentiate between water overload “primary polydipsia” and SIADH is by urine osmolality. Low osmolality means low ADH. High osmolality means high ADH.

A 60-year-old man presents to Emergency Department with altered sensorium. He was recently diagnosed to have inoperable small cell carcinoma of the lung. On examination, he is euvolemic (see lab results).
Blood pressure 110/70 mmHg
Heart rate 76 /min
Test Result Normal Values
Sodium 115 134-146 mmol/L
Potassium 3.8 3.5-5.1 mmol/L
Urea 4.3 2.75-7.4 mmol/L
Creatinine 76 44-115 mol/L
Which of the following is the best initial management?
A. 5% destrose
B. Normal saline
C. Hypertonic saline
D. Half normal saline

Answer:C

A 70-year-old man presented to Emergency Department with generalized fatigability. 2 days ago, he had CT brain with contrast for possible transient ischemic attack. Physical examination is unremarkable (see lab results).
Blood pressure 130/70 mmHg
Heart rate 76 /min
Test Result Normal Values
Hb 170 130-170 g/L (Male)
120-160 q/L (Female)
Urea 30.4 2.75-7.4 mmol/L
Creatinine 378 44-115 mol/L
Which of the following is the most likely mechanism for the renal impairment?
A. Pre-renal azotemia
B. Acute tubular necrosis
C. Acute interstitial nephritis
D. Acute glomerular nephritis

Answer:B (Contrast Induced Nephropathy)
A 30-year-old man is found to have a repeated BP of 170/110 mmg. He is not on any medications. Physical examination is unremarkable (see lab results).
Test Result Normal Values
Sodium 138 134-146 mmol/L
Potassium 2.1 3.5-5.1 mmol/L
Urea 3.67 2.75-7.4 mmol/L
Creatinine 66 44-115 mol/L
Which of the following is the most likely diagnosis?

A. Primary hyperaldosteronism
B. Polycystic kidney disease
C. Coarctation of the aorta
D. Primary hypertension

Answer: A
A 20-year-old man presents with hematuria, 1-day following a throat infection. Which of the following is the most likely diagnosis?
A. IgA nephropathy
B. Membranous nephropathy
C. Post-infectious glomerular nephritis
D. Focal segmental glomerulosclerosis

Answer: A
A 22-year-old woman is admitted with pulmonary hemorrhage and rapidly progressive glomerulonephritis. She has history of recurrent sinusitis and numbness in her right upper limb and left lower limb. Which of the following is the most likely diagnosis?
A. Giant cell arteritis
B. Polyarteritis nodosa
C. Henoch-Schonlein purpura
D. Granulomatosis with polyangitis

Answer: D
A 50-year-old man with diabetes, hypertension and chronic kidney disease. presents to the Nephrology Clinic for follow-up (see image and lab results).
Blood pressure 160/98 mmHg, Heart rate 80 /min
Test Result Normal Values
Potassium 6.5 3.5-5.1 mmol/L
Creatinine 440 44-115 mol/L
Which of the following is the best immediate step in management?
A. Intravenous calcium gluconate
B. Intravenous NAHCO3
C. Insulin with dextrose
D. Dialysis

Answer: A
An 18-year-old man is complaining of red urine for 5 days. He has no abdominal pain and no dysuria, urgency or frequency. His history is significant for tonsillitis 3 weeks ago. Physical examination is normal except for elevated blood pressure. Blood pressure 160/100 mmH, Heart rate 76 /min
Which of the following is the most likely diagnosis?
A. Kidney stone
B. Hypernephroma
C. IgA nephropathy
D. Post streptococcal glomerulonephritis

Answer: D
A 68-year-old woman known to have dilated cardiomyopathy with an ejection fraction of 30% one month ago is admitted with a diagnosis of acute gastroenteritis. She had repeated vomiting and watery diarrhea for 3 days. Her urine output is reduced in volume. She is on furosemide, lisinopril, spironolactone, and carvedilol. On examination, she is dehydrated, confused, has cold clammy skin and the JVP is 1 cm above the sternal angle (see lab results and reports).
Blood pressure 88/55 mmH
Heart rate 70 /min

Test Result Normal Values
Sodium 142 134-146 mmol/L
Potassium 5.5 3.5-5.1 mmol/L
Urea 230 2.75-7.4 mmol/L
Creatinine 220 44-115 mol/L
Sodium 15 40-230 mmal/day
Osmolality 1300 280-910 mOsm/Kg H20
Central venous line: inserted and is 3 cm of H20.
Fractional excretion of sodium: 0.6%.
What is the best option in management?
A. Increase the dose of furosemide
B. Intravenous dobutamine infusion
C. Careful intravenous saline infusion
D. Increase the dose of spironolactone

Answer:C

A 62-year-old man presents with progressive lower limbs edema for the last 4 months otherwise he is asymptomatic. He is known to have diabetes complicated with diabetic nephropathy and hypertension. His current medications are insulin, thiazide, ramipril and amlodipine. On examination blood pressure is 144/78 mmg, JVP not raised, +3 lower limbs edema with normal cardiac and respiratory examination. Renal function was similar to 3 months ago (see lab results).
Test Result Normal Values
Sodium 138 134-146 mol/L
Potassium 4.8 3.5-5.1 mmol/L
Urea 12.9 2.75-7.4 mmol/L
Creatinine 230 44-115 mol/L
Creatinine clearance 22 70-103 mol/L
which of the following is the most appropriate action?
A. Switch thiazide to furosemide
B. No change to his medication
C. Add a beta-blocker
D. Stop ramipril

Answer: A
A 55-year-old man who has developed chronic kidney disease secondary to focal segmental glomeruloscierosis is reviewed in the clinic. His renal function is deteriorating. Haemodialysis is planned to start soon (see lab result).
Test Result Normal Values
Creatinine clearance 10 70-103 mol/L
What is the preferred method of access to start Haemodialysis?
A. Arteriovenous graft
B. Arteriovenous fistula
C. Tunelled venous catheter
D. Non-tunelled venous catheter

Answer: B
which of the following scenario with acute kidney injury should be referred for urgent Haemodialysis?
A. Patient with eGFR < 5
B. Potassium of 7.5 after 3 courses of medical management
C. Patient who has had 10ml of urine output over the last 24 hours
D. Patient who has previously had acute kidney injury three times

Answer: B
A 65-year-old man has been admitted with abdominal pain and he need urgent contrast- enhanced CT but there is a concer regarding his kidney status as he has chronic kidney disease stage 3 (see lab results).
Test Result Normal Values
Sodium 143 134-146 mmol/L
Potassium 4.5 3.5-5.1 mmol/L
Blood urea nitrogen 9 2.8 to 8.9 mmol/L
Creatinine 160 44-115 mol/L
Which of the following is the most important step in reducing the risk of contrast-induced nephropathy?
A. Oral N-acetylcysteine
B. Oral sodium bicarbonate
C. Intravenous furosemide pre-procedure
D. Intravenous 0.9% sodium chloride pre- and post-procedure

Answer: D
A 55-year-old man presents in the Emergency Department with nausea, pallor and lethargy. He has no past medical history of note. On examination, he has normal vital signs and unremarkable systemic examination (see lab results and report).
Test Result Normal Values
Sodium 139 134-146 mol/L
Potassium 6.9 3.5-5.1 mmol/L Bicarbonate 14 21-28 mmol/L.
Blood urea nitrogen 14 2.8 to 8.9 mmol/L Creatinine 240 44-115 mol/L
ECG: Showed tall peaked T waves.
Which of the following is the most appropriate initial management?
A. Haemodialysis
B. Nebulised salbutamol
C. Insulin dextrose infusion
D. Intravenous calcium gluconate

Answer: D
A 55-year-old man is seen in the Emergency Department complaining of lethargy for the last 2 days. On physical examination, vital signs are stable with unremarkable physical examination. ECG shows no acute changes (see lab results).
Test Result Normal Values
Sodium 139 134-146 mmol/L
Potassium 6.6 3.5-5.1 mmol/L Bicarbonate 15 21-28 mmol/L
Blood urea nitrogen 9 2.8 to 8.9 mmol/L Creatinine 118 44-115 mol/L
Which of the following is the most appropriate initial treatment?
A. Haemodialysis
B. Intravenous bicarbonate
C. Insulin dextrose infusion
D. Intravenous normal saline

Answer: C
A 26-year-old woman is recently diagnosed as nephrotic syndrome secondary to minimal change glomerulonephritis after being presented with proteinuria.
Which of the following is the most appropriate initial treatment to reduce proteinuria?
A. Prednisolone
B. Low protein diet
C. No treatment required

D. Angiotensin-converting-enzyme inhibitor

Answer: A
A 30-year-old woman with Crohn's disease presents with hypokalemia refractory to aggressive paranteral and oral KCl supplementation. Which of the following is the next step in managment?
A. Assess her serum plasma aldosterone to plasma renin ratio
B. IV magnesium sulfate
C. Start spironolactone
D. Observation

Answer: B
A 60-year-old woman presents to the clinic with a 10-year history of type 2 diabetes treated with insulin. Recently diagnosed to have hypertension and proteinuria (see lab results).
Blood pressure 150/90 mmHg
Heart rate 87 /min
BMI 29 kg/m2
Test Result Normal Values
Blood urea nitrogen 4 2.8 to 8.9 mmol/L
Creatinine 80 44-115 mol/L
Glucose, fasting 6.4 3.5-6.5 mmol/L
HbA1C 7 4.7-5.6 %
Which of the following is the most appropriate management?
A. Life style modification for 6 months
B. Increase Insulin doses
C. Start ACE inhibitors
D. Start atenolol

Answer: C

Which of the following is the most useful therapy to reduce stone formation in a patient with hypercalciuria?
A. Allopurinol
B. Pencillamine
C. Thiazide diuretics
D. Dietary calcium restriction

Answer: C
A 50-year-old man presented to the clinic with recent onset of numbness of his feet and hands History revealed he has chronic kidney disease for 3 years. Examination showed decrease sensation to pinprick and vibration stimuli below the knee with absent ankle reflexes (see lab results).
Blood pressure 140/90 mmHg
Test Result Normal Values RBC 4 4.7-6.1 x 1012/L (Male) 4.2-5.4 x 1012/L (Female)
Hb 9 130-170 g/L (Male)
120-160 g/L (Female)
MCV 73 80-95 fl
Potassium 5.3 3.5-5.1 mmol/L
Urea 46 2.75-7.4 mmol/L
Creatinine 674 44-115 umol/L
Which of the following is the most appropriate next step in the management?
A. Erythropietin
B. Oral bicarbonate
C. Vitamin B complex
D. Initiation of dialysis

Answer: D

A 45-year-old woman with systemic lupus erythematosus comes for routine follow-up visit. She was asymptomatic and currently on no medications. Her condition is stable for several years (see lab results).
Blood pressure 140/90 mmH
Test Result Normal Values
24 hr urine protein 35 0.8-14 mg/di (Male)
2.5-11 mg/di (Female)
Protein 500 0-150 mg/24 hr
RBC 5000 up to 4,000 in 1 ml
Urea 20 2.75-7.4 mmol/L
Creatinine 160 44-115 mol/L
Which of the following is the most appropriate next step in management?
A. Anti Ds-DNA
B. Renal biopsy
C. Doppler renal ultrasound
D. Repeat urine analysis in next visit

Answer: B
A 58-year-old man with diabetes mellitus came for follow-up. He is on metformin (Glucophage), 1000 mg/day, as well as atorvastatin (Lipitor), 40 mg daily for hypertipidaemia. There is no diagnosis of hypertension. His urine albumin/creatinine ratio is in the microalbuminuria range for the first time (see lab results). Blood pressure 128/77 mmH
Test Result Normal Values
HA1C 6.3 4.7-5.6 %
LDL (cholesterol) 2.1 <4.0 mol/L
Which of the following would be most appropriate next step?
A. Renal ultrasonography
B. Repeat urine albumin/creatinine ratio
C. Stopping metformin
D. 24-hour urine for microalbumin

Answer: B
A 25-year-old woman presents with symptoms of extreme fatigue, no appetite, and bloody urine She developed a very sore throat 3 weeks ago but did not have it examined or treated. 3 days ago, she began to have bloody urine and swelling of her hands and feet. Her past health has been excellent. There is no family history of significant illness. She has no allergies. On examination, she has significant edema of both lower extremities. Blood pressure 166/108 mmHg, Heart rate 82 /min
Which of the following is the most likely diagnosis?
A. Berger disease
B. Immunoglobulin A nephropathy
C. Poststreptococcal glomerulonephritis
D. Membranous glomerulonephritis

Answer: C
A 51-year-old man with chronic and progressing renal disease. His GFR is decreasing and most likely he will need dialysis sometime within the next year. Which of the following will be the most likely cause of death in his case?
A. Renal failure
B. Coagulopathy
C. Pulmonary edema

D. Cardiovascular disease

Answer: D

A 77-year-old man with a history of ischemic heart disease is seen in the Hypertension Clinic. He is taking 4 anti-hypertensives drugs. Physical examination is unremarkable. Abdominal ultrasound shows asymmetrical kidneys. Blood pressure 169/99 mmHg, Heart rate 79 /min

Which of the following is the most likely diagnosis?

A. Renal artery stenosis
B. Cushing's syndrome
C. Primary hyperaldosteronism
D. Adult polycystic kidney disease

Answer: A

A 44-year-old man who was recently diagnosed as chronic kidney disease stage 4, presented to the clinic to discuss the diet in regard to his condition. He has read that potassium rich food should be consumed in moderate amount. Which of the following foods should he eat in moderation?

A. Grapes
B. Tomatoes
C. Green beans
D. Cranberry juice

Answer: B

A 28-year-old woman who is 38 weeks pregnant presents with dysuria and urinary frequency. A urine dipstick is positive for nitrites and leucocytes. Which of the following is the most appropriate management?

A. Cefalexin
B. Doxycycline
C. Nitrofurantoin
D. Advise to increase fluid intake

Answer: A

A 27-year-old man who has a sister with adult polycystic kidney disease came to screen for the disease. Which of the following is the most appropriate screening test?

A. CT abdomen
B. Urine microscopy
C. Ultrasound abdomen
D. Anti-polycystin 1 antibodies levels

Answer: C

A 61-year-old man with stable chronic kidney disease stage 3. Which of the following drugs is it safe to prescribe for this patient?

A. Nitrofurantoin
B. Metformin
C. Warfarin
D. Lithium

Answer: C

A 58-year-old man with chronic kidney disease has been on haemodialysis for chronic kidney disease for the past 8 years. What is most likely cause of mortality in such case?
A. Ischaemic heart disease
B. Dialysis related sepsis
C. Hyperkalaemia
D. Malignancy

Answer: A

A 23-year-old pregnant woman presented with frequency and dysuria for 5 days (see lab result), Blood pressure 110/70 mmHg, Heart rate 76 /min, Temperature 37.6 °C
Test Result Normal Values
Leukocytes 30 0-3 per high power field
Which of the following is the best treatment option?
A. Flucioxacillin
B. Augmentin
C. Ciprofioxacin
D. Nitrofurantion

Answer: B

A 70-year-old man admitted with left lower limb swelling and pain diagnosed as DVT where heparin was started. He has a history of diabetes mellitus, hypertension and chronic heart failure. He is on insulin; furosemide and enalapril (see lab results). Blood pressure 110/70 mmHg, Heart rate 76 /min, Oxygen saturation 94 %
Test Result Normal Values
Sodium 135 134-146 mmol/L
Potassium 6,0 3.5-5.1 mmol/L
Chloride 107 97-108 mmol/L
Urea 7.9 2.75-7.4 mmol/L
Creatinine 120 44-115 mol/L
APTT 50 30-40 sec
Prothrombin time 14 10-13 sec
Which of the following medications should be discontinued at this stage?
A. Insulin
B. Enalapril
C. Heparin
D. Furosemide

Answer: B

A 78-year-old man was admitted to hospital with lethargy and back pain. He had a history of type 2 diabetes mellitus and rheumatoid arthritis (see lab results).
Test Result Normal Values
Calcium 2.72 2.15-2.62 mol/L Creatinine 196 44-115 mol/L
Urinary protein: creatinine ratio 154 mg/mmol (<30)
Which of the following is most likely diagnosis?
A. Primary hyperparathyroidism
B. Membranous nephropathy
C. Diabetic nephropathy
D. Primary amyloidosis
Answer: C

Neurology

A 58-year-old hypertensive patient presented with a 10-day history of a left-sided hemiparesis. The CT scan of the brain confirmed an area of infarction. There was no significant finding on a clinical examination apart from his hemiparesis. The patient had already started physiotherapy and is taking 20 mg lisinopril and hydrochlorothiazide once daily.
Blood pressure 140/90 mmHg
Heart rate 76 /min
Respiratory rate 18 /min
Temperature 36.6° C
Oxygen saturation 95% on room air

Which of the following is the best management?
A. Apixaban
B. Warfarin
C. Aspirin
D. T-PA1

Answer: C

A 55-year-old woman presents with sudden left eye visual loss for 20 minutes. The vision returned to normal after this episode. History reveals, she has diabetes mellitus. The physical examination is normal.

Blood pressure 125/75 mmHg
Heart rate 75 /min
Respiratory rate 16/min
Temperature 37.2° C

Which of the following is the most likely diagnosis?
A. Multiple sclerosis
B. Retinal detachment
C. Conversion disorder
D. Transient ischemic attack

Answer: D

A 45-year-old patient presents to the clinic with a 3-month history of stumbling, weak grip, dysphagia, and generalized weakness, and also had 2 episodes of aspiration pneumonia in the last 2 months. Examination confirms brisk reflexes, spastic muscles, and fasciculation of the tongue and thigh.

Which of the following is the most likely diagnosis?
A. Mononeuropathy
B. Myasthenia gravis
C. Myasthenic syndrome
D. Motor neuron disease

Answer: D

Explanation: signs of both upper and lower motor lesions on Px.
UMN: Spasticity, brisk reflexes
LMN: Fasciculation

A man on treatment for multiple sclerosis presents with acute onset diplopia, ataxia, and weakness of the right hand (see report).

MRI brain scan:
New active lesions compared to previous scans.

Which of the following is the best treatment for this development?
A. Interferon
B. I.V. corticosteroids
C. I.V. immunoglobulin
D. Oral corticosteroids

Answer: B

A 71-year-old man comes to the Outpatient Clinic with difficulty walking and a hand tremor at rest for the last 3 months. There is no history of trauma or past illness. General examination reveals an expressionless face, slow and slurred speech, and a hand tremor at rest. There is no sensory abnormality, but the motor system showed brisk reflexes and hypertonia. The gait is quick, with short steps and diminished associated movements, such as no arm swinging. Which of the following is the most probable location of the defect?
A. Cerebellum
B. Frontal lobe
C. Lateral ventricle
D. Substantia nigra

Answer: D

A 70-year-old patient presents with a 6-month history of decline in cognitive capacity and conscious level. On examination, the patient has bradykinesia, resting tremors, a mask-like face, and a shuffling gait.

Which of the following is the most likely diagnosis?
A. Parkinson disease
B. Tardive dyskinesia
C. Alzheimer disease
D. Huntington's disease

Answer: A

A 32-year-old patient is having repetitive generalized seizures that continue for 35 minutes. The patient received 20 mg of diazepam intravenously, but there has been no response. Which of the following is the best next step in management?
A. Phenytoin IV
B. Phenobarbital IV
C. Ethosuximide orally
D. Carbamazepine orally

Answer: A

A 44-year-old woman comes to her physician as she is unable to close her left eye. There is a facial asymmetry when she smiles and inability to wrinkle the left side of her forehead. Which of the following nerves is mostly affected?
A. Ophthalmic
B. Maxillary

C. Facial
D. Optic

Answer: C

A 35-year-woman comes to the clinic complaining of severe "pins and needles" sensations on the dorsum of her left hand over parts of the thumb, index and middle fingers. History reveals, she works as a typist. On physical examination, the patient has weakness of wrist dorsiflexion and fingers extension.
Which of the following nerves is most likely to be injured?
A. Axillary
B. Median
C. Radial
D. Ulnar

Answer: C

A 45-year-old man presents to the clinic with bilateral ptosis and weakness of the proximal upper and lower limb muscles. Physical examination is normal except for a positive Simpson test. Edrophonium test is performed that shows transient improvement in the ptosis (see report).

Chest X-ray: Normal.

Which of the following would be the most suitable treatment?
A. Rivastigmine
B. Pyridostigmine
C. Physostigmine
D. Echothiophate

Answer: B

Explanation: Pyridostigmine is the first line for MG because it doesn’t cross the brain barrier.
Rivastagmine used for alzheimer's
Physostigmine used for atropine toxicity

A 77-year-old woman with poorly controlled long-standing type 2 diabetes and hypertension is complaining of dizziness when standing up rapidly. Current medications include aspirin, atenolol, insulin, metformin, and multivitamins.
Sitting:
Blood pressure 140/85 mmHg

Standing:
Blood pressure 115/80 mmH
Heart rate 95 /min (unchanged on standing)
Respiratory rate 18 /min
Temperature 36.6° C

Which of the following is the most likely explanation?
A. Hypoglycemia
B. Medication side effect
C. Autonomic neuropathy
D. Left ventricular dysfunction

Answer: C

A 43-year-old woman with complex regional pain syndrome comes to the clinic. She suffers from pain around her hand and wrist, which started after carpal tunnel release surgery. She was reviewed by orthopaedics and a specialist in Pain Clinic. Which of the following is the most appropriate management option?
A. Opiate analgesia
B. Physiotherapy
C. Counselling
D. Triptans

Answer: B

A 41-year-old man complains of double vision. Whilst looking forward the patients left eye turns towards the nose. On looking to the patient's right there is no obvious squint. However, on looking to the left, the patient is unable to abduct the left eye and double vision worsens. Which of the following is the most likely cranial nerve palsy?
A. Left 3rd
B. Left 6th
C. Right 3rd
D. Right 6th

Answer: D
A 42-year-old man comes to the clinic with his brother. The brother is concerned that he has been acting strangely over the past 4 months. In particular, he has been irritable and criticizes everyone at home. The changes were subtle at first but now arguments arise every other day. He has difficulty initiating conversation and even pauses halfway through sentences. There is no family history of note and he is not taking any medications. Which of the following is the most likely diagnosis?
A. Alzheimer's disease
B. Vascular dementia
C. Huntington's disease
D. Frontotemporal dementia

Answer: D
A 71-year old man with history of osteoarthritis and previous cervical laminectomy for degenerative cervical myelopathy presents with a 2-month history of worsening gait instability and urinary urgency. Which of the following is the most likely explains his symptoms?
A. Multiple sclerosis
B. Transverse myelitis
C. Cauda equina syndrome
D. Recurrent degenerative cervical myelopathy

Answer: D
A 66-year-old man is referred to Neurology Outpatients with arm pain, stiffness and imbalance. Following investigations, he is diagnosed with degenerative cervical myelopathy. He misses his next Outpatient Clinic due to admission with acute coronary syndrome. He attends his appointment 2 months later and mentions his ongoing neurological symptoms. Which of the following is the most important next step in his care?
A. Refer to spinal surgery
B. Refer for physiotherapy
C. Commence neuropathic analgesia
D. Reassure the patient of his diagnosis

Answer: A

A 50-year-old man with diabetes and hypertension, presents to the Emergency Department with sudden onset left sided body weakness for 4 hours. On examination, he has left sided upper motor neuron facial palsy and left sided hemiparesis.
Which of the following is the best next step in management?
A. Aspirin
B. Brain MRI
C. Clopidogrel
D. Brain CT-scan

Answer: D
A 16-year-old boy presents with a several days history of progressive arm and leg weakness.
No diplopia or loss of sphincter control. He has been well except for an upper respiratory infection 2 weeks ago. On examination, his breathing is slow and shallow. His heart rate and blood pressure are fluctuant. There is symmetrical weakness of the face and all four extremities. Deep tendon reflexes are absent, and sensations are intact.
Which of the following is the most likely diagnosis?
A. Polymyositis
B. Myasthenia gravis
C. Transverse myelitis
D. Guillain-Barre syndrome

Answer: D
A 45-year-old man presents with an 8-week history of severe headache. The headaches have occurred at the same time each day, about 2 a.m. and last around 30 minutes. Headaches are accompanied by lacrimation and redness of the right eye.
Which of the following is the best to prevent this type of headache?
A. Verapamil
B. Propranolol
C. Sumatriptan
D. 100% oxygen

Answer: A; While 100% is best abortive treatment for cluster headache.
A 55-year-old man with diabetes and hypertension complains of pain in the right eye and double vision when looking to the left. On examination, there is ptosis, limited adduction, elevation, and depression of the right eye. The pupils are equal and reactive in both sides. Fundoscopy and other cranial nerve are normal as well as the rest of neurological examination of the upper and lower limbs.Which of the following is the most likely cause of these findings?
A. Posterior communicating artery aneurism
B. Diabetic third nerve neuropathy
C. Cavernous sinus thrombosis
D. Mid brain tumor

Answer: B
An 18-year-old boy known to have grand mal epilepsy since childhood is brought to the Emergency Department after having repeated tonic-clonic seizures at home. He is compliant
with his medication, which is sodium valproate. In the Emergency Department, he is unconscious and continuously convulsing. After oxygenation and securing the airway, which of the following intravenous drugs should be administered?
A. Phenytoin
B. Diazepam
C. Thiopental
D. Phenobarbitone

Answer: B
In a patient with Guillain Barre Syndrome, which of the following is the best treatment option?
A. Pulse steroid
B. Cyclosporine
C. Plasmapheresis
D. Cyclophosphamide

Answer: C
A 30-year-old woman presents with pain in the right eye associated progressive visual loss over 2 days. She also reported a numbness in the left side of the face and weakness in both lower limbs. Physical examination revealed reduced visual acuity, loss of direct light reflex with preserved indirect light reflex in the right eye. Lower limbs examination showed weakness, clasp knife rigidity, hyperrefiexia in both knee and ankle jerks. There is loss of all modalities of sensation in both lower limbs.
Which of the following has the highest diagnostic value?
A. Electromyography
B. Brain and spine MRI
C. Thoracic spine CT scan
D. Nerve conduction study

Answer: B
An 80-year-old man has 2-month history of progressive gait disturbance and incontinence Lately he has been forgetful, confused, and withdrawn. His gait is slow, short-stepped and with tendency to fall backward (see report).
Brain CT scan: shows dilated ventricles.
Which of the following is the most likely diagnosis?
A. Alzheimer's disease
B. Creutzfeldt-Jakob disease
C. Fronto-temporal dementia
D. Normal-pressure hydrocephalus

Answer: D
An elderly patient sustained a major stroke and he is intubated in the intensive Care Unit. However, even he was in a very critical state, he managed to write in the bed clothes that he wants to die in peace. Few days later, he passed away without resuscitation according to what he wrote. His son was very angry and said that his father had no mental capacity to decide. What is the most appropriate stand when dealing with this situation?
A. Find out whether the son is the real next of kin
B. Tell the son that in anyway his father would have never made it
C. Inform the son that the doctors respected the wish of your father
D. Inform the son that we do not resuscitate patients who are brain dead

Answer: A
A young woman is diagnosed to have multiple sclerosis. She went through an emotional breakdown during breaking bad news. Which of the following is the most appropriate way to express empathy?
A. I share you your feelings and will support you"
B. "Pull yourself up and follow our instructions"
C. "I am very sorry for what happened"
D. "Do not worry, you will be fine"

Answer: A

A 65-year-old man presented with 6-month symptoms of Parkinsonism, early dementia. fluctuations in cognition and visual hallucinations. Which of the following is the most likely diagnosis?
A. Parkinson disease
B. Alzheimer disease
C. Corticobasal syndrome
D. Dementia with Lewy bodies

Answer: D
An 81-year-old man with memory loss and difficulty recognizing his grandsons. He has no cardiovascular risk factors. Which of the following is the most common cause of his cognitive impairment?
A. Alcohol-induced encephalopathy
B. Parkinson's associated dementia
C. Multi-infarct dementia
D. Alzheimer's disease

Answer: D
An 82-year-old woman at long term nursing facility who wears hearing aids and is found to have bilateral cerumen impaction, a Weber test that lateralizes to the right. and a bilaterally positive Rinne test. Which of the following is the most likely explanation?
A. Right-sided acoustic neuroma
B. Prebycusis that is greater on the right
C. Conductive hearing loss that is greater on the right
D. Chronic serous otitis media that is greater on the left

Answer: B
A 79-year-old man comes to the clinic for a follow-up. He has a 1-year history of progressive memory impairment and clinical evidence of early Alzheimer dementia. Which of the following is the most appropriate initial diagnostic tests?
A. Brain imaging
B. Lumbar puncture
C. Carotid ultrasound
D. Electroencephalogram

Answer:A
A 75-year-old man comes to the clinic with hypertension and gradual loss of cognitive function. He has predominant executive function impairment (see report).
MRI: Reveals diffuse periventricular white-matter hyperintensities
Which of the following is most likely diagnosis?
A. Alzheimer disease
B. Dementia with Lewy bodies
C. Vascular cognitive impairment
D. Normal pressure hydrocephalus

Answer: C
A 77-year-old man with a 7-years history of Parkinson disease reports hallucinations, confusion and worsening motor function. On physical examination, he is kinetic and rigid with no tremor. He has prominent gait disorder and postural instability.
Which of the following symptoms is most strongly associated with an increased risk for dementia?
A. Increase irritability
B. Word-finding difficulty

C. Trouble recalling appointment times
D. Difficulty reading an article in a single sitting

Answer: C
A son of 78-year-old patient brought a medical report of his father cognitive function status stating that he has" mild cognitive impairment'.
Which of the following is indicating mild cognitive impairment?
A. Low mood for 2 weeks
B. Localized motor dysfunction
C. Objective evidence of memory decline
D. Impairment in at least 1 activity of daily living

Answer: C
A 67-year-old man comes to the clinic complaining that he has a tremor in the right hand. It is best displayed when he reaches for a pen and is manifested easily in the finger-to-nose test. In which of the following locations is the lesion?
A. Left basal ganglia
B. Right cerebellum
C. Left motor cortex
D. Left cerebellum

Answer: B
A 72-year-old woman with Parkinson's disease, complaining of a very annoying tremor of the dominant right hand. She is not on any medication. She states that her tremor is the only feature that disturbs her functioning. Which of the medications would be most likely to improve her tremor?
A. Procyclidine
B. Amitriptyline
C. Bromocriptine
D. Carbidopa/levodopa

Answer: A
A 69-year-old man comes to the clinic complaining of increasing fatigue and drowsiness and difficulty to arouse. History reveals he was in a vacation 2 weeks ago. Although he admits to taking several falls, he is unaware of any head injury. Neurological examination showed no lateralizing or focal findings. Head and scalp examinations are negative for focal tenderness and deformity. Plain X-ray of the skull is negative (see lab results).
Test Result Normal Values
RBC 5.1 4.7-6.1 x 1012/L (Male) 4.2-5.4 x 1012/L (Female)
Hb 132 130-170 g/L (Male) 120-160 g/L (Female)
HCT 0.40 0.42-0.52 (Male) 0.37-0.48 (Female)
MCH 30 28-33 pg/cell
MCHC 322 320-360 g/L
MCV 88 80-95 fl
Reticulocyte 0.9 0.2-1.2 %
Platelets count 200 150-400 x 109/L
WBC 5.5 4.5-10.5 x 109/L
Neutrophils 50 40-60%
Eosinophils 2 1-4 %
Basophils 0.8 0.5-1 %
Lymphocytes 32 20-40 %
Monocytes 4.1 2-8 %
Calcium 2.3 2.15-2.62 mmol/L
Calcium ionized 1.2 1.1-1.3 mmol/L

Magnesium 0.8 0.7-1.2 mol/L
Thyroid-Stimulating Hormone 4.4 0.4 - 5.0 p/mL Thyroxine (T4 free serum) 12 8.5 - 15.2 pmol/L
Sodium 136 134-146 mmol/L
Potassium 4.2 3.5-5.1 mmol/L
Which of the following is the most likely diagnosis?
A. Stroke
B. Acute bacterial meningitis
C. Post-concussion syndrome
D. Chronic sub-dural hematoma

Answer: D
A 73-year-old diabetic man presents with right-sided weakness. The weakness is equal in the right face, arm and leg. Sensation, speech and comprehension are intact. Which of the following arteries is affected?
A. Vertebral artery
B. Mid-basilar artery
C. Anterior cerebral artery
D. Penetrating branch, middle cerebral artery

Answer: D
A 70-year-old man complains of a sudden severe headache while bending down to pick up his keys this morning. He describes it as the worst headache he has ever had; he says it was like being hit on the back of the neck. Which of the following is the most likely diagnosis?
A. Meningitis
B. Cluster headache
C. Tension headache
D. Sub-arachnoid hemorrhage

Answer: D
A 48-year-old woman has a 2-year history of a slowly progressive bilateral tremor. The tremor interferes with her writing and eating. She has recently noted head bobbing and a change in her voice. Which of the following would be an appropriate first-line medication for her problem?
A. Carbidopa/levodopa
B. Alprazolam
C. Propranolol
D. Olanzapine

Answer: C; Hyperthyroidism
A 45-year-old man presents with a sudden onset of unilateral peripheral facial nerve weakness 1 hour ago. He has no medical illness and not on any medication. Which of the following is most likely to shorten his symptoms?
A. Antiviral therapy
B. Thrombolytic therapy
C. Corticosteroid therapy
D. Hyperbaric oxygen therapy

Answer: C
A 36-year-old woman presented with 2-year history of recurrent headaches. These headaches occur 3 to 4 times per month and last 12 to 24 hours. The headaches are usually located over the right or left frontal sinuses. This particular headache has lasted 2 days now, and she has some nausea associated with it. The patient has been using an over-the counter analgesic-decongestant combination without significant relief. Neurological examination is within normal limits. The optic fundi are normal, Blood pressure 110/70 mmHg Heart rate 76 /min Temperature 36.6 °C

What is the appropriate next step to determine the etiology of this headache?
A. MRI scan of the brain
B. CT scan of the paranasal sinuses
C. Erythrocyte sedimentation rate
D. Careful history and physical examination

Answer:D